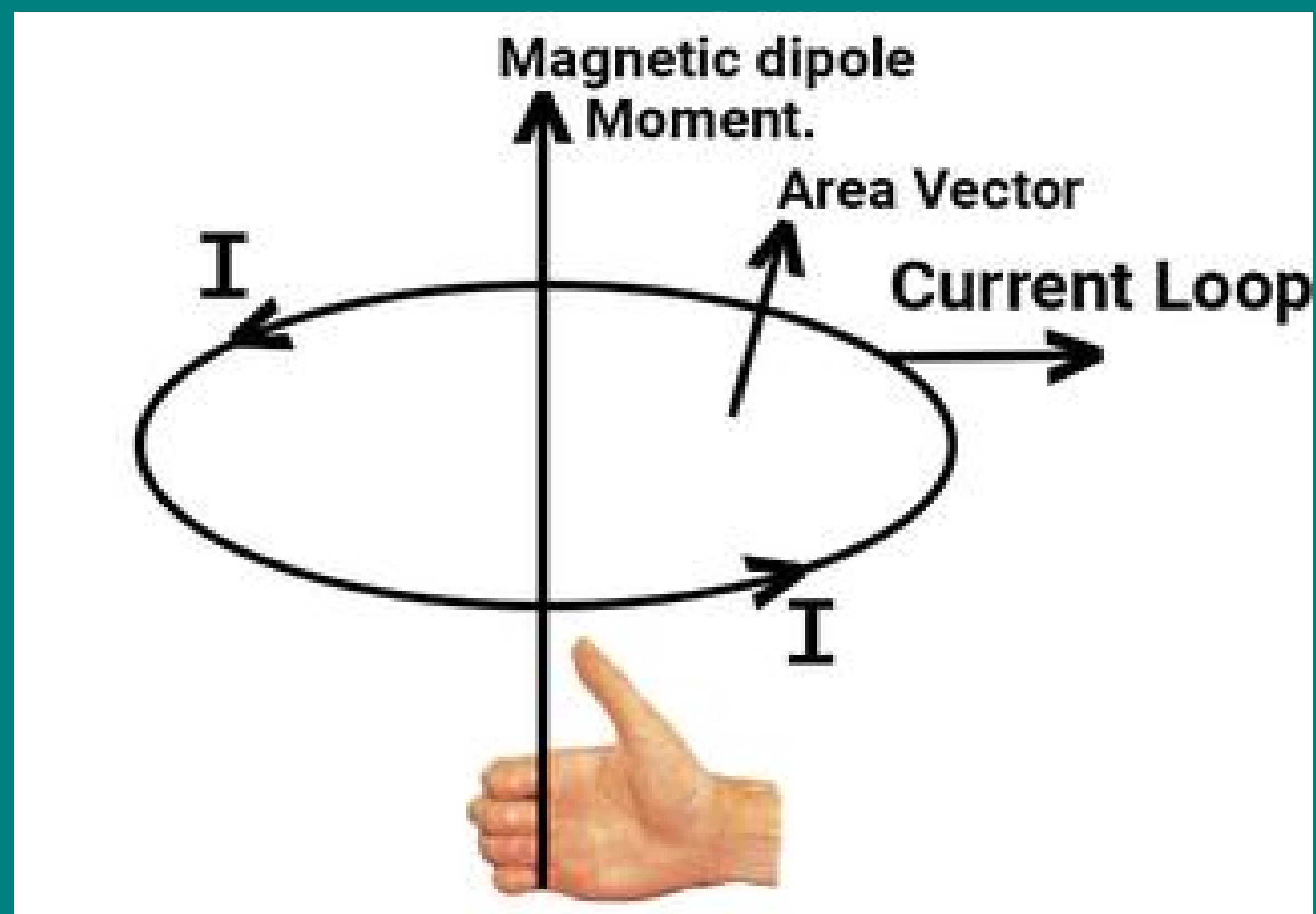
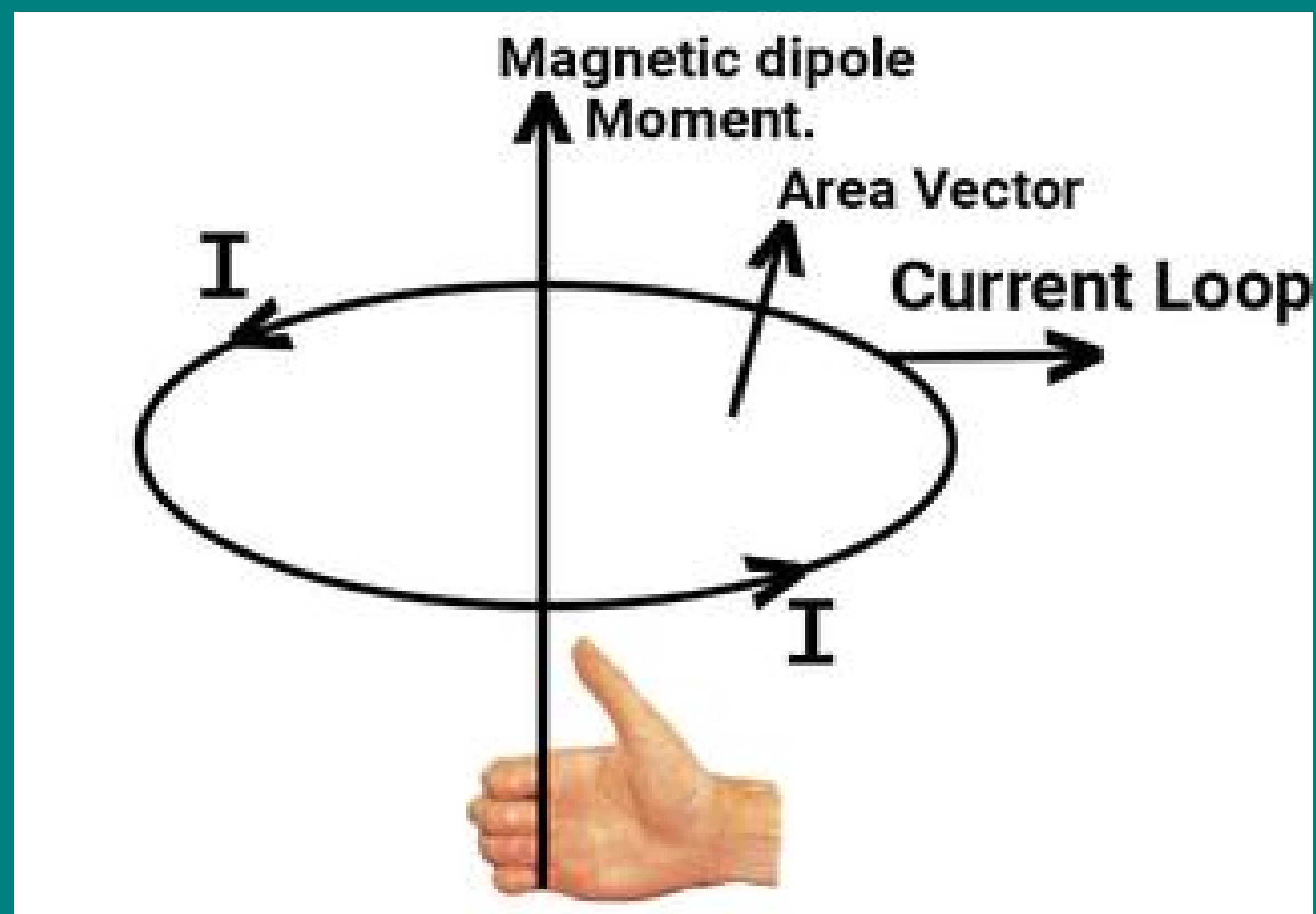


Circular current loop as a magnetic dipole



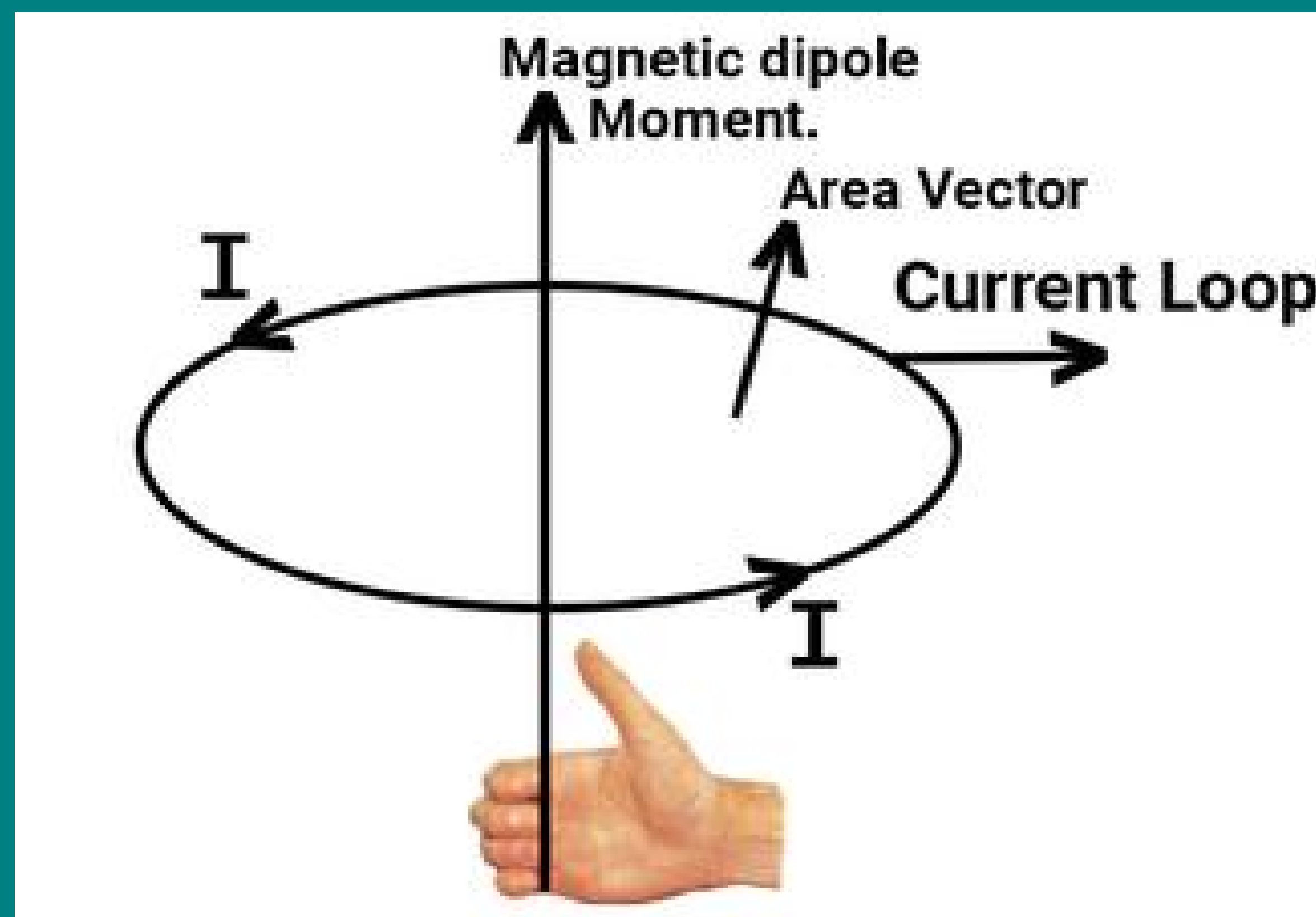
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Circular current loop as a magnetic dipole

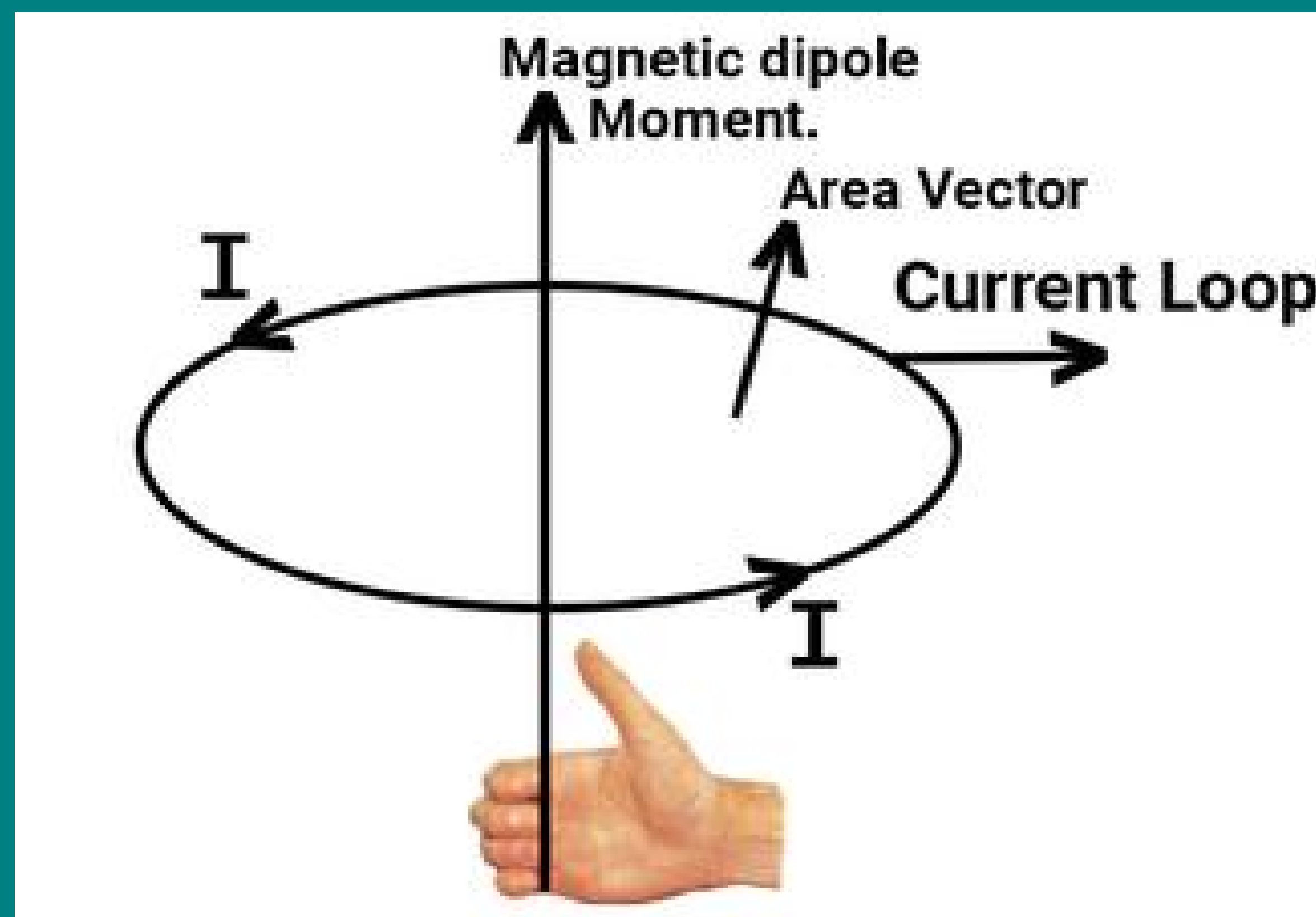


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Circular current loop as a magnetic dipole



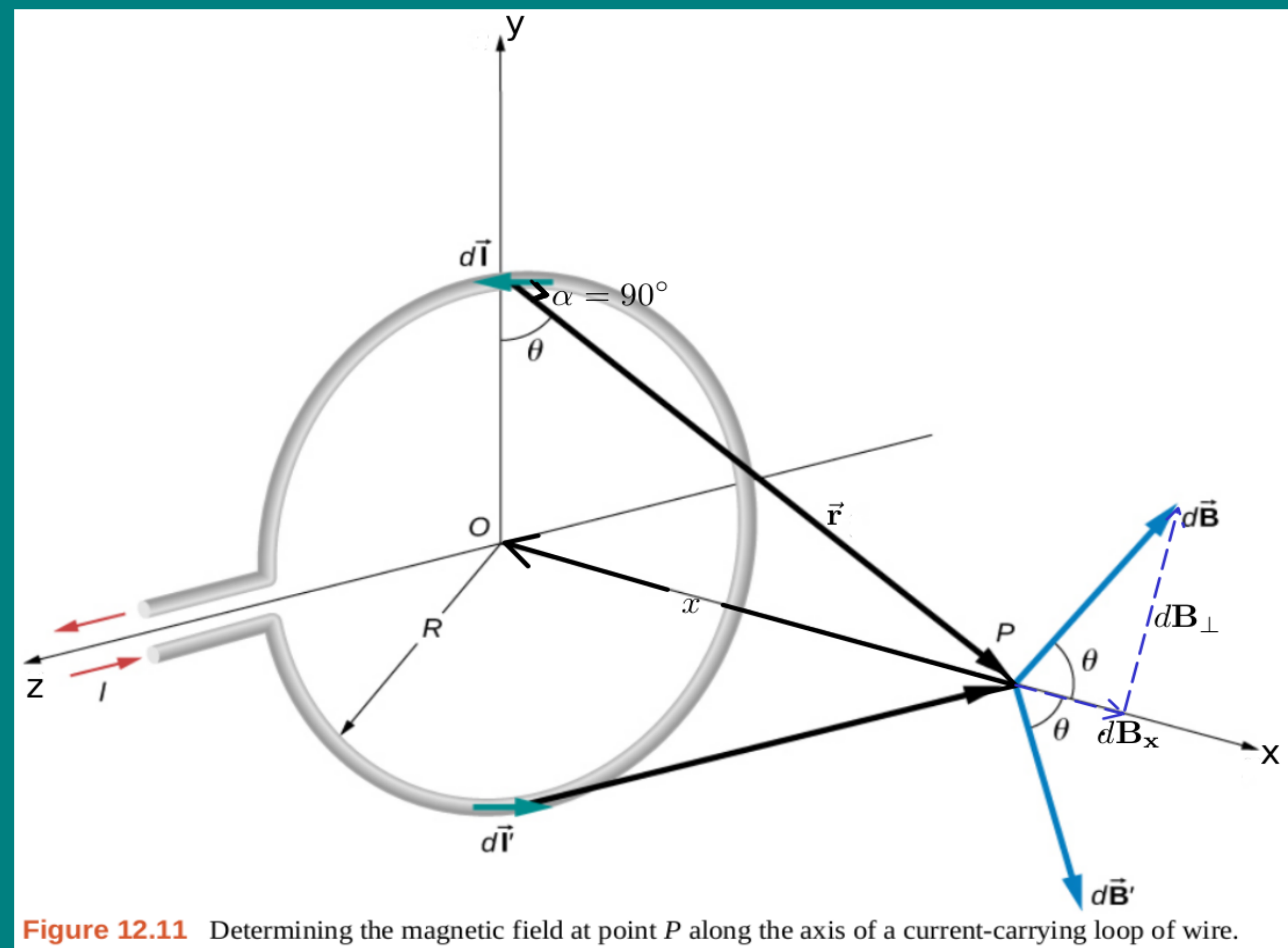
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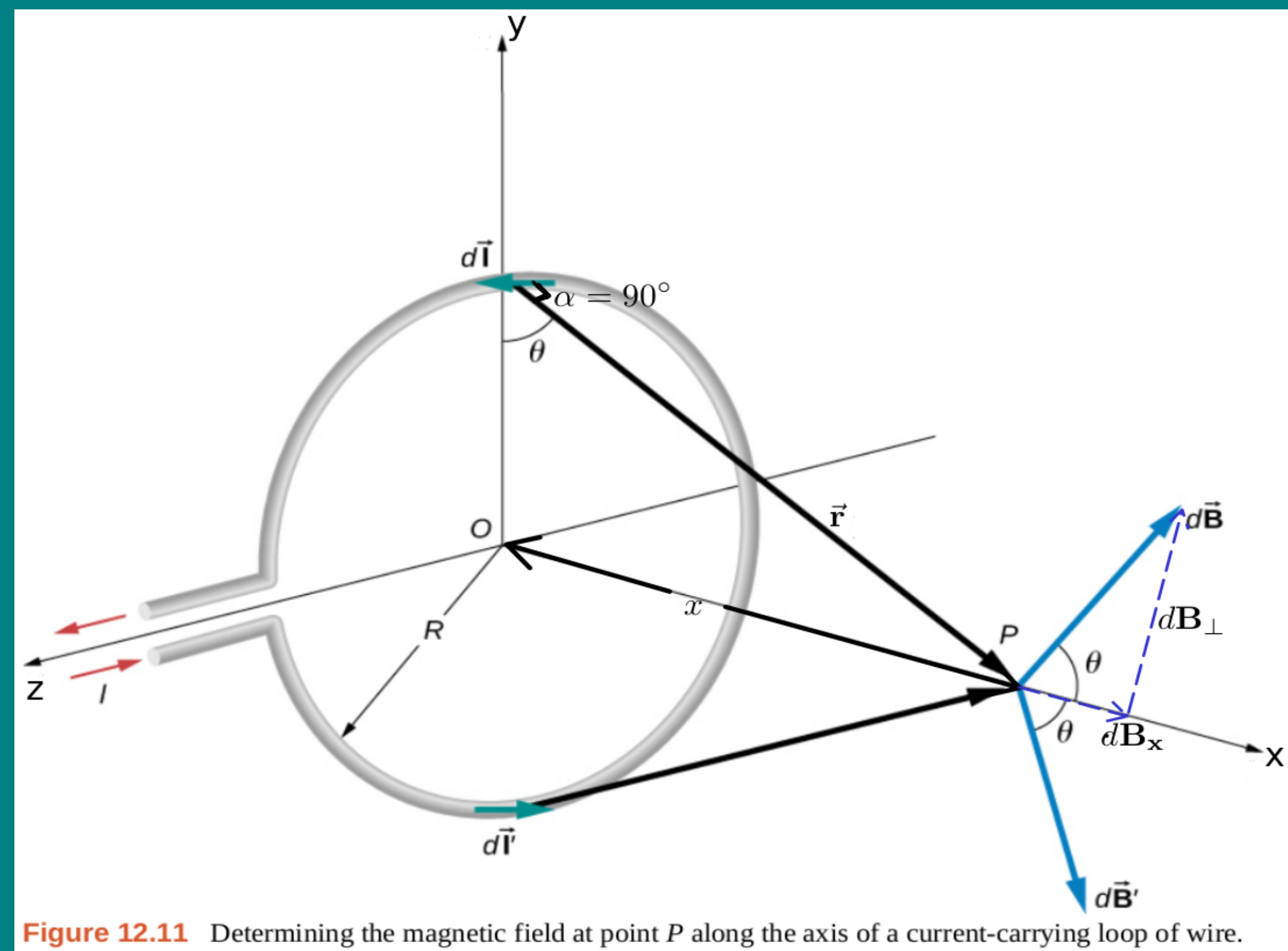
- The **direction of magnetic dipole is determine** according to **right hand thumb rule**.

Circular current loop as a magnetic dipole



- **Magnetic field on the axis of a circular current loop of radius R and carrying steady current (I) is**

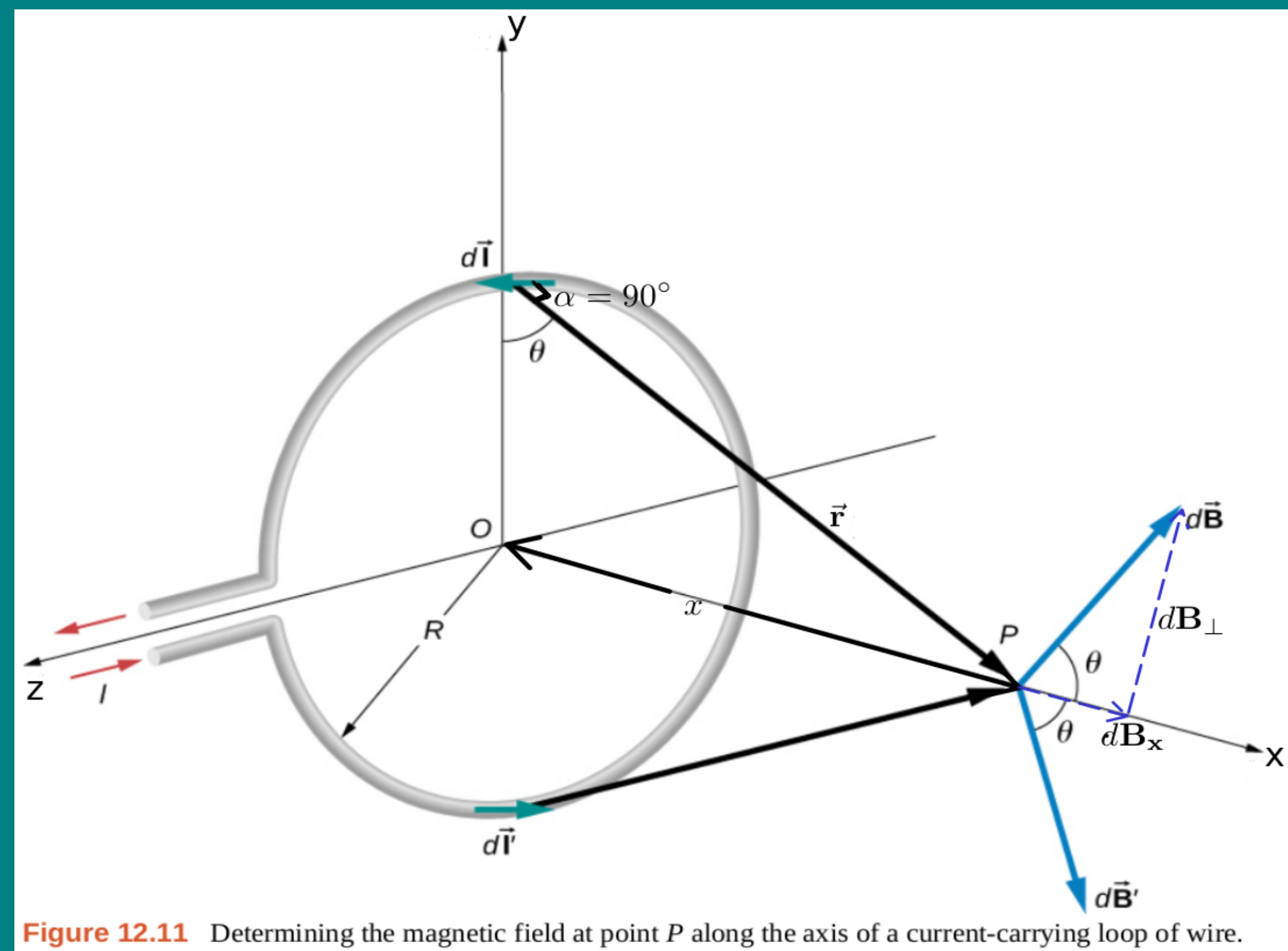
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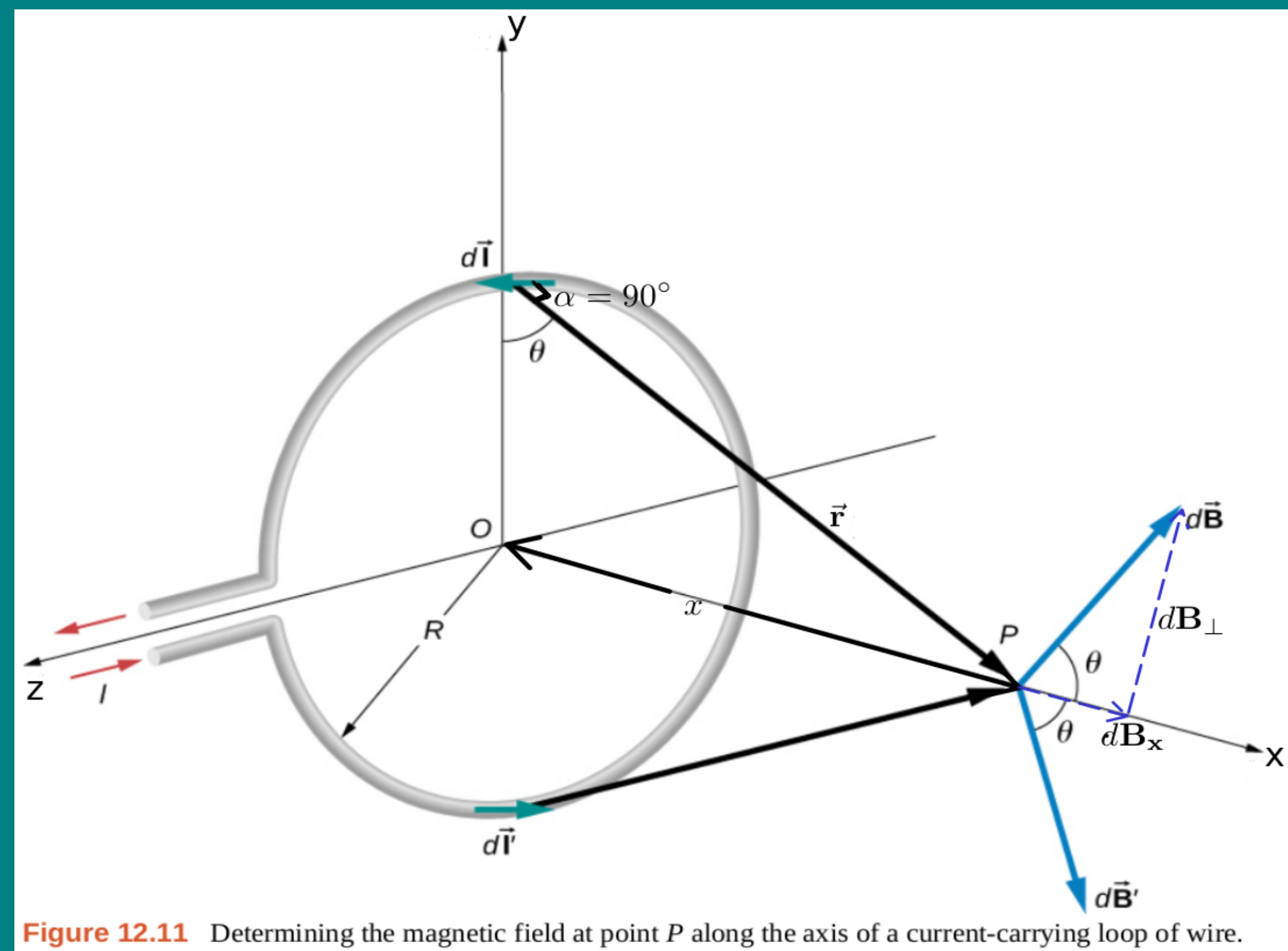


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

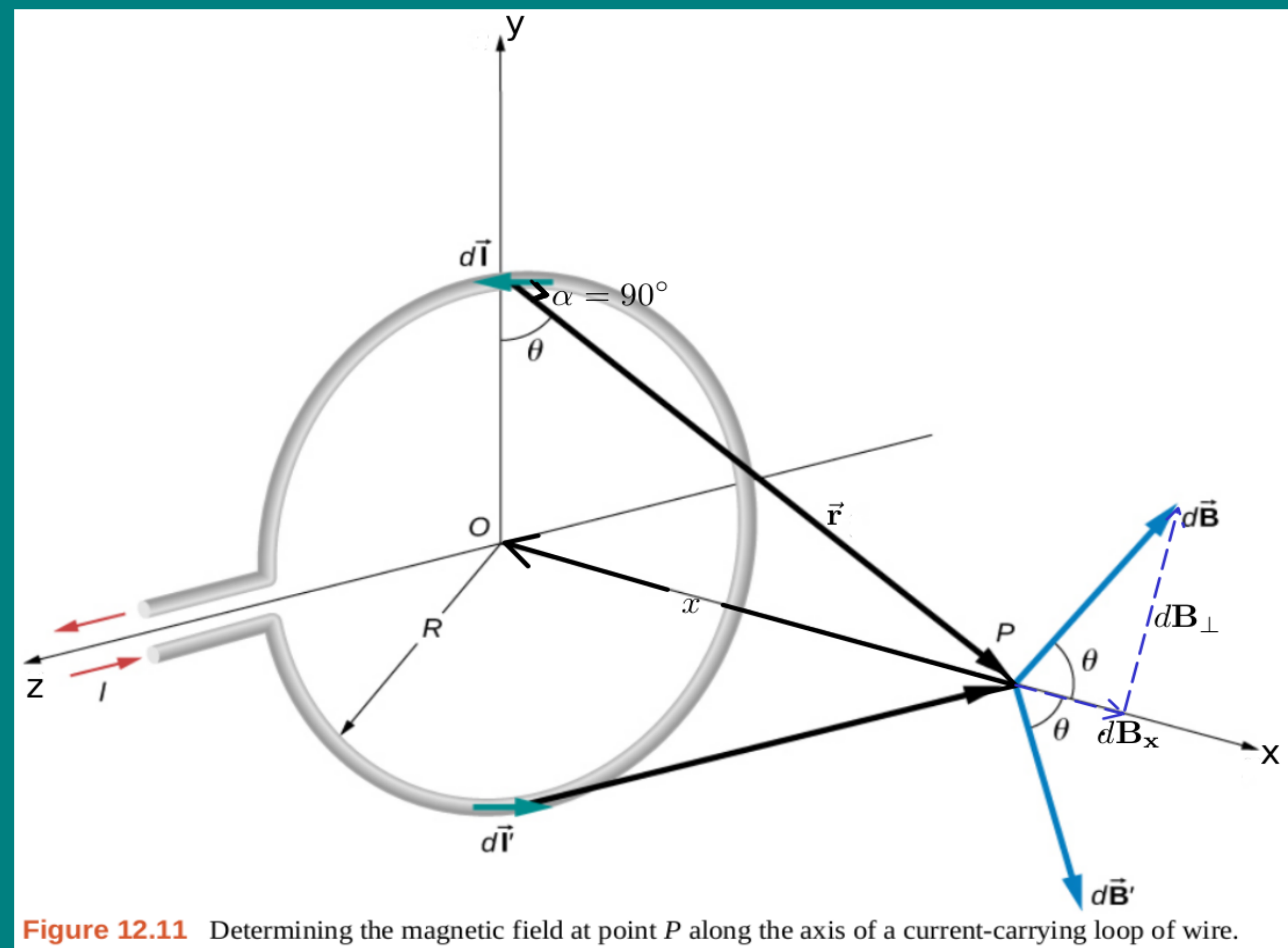
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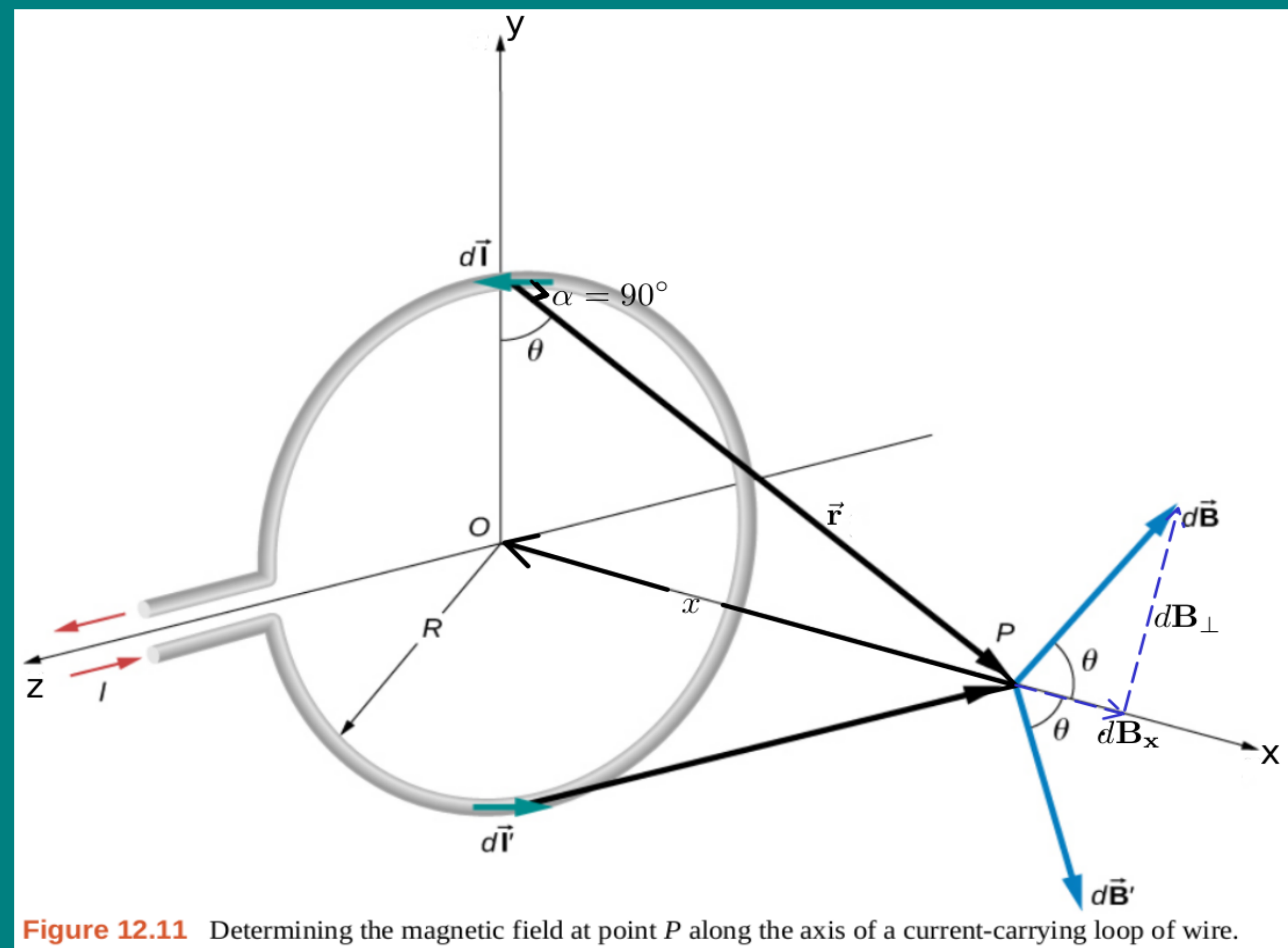


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Circular current loop as a magnetic dipole

- This above expression is similar to electric field due to an electric dipole on its axis.

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$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3}$$

Circular current loop as a magnetic dipole

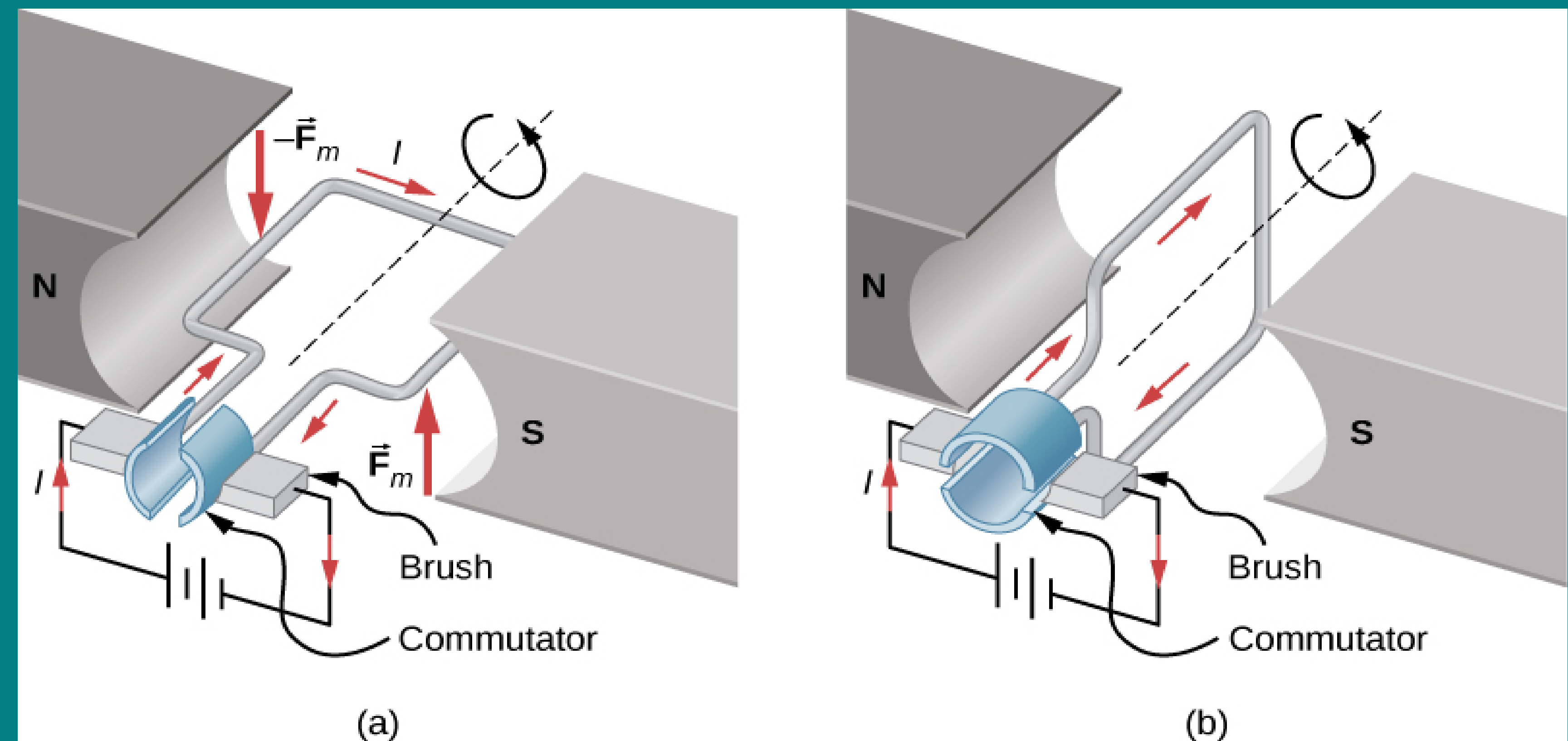
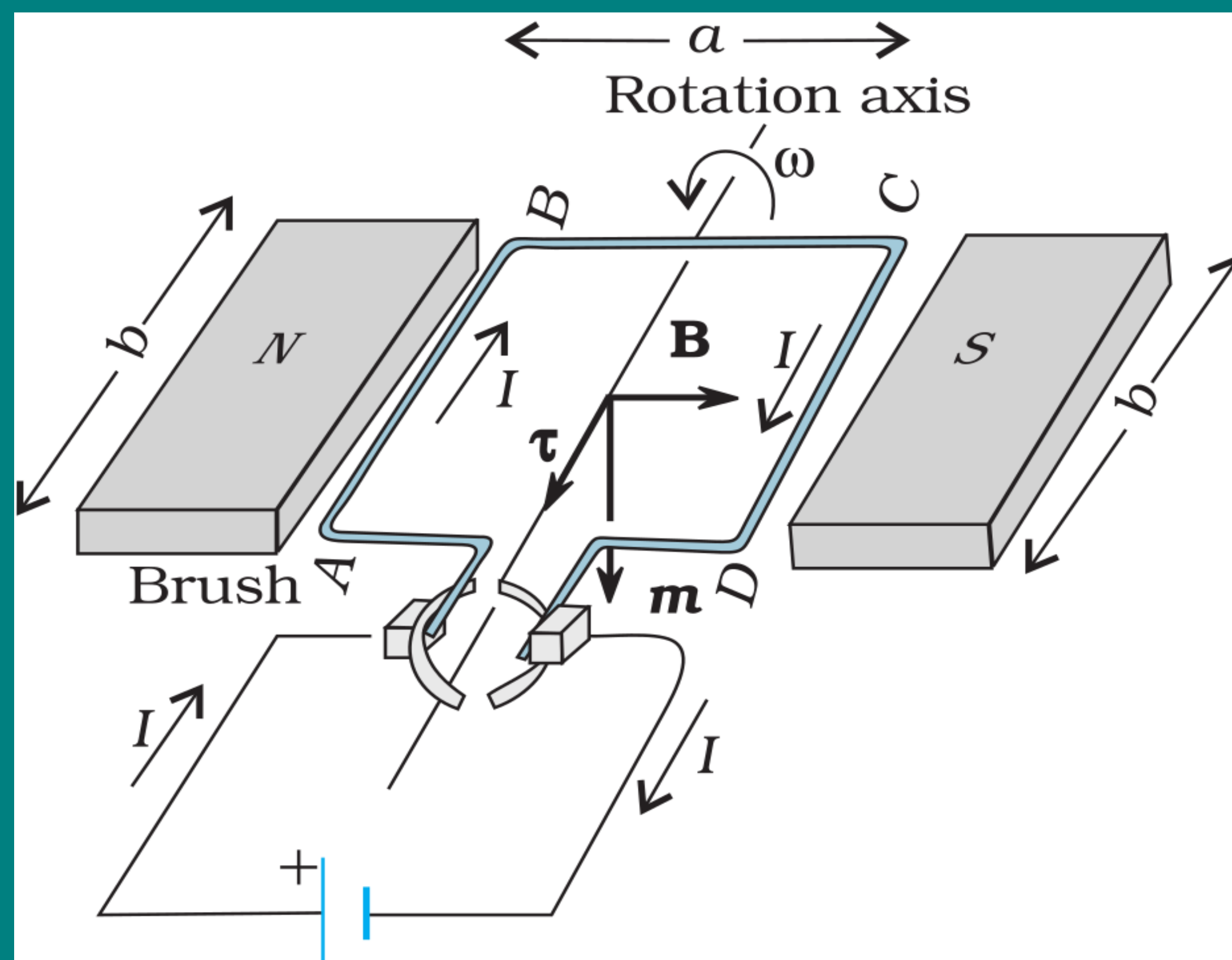
- **If we replace p by m , $\frac{1}{\epsilon_o}$ by μ_o and E by B , We obtain the result for magnetic field B for point on the plane of the current loop.**

Circular current loop as a magnetic dipole

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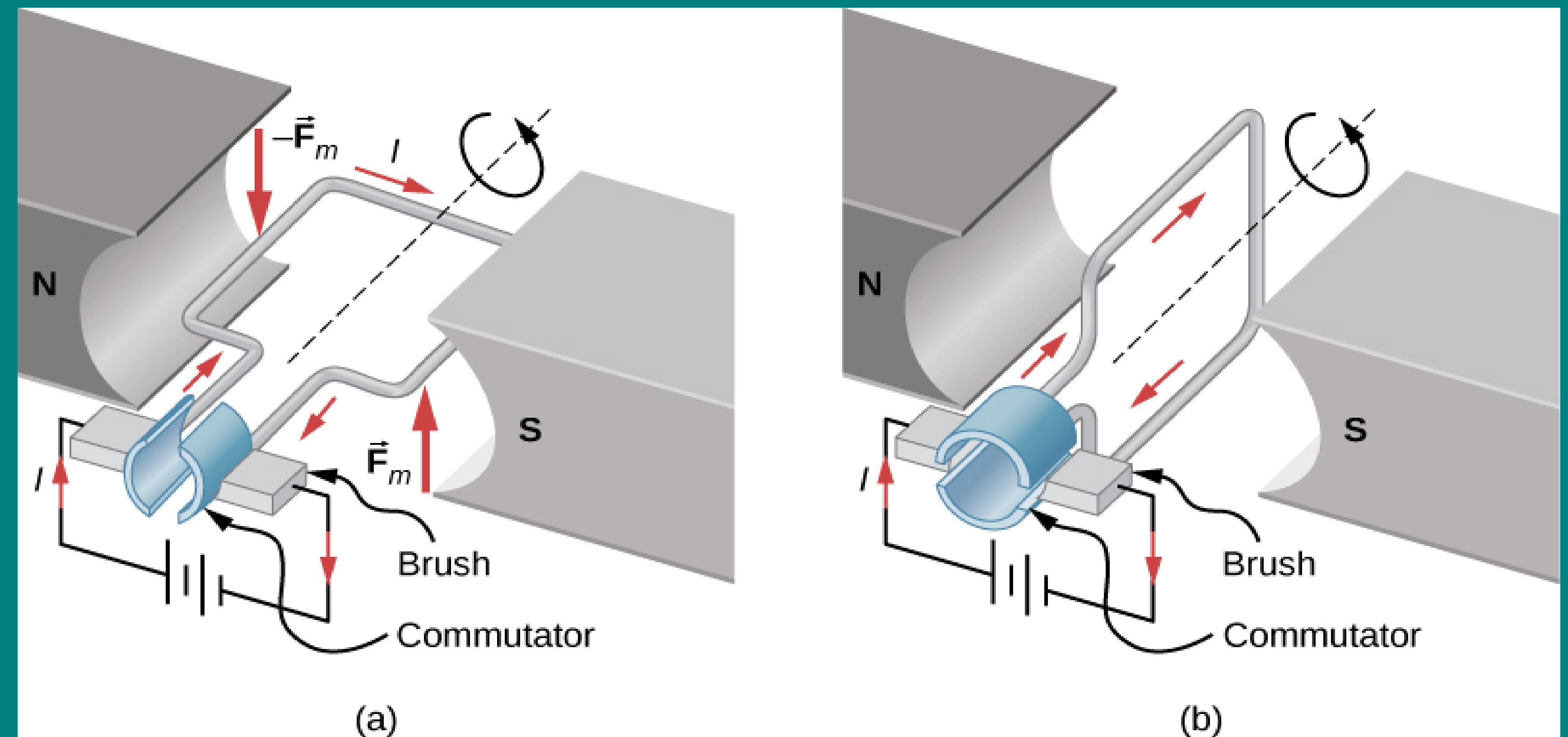
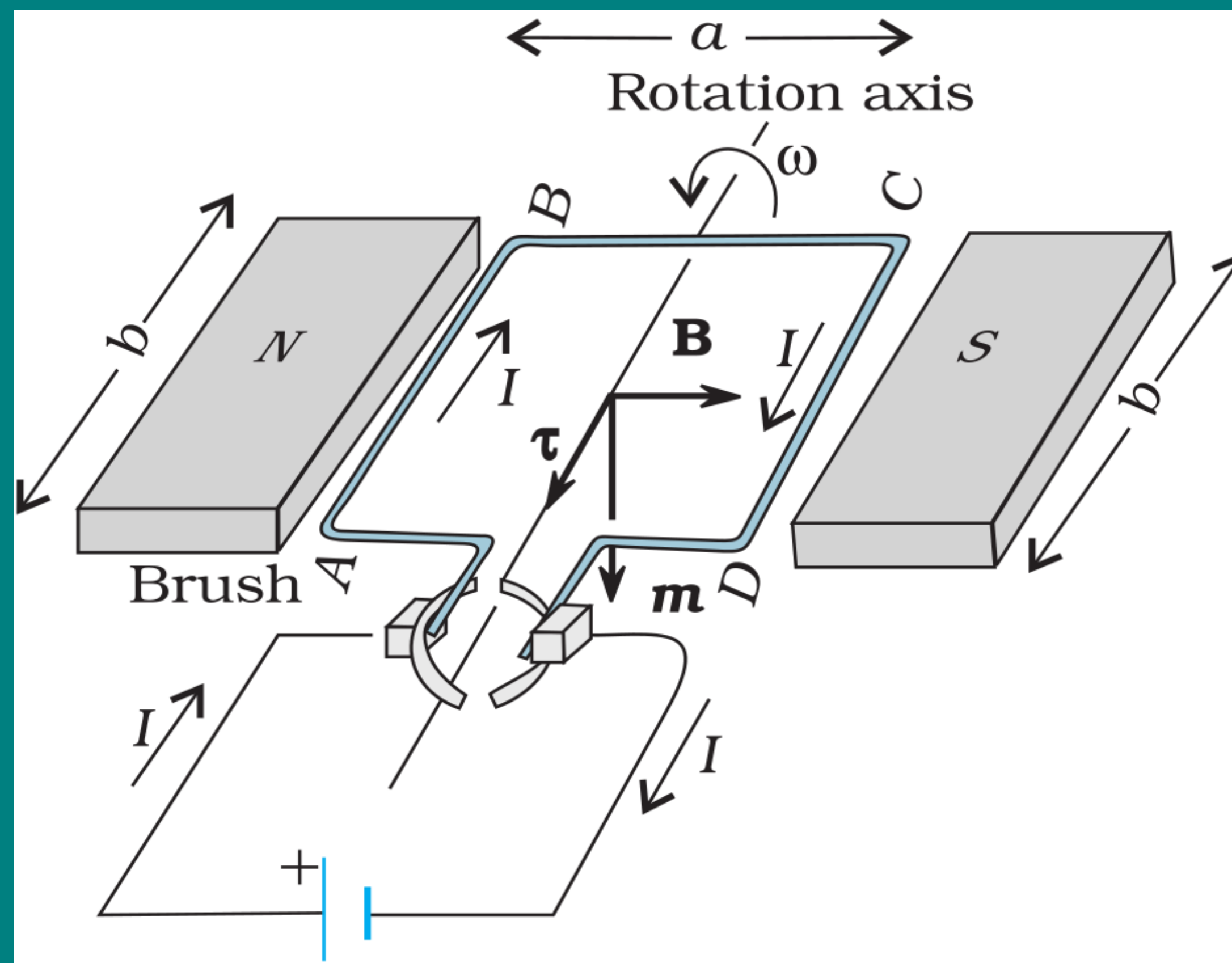
$$B = \frac{\mu_o}{4\pi} \frac{m}{x^3}$$

Torque on a Current Loop In a Uniform Magnetic Field



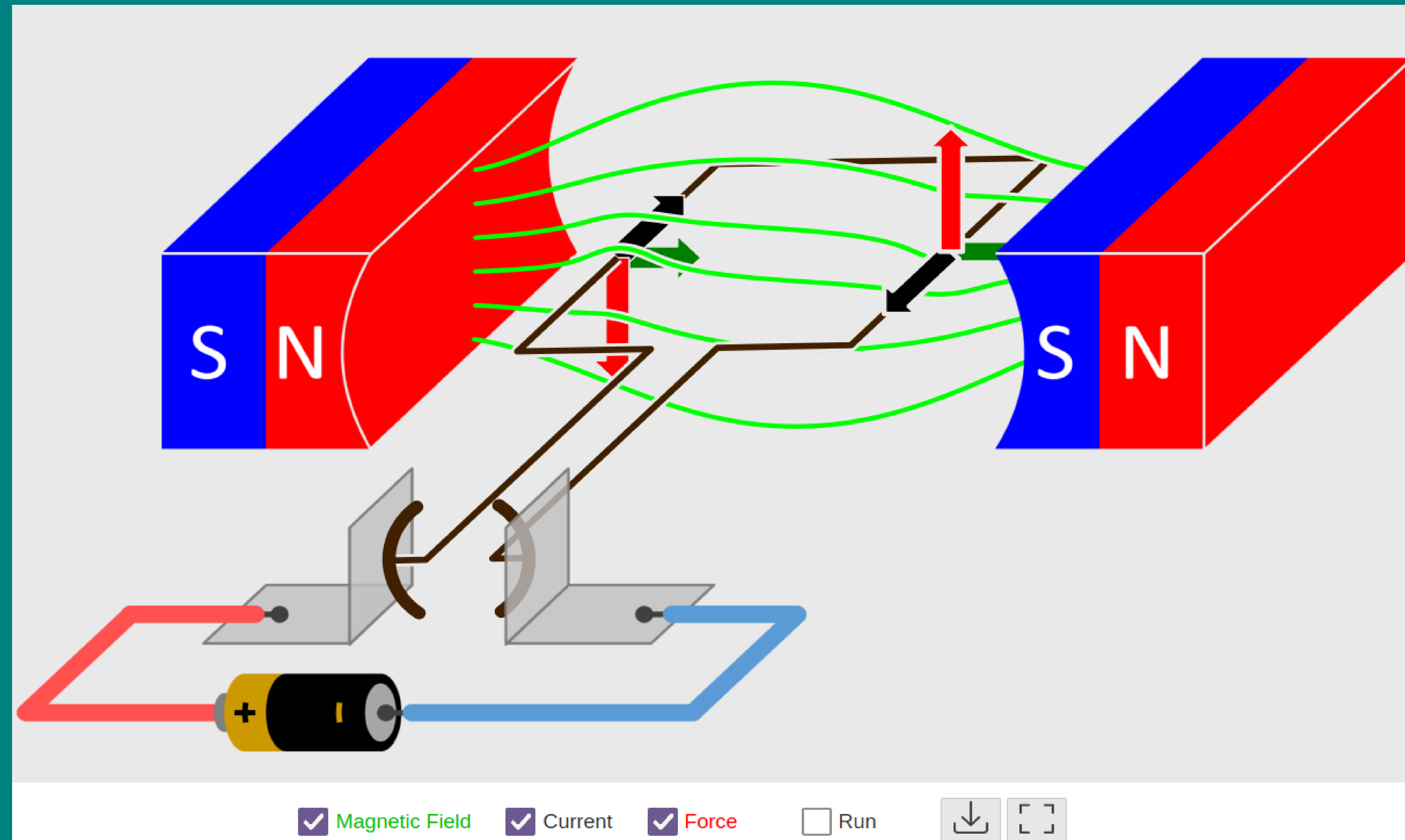
- A rectangular loop carrying a steady current I and placed in a uniform magnetic field experiences a torque. It does not experience a net force.

Torque on a Current Loop In a Uniform Magnetic Field

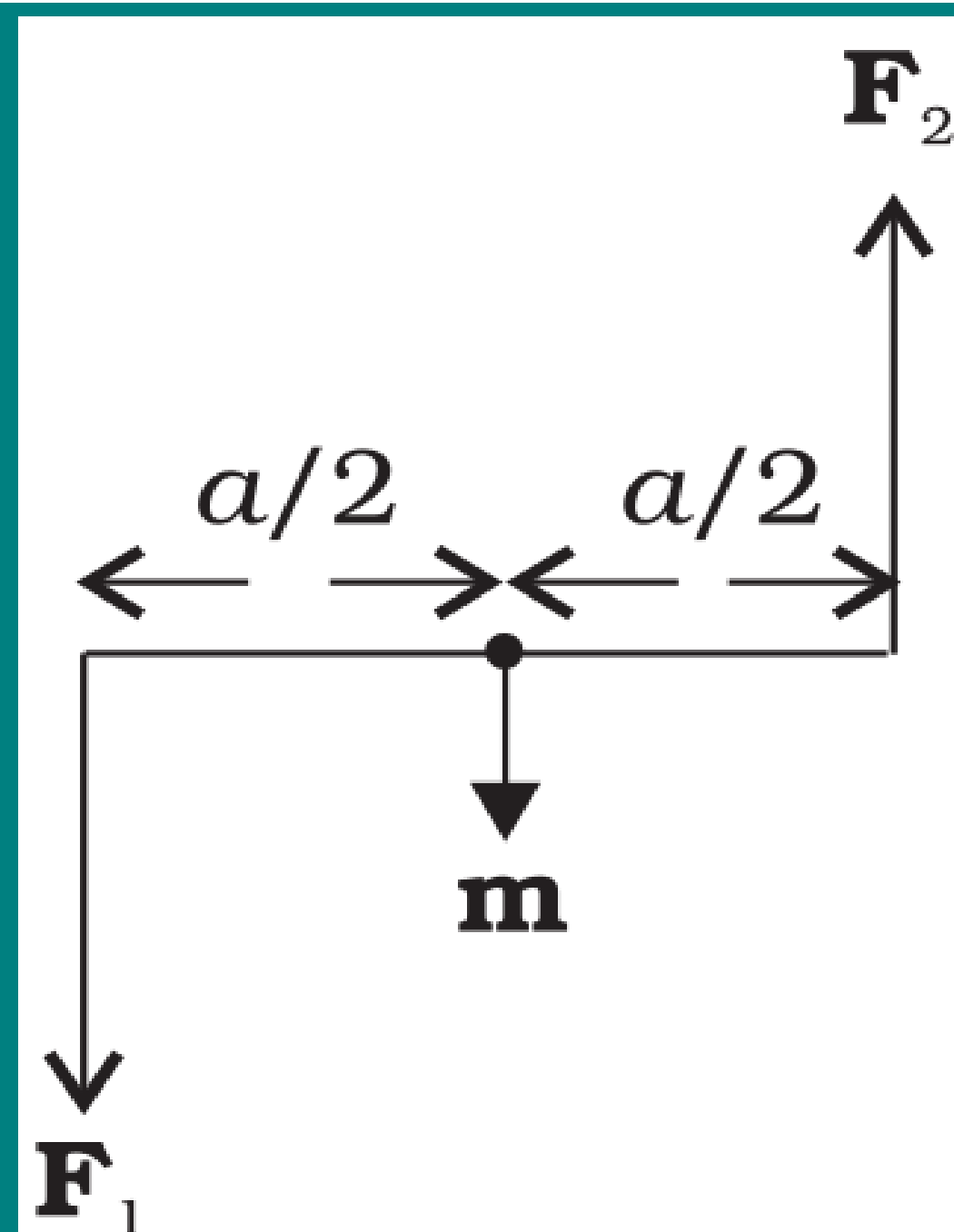
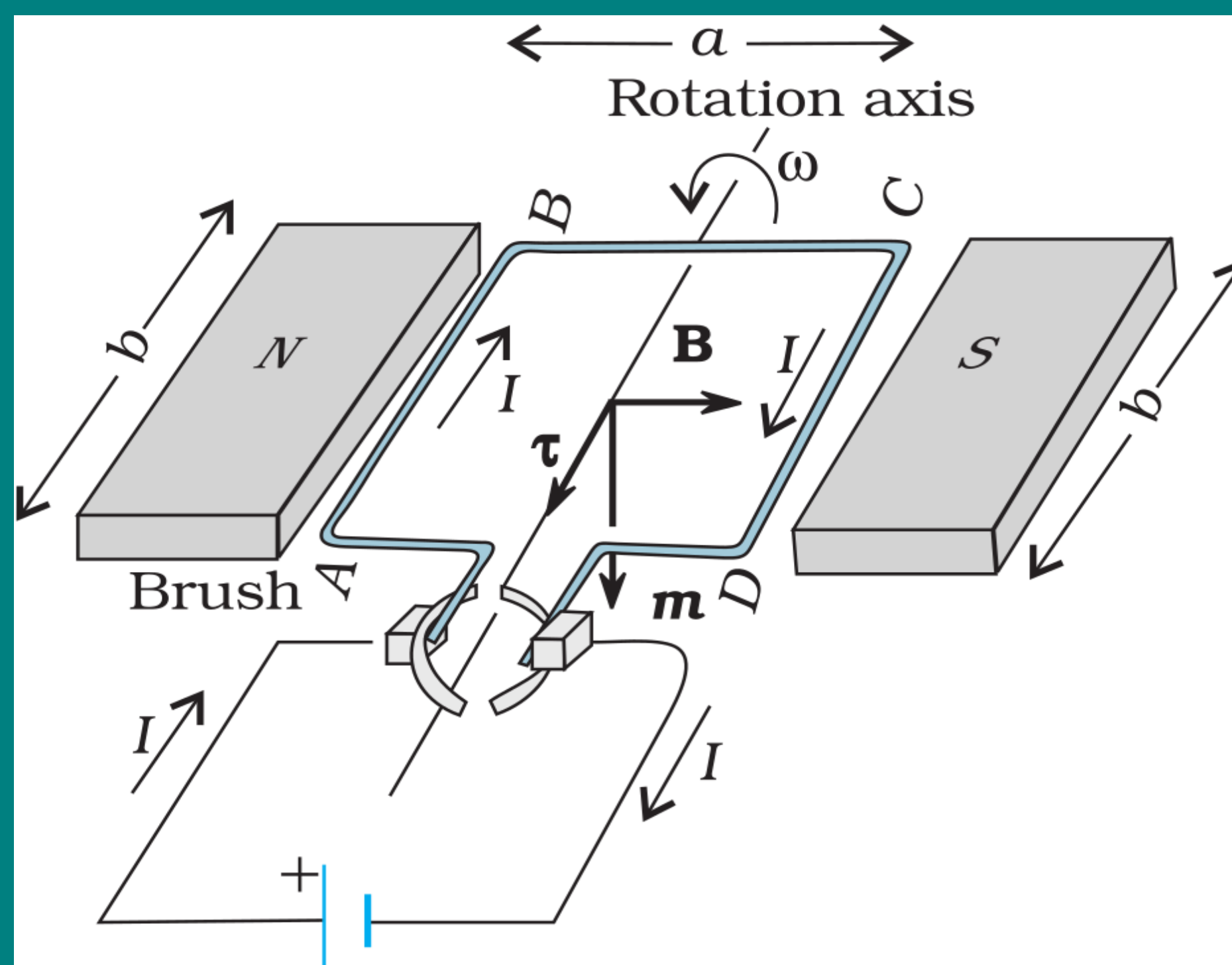


- A rectangular loop carrying a steady current I and placed in a uniform magnetic field experiences a torque. It does not experience a net force.
- This behaviour is analogous to that of electric dipole in a uniform electric field.

(1) When uniform magnetic field B is in the plane of the loop

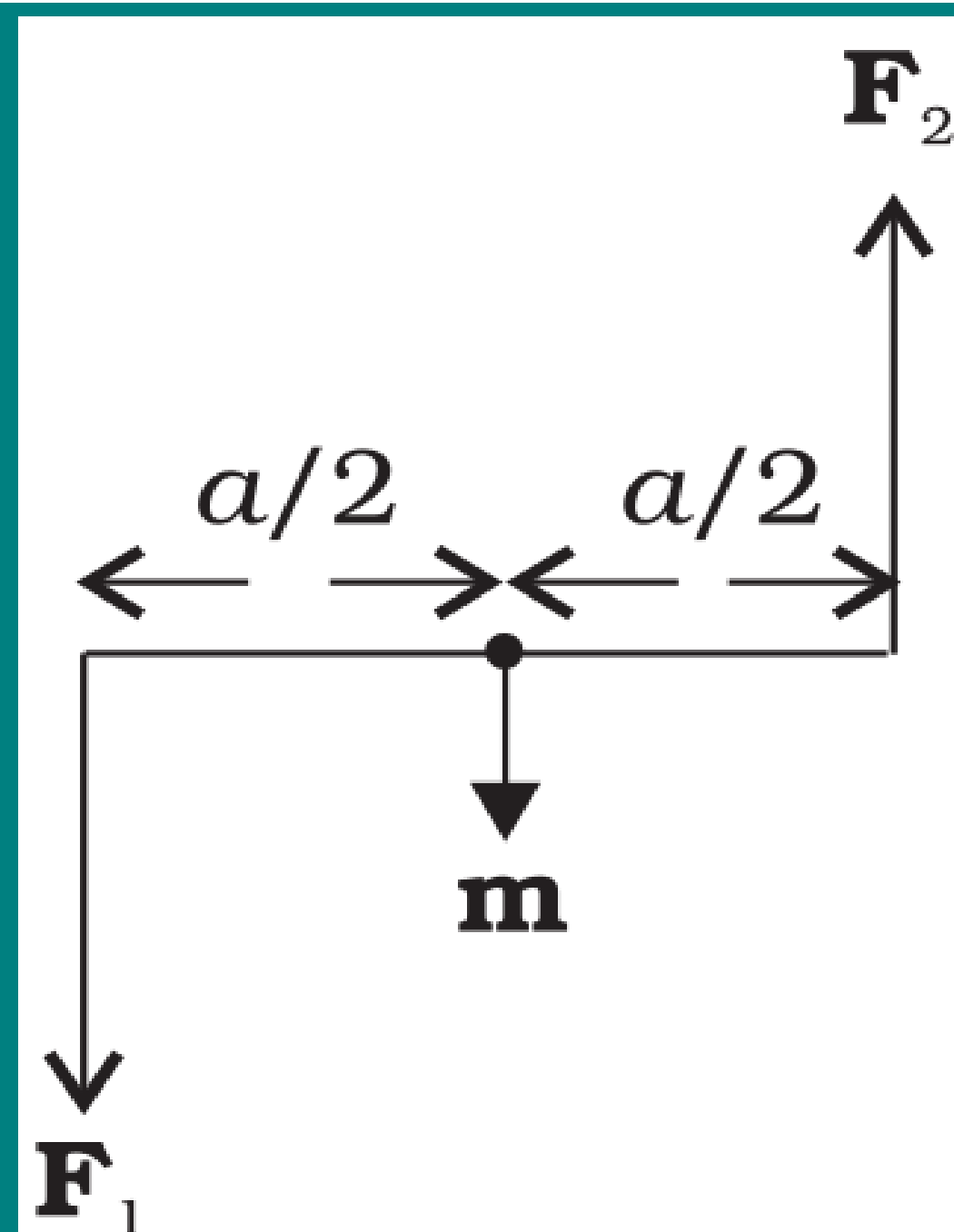
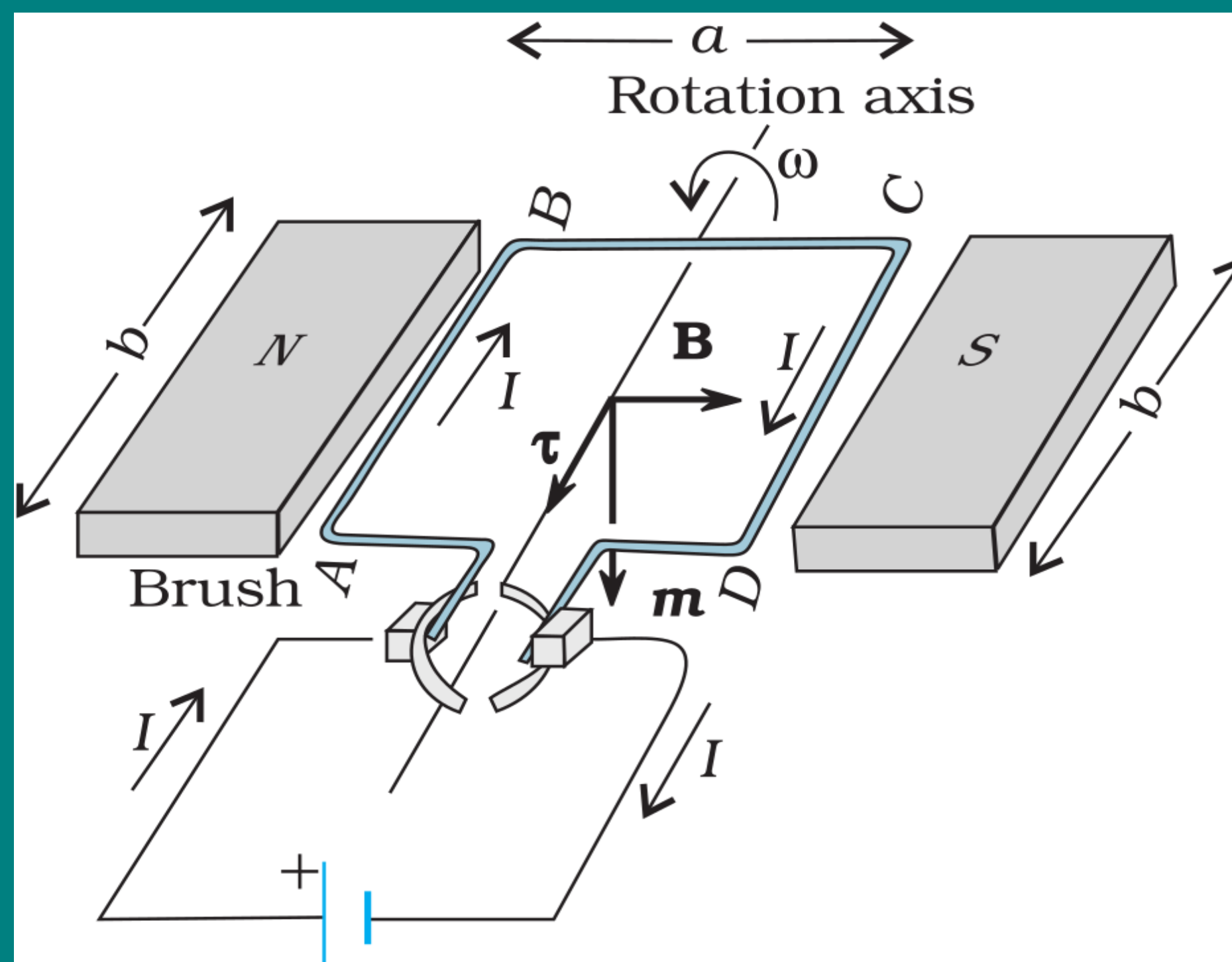


Torque on a Current Loop In a Uniform Magnetic Field



Case 1: When uniform magnetic field $\mathbf{B}$ is in the plane of the loop

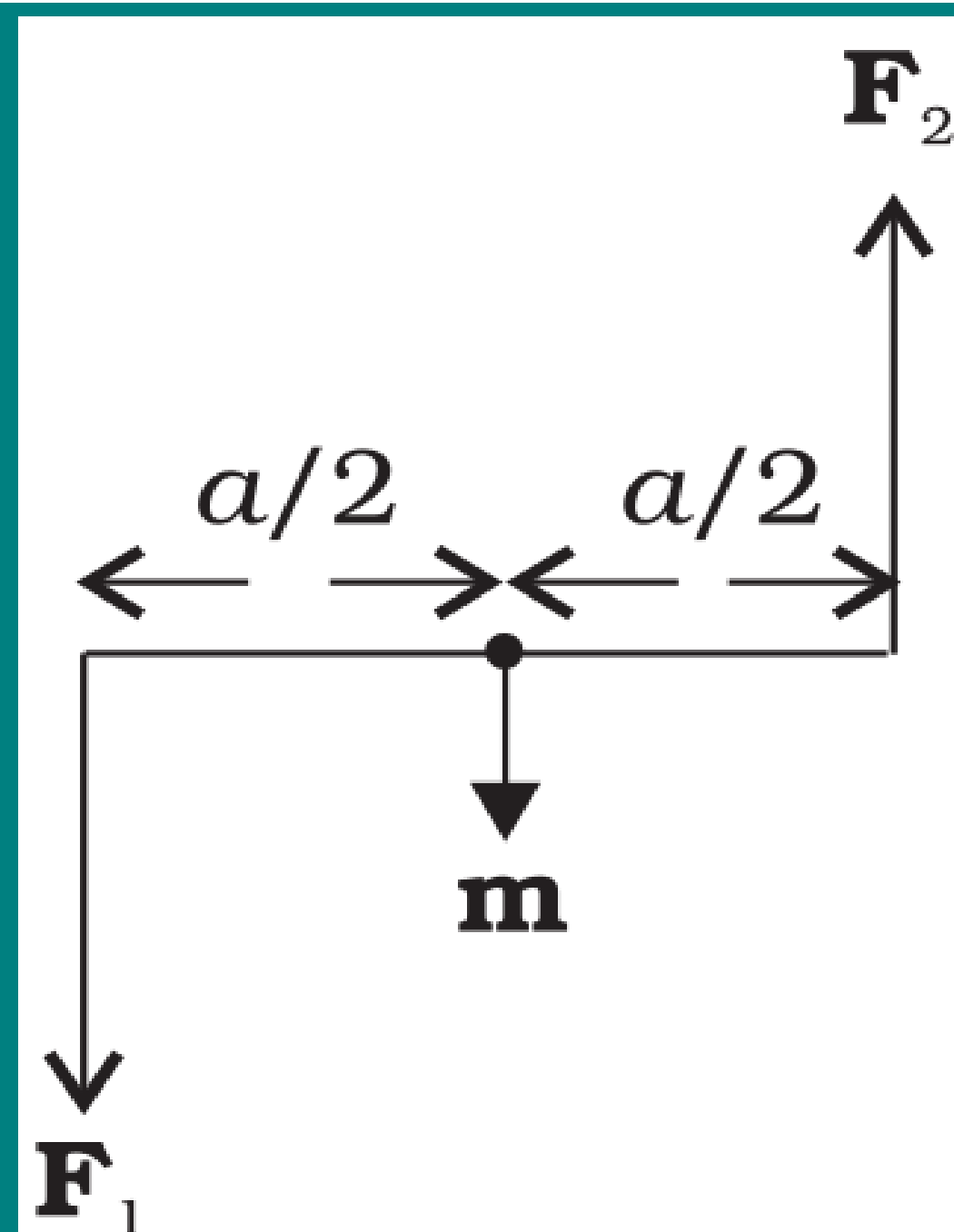
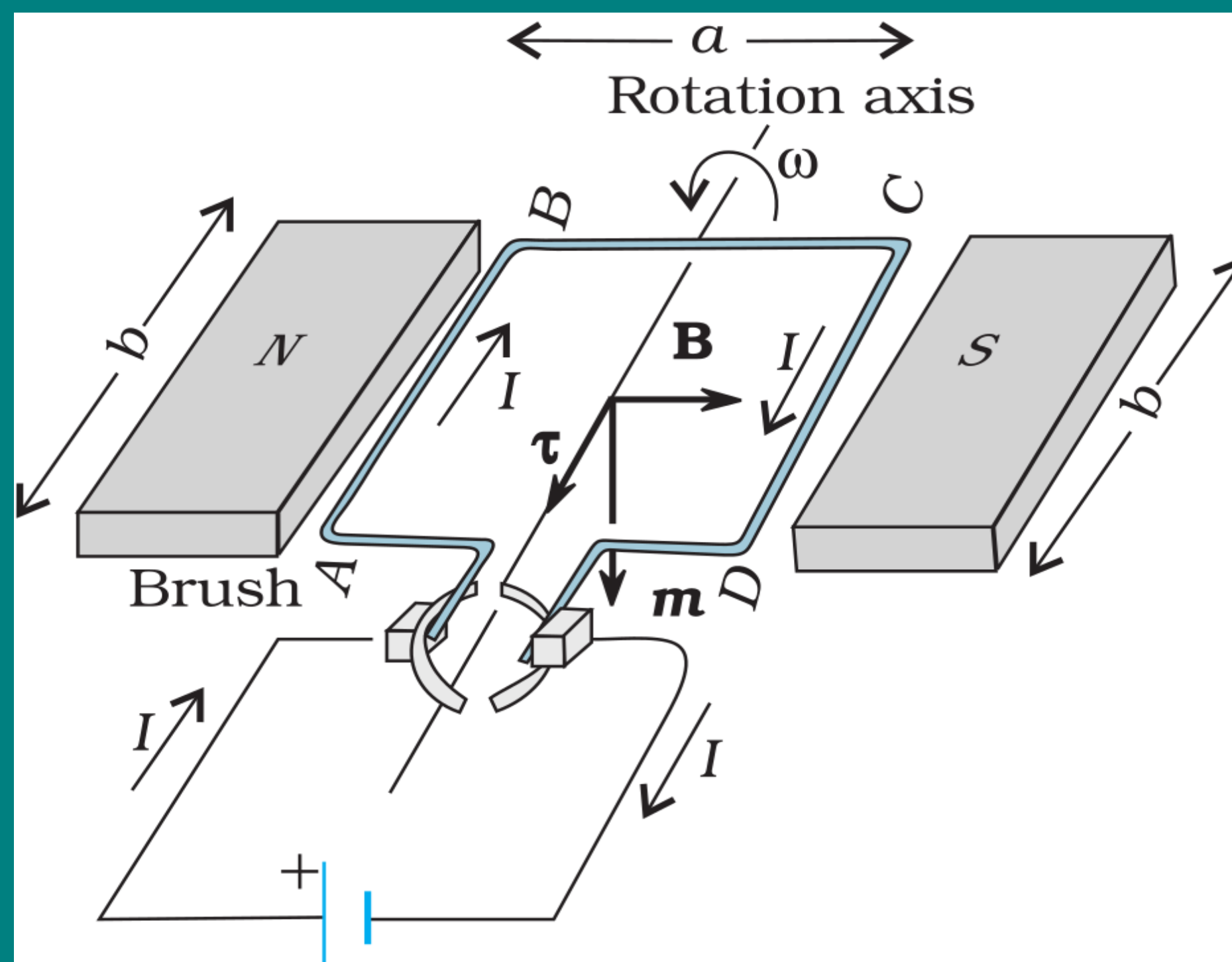
Torque on a Current Loop In a Uniform Magnetic Field



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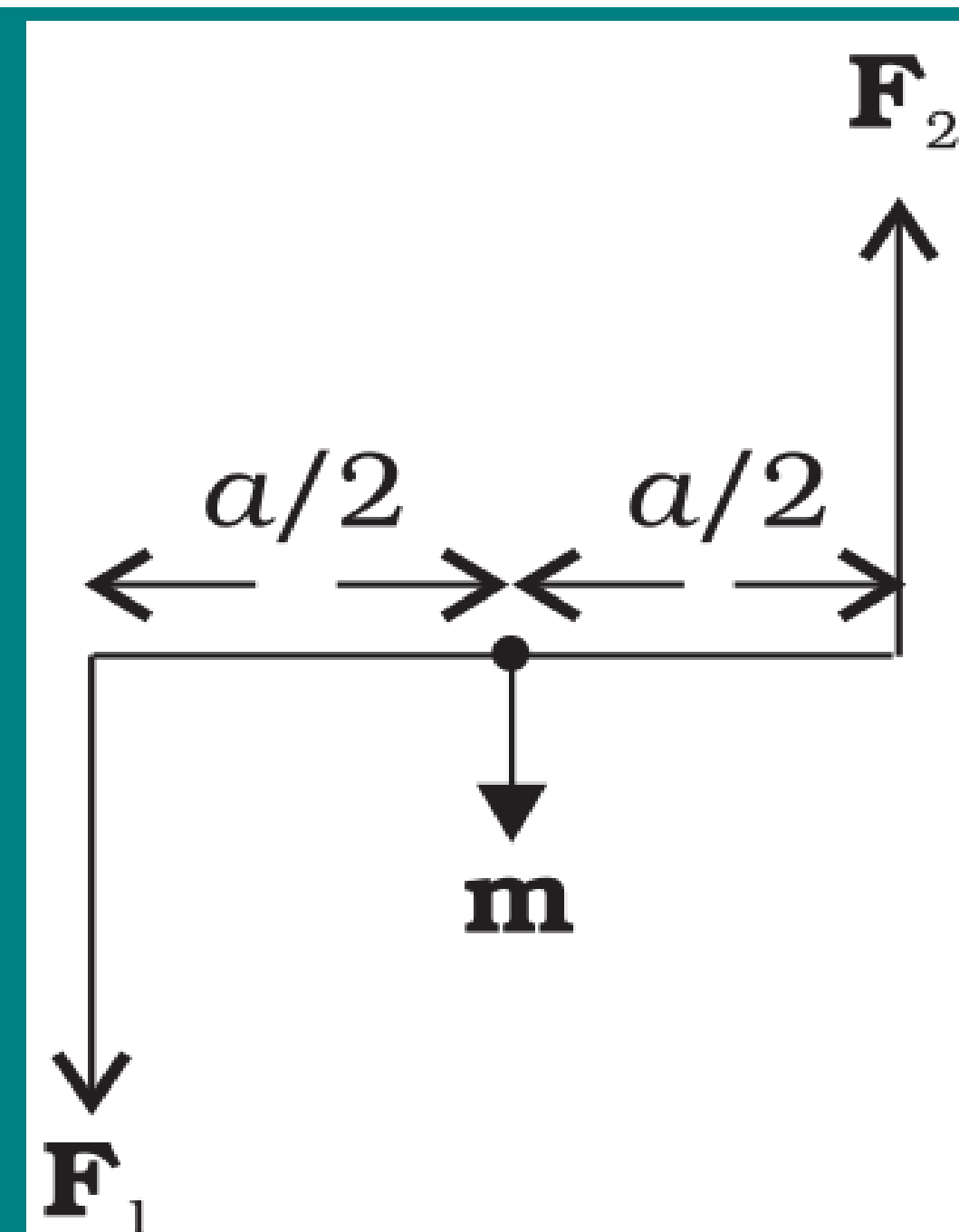
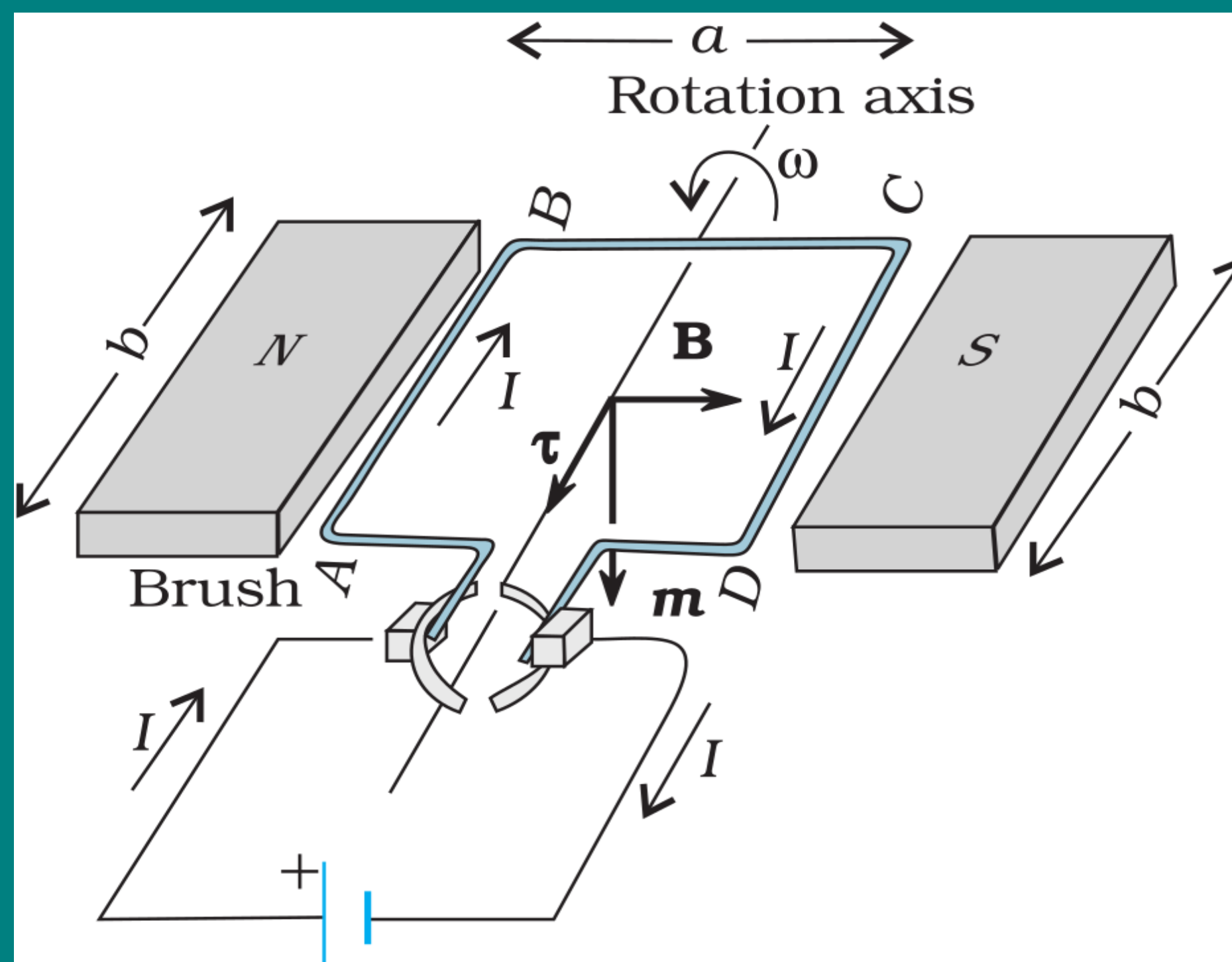
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Torque on a Current Loop In a Uniform Magnetic Field



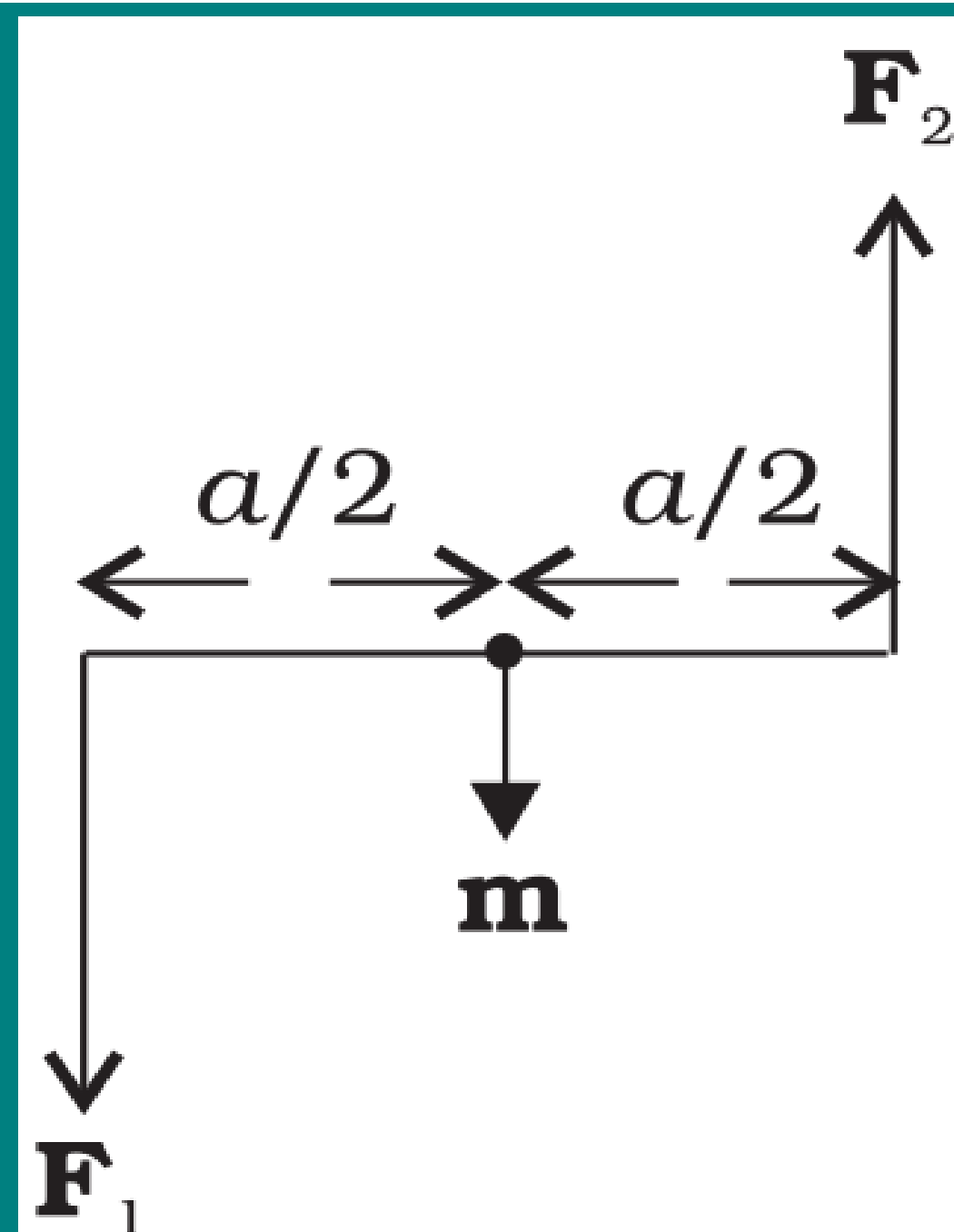
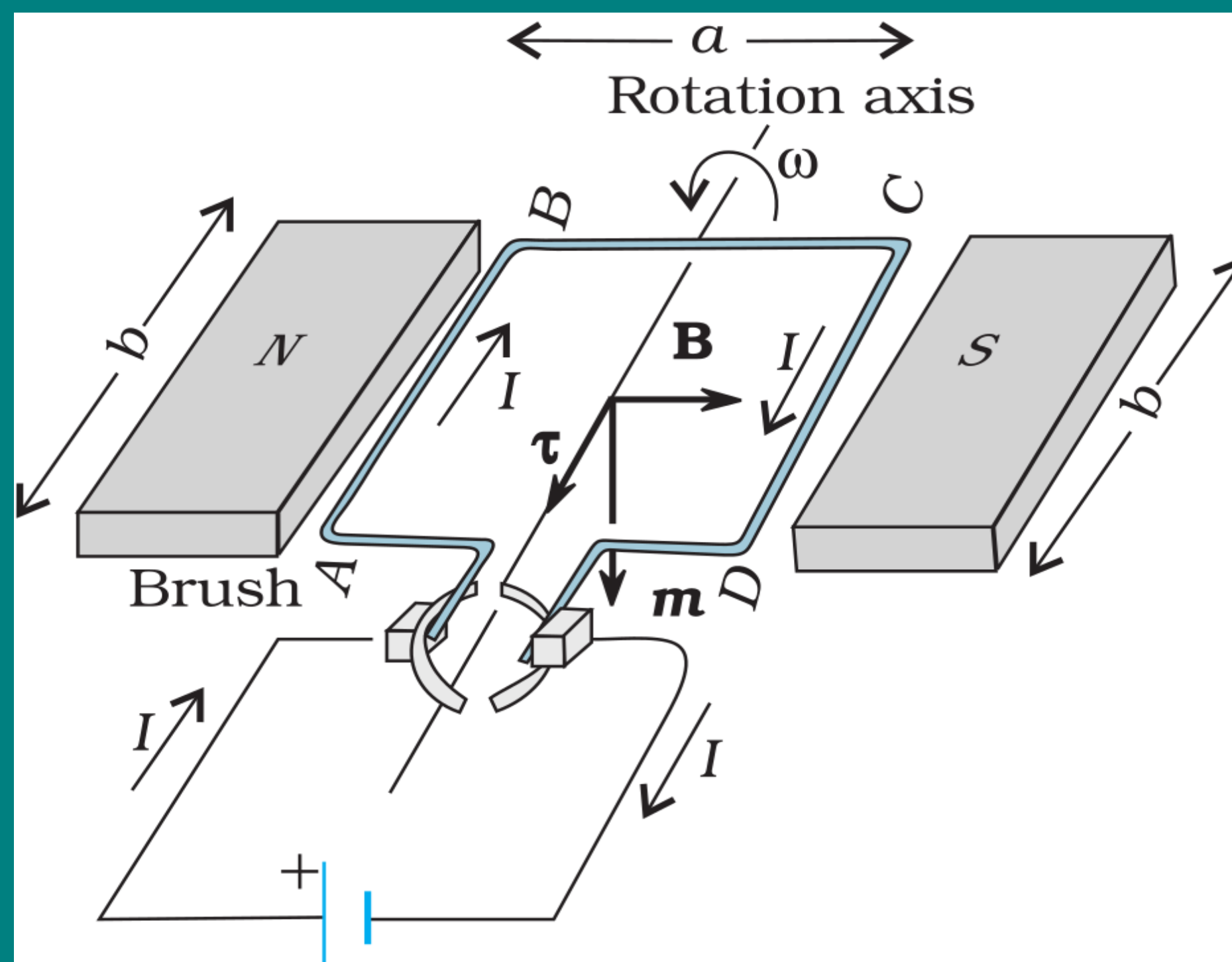
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Torque on a Current Loop In a Uniform Magnetic Field

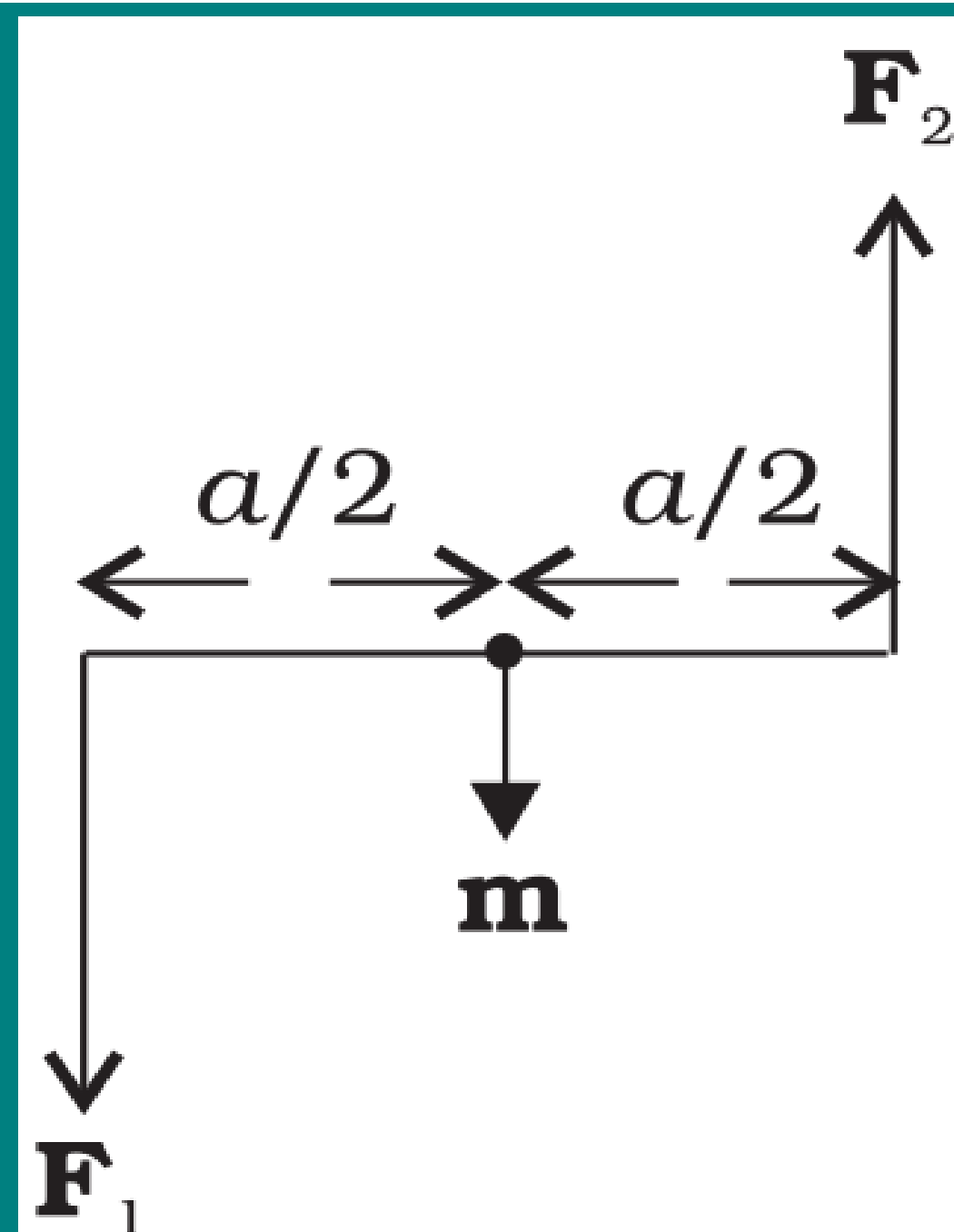
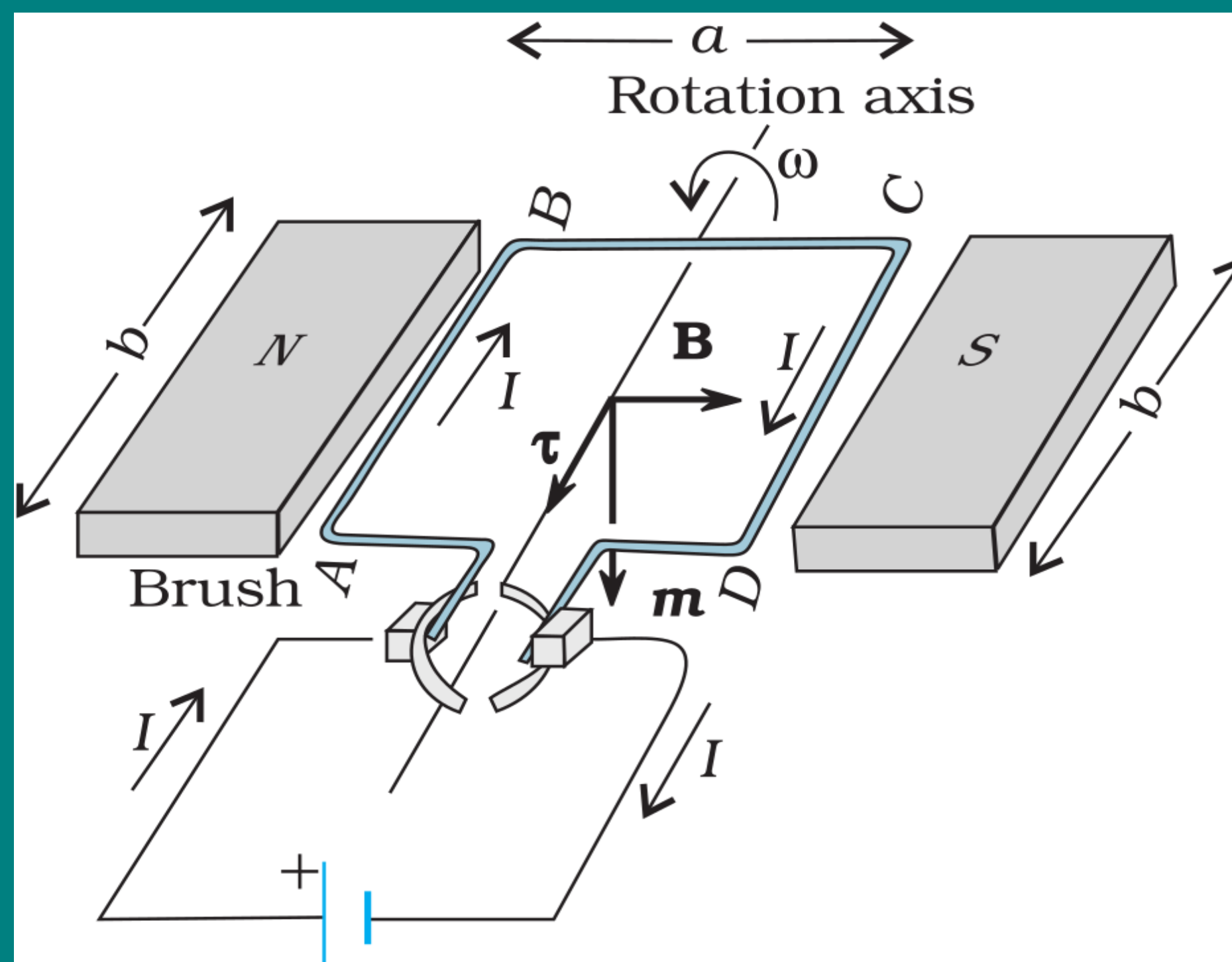


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Torque on a Current Loop In a Uniform Magnetic Field



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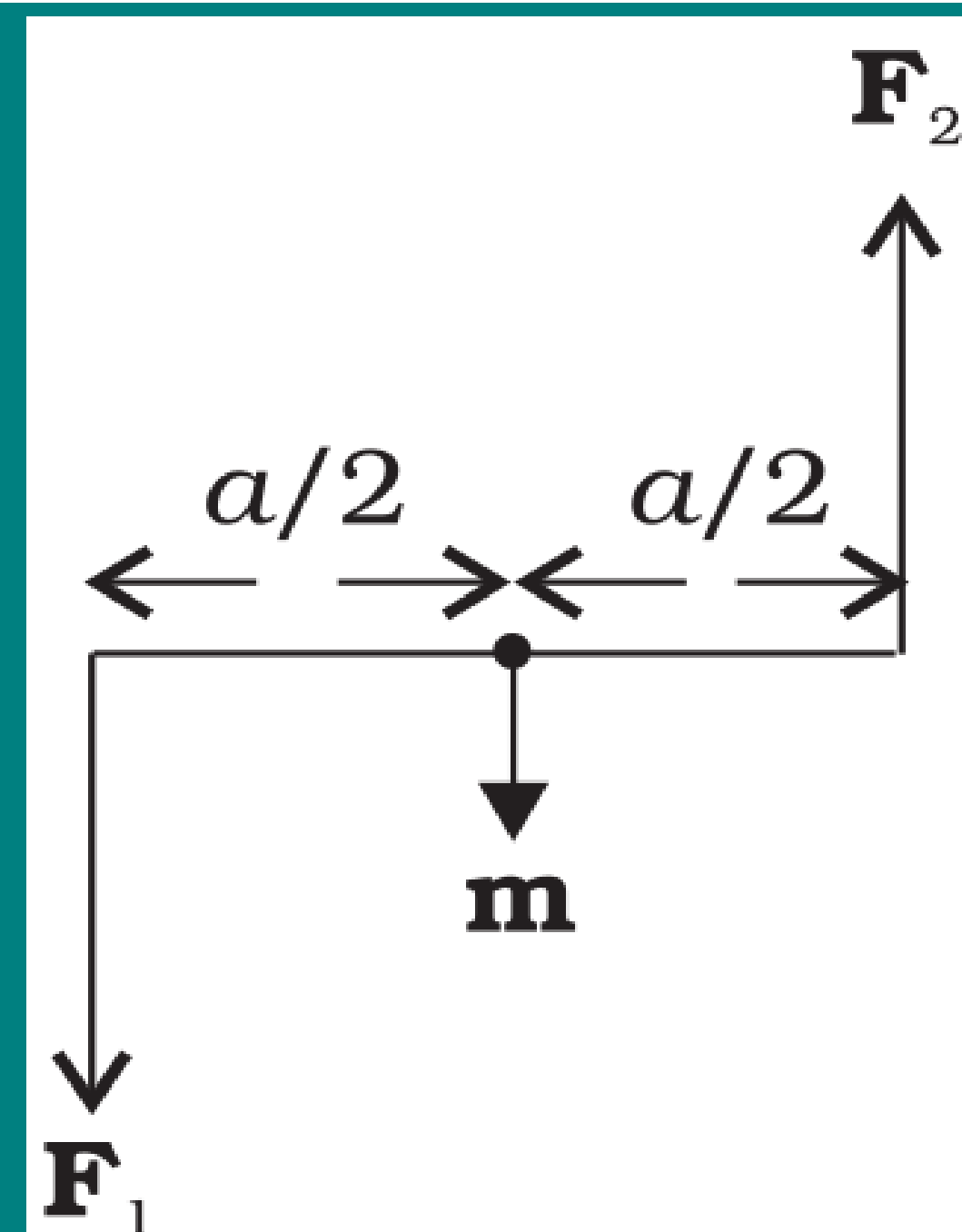
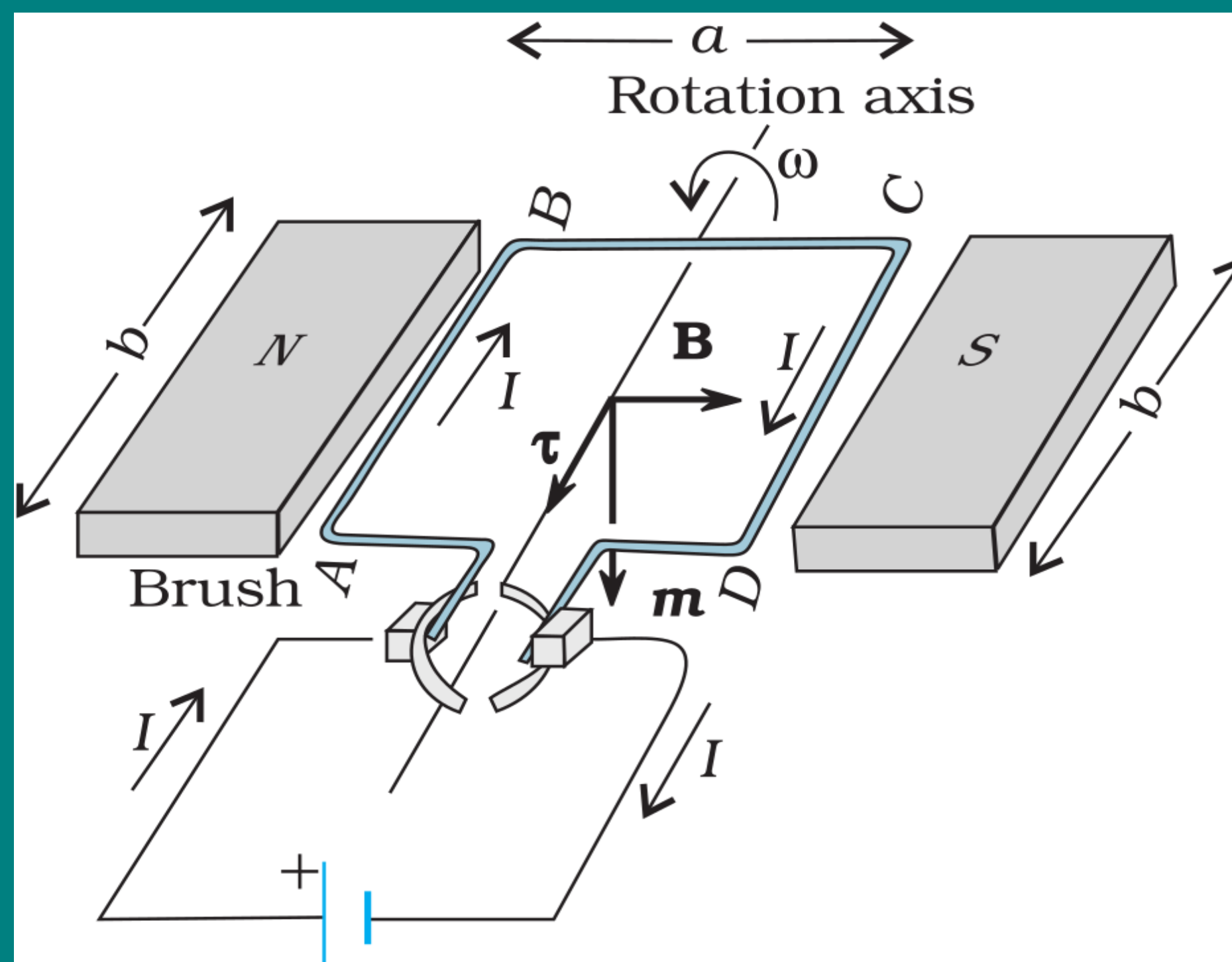
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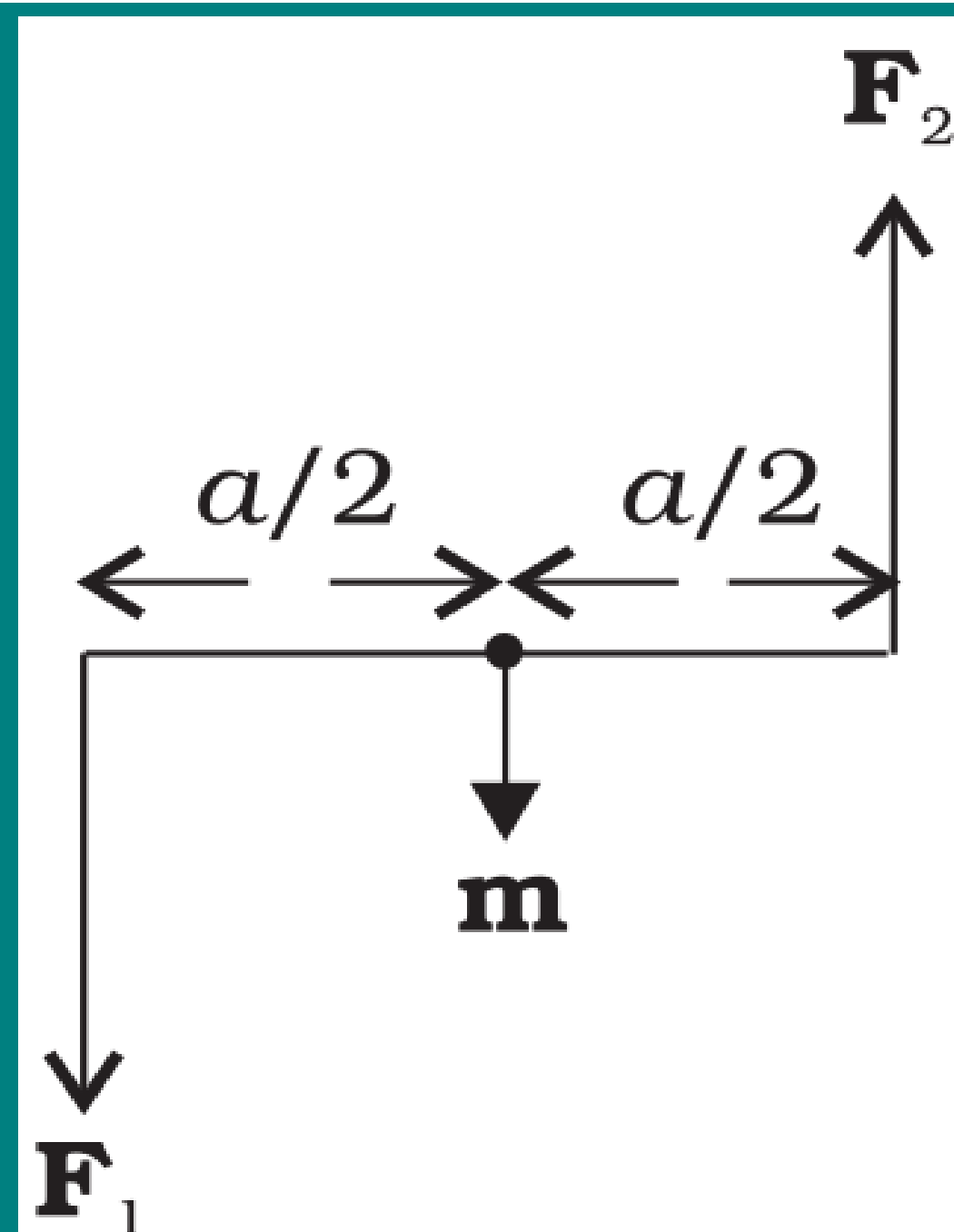
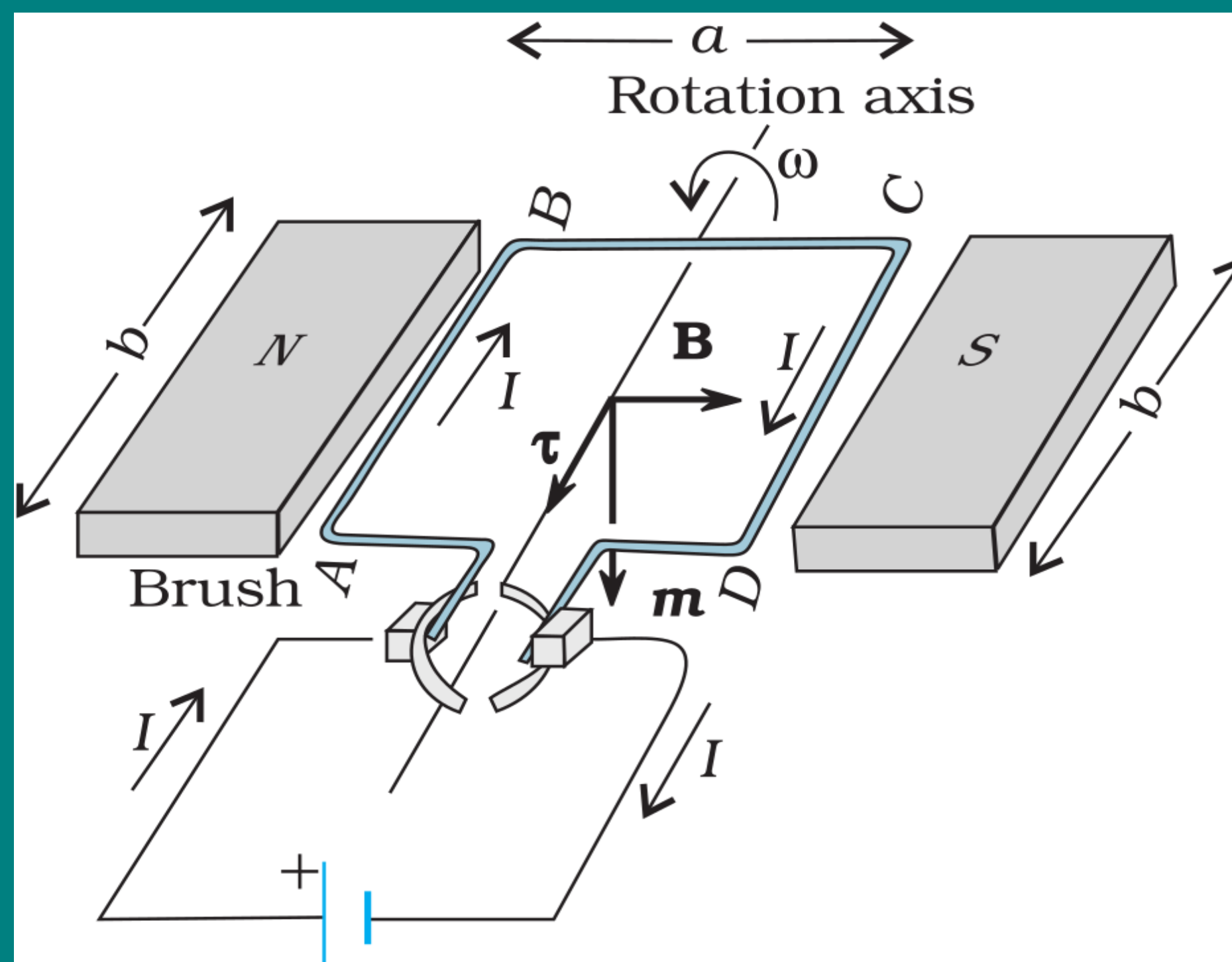
Torque on a Current Loop In a Uniform Magnetic Field



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- Thus, the **net force on the loop is zero**.

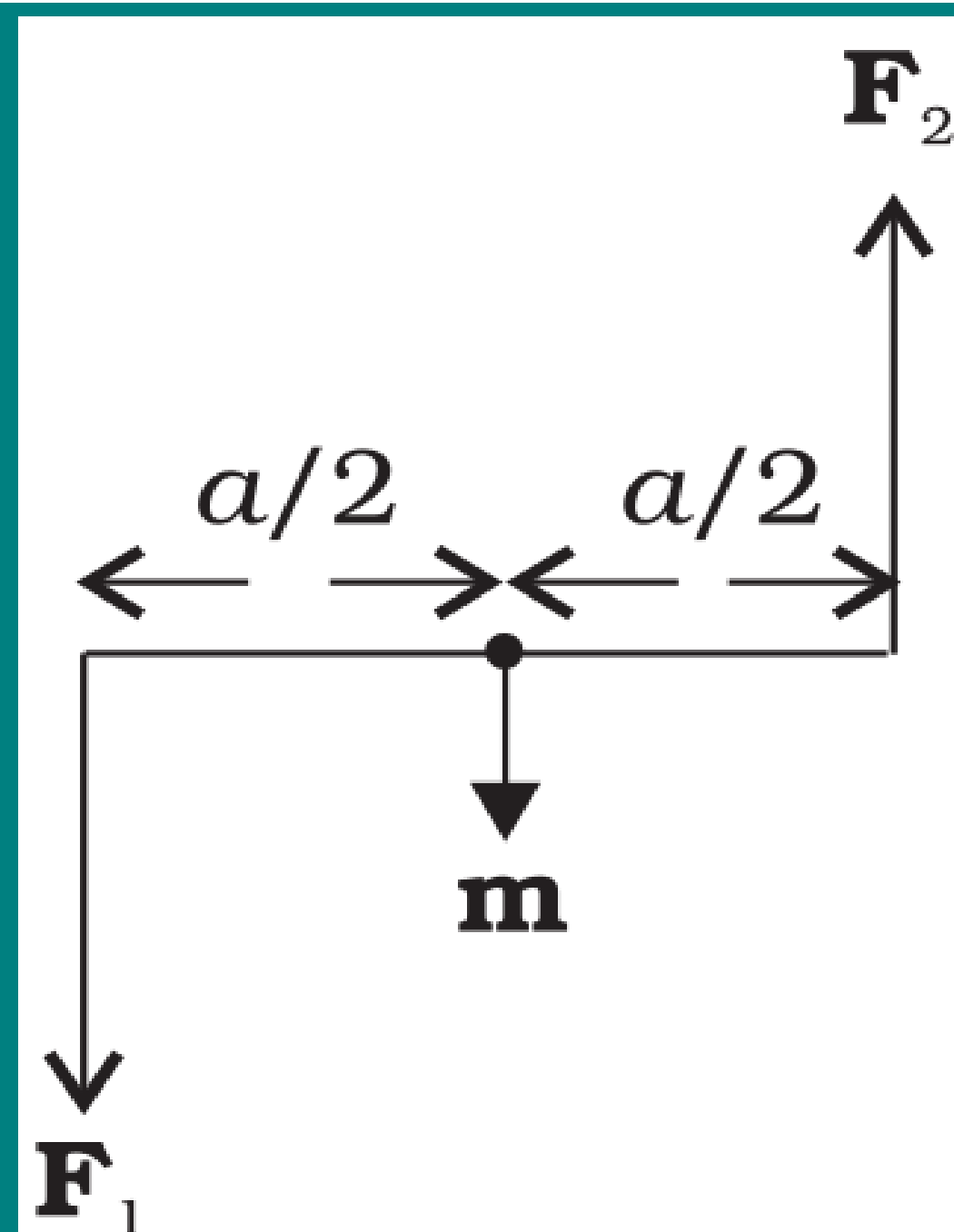
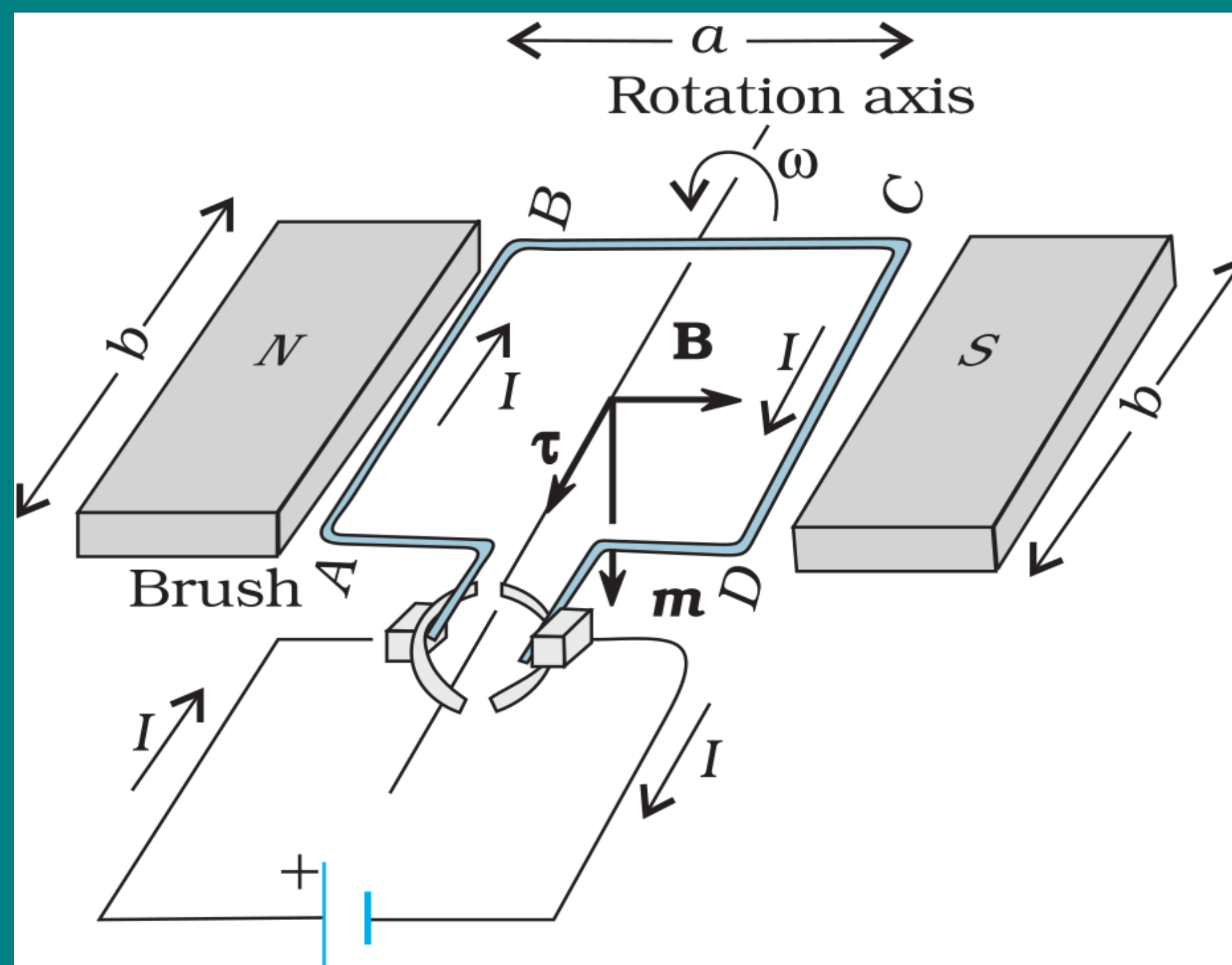
Torque on a Current Loop In a Uniform Magnetic Field



Case 1: When uniform magnetic field $\mathbf{B}$ is in the plane of the loop

- There is a **torque on the loop** due to the pair of **forces $\mathbf{F}_1$ and $\mathbf{F}_2$** .

Torque on a Current Loop In a Uniform Magnetic Field

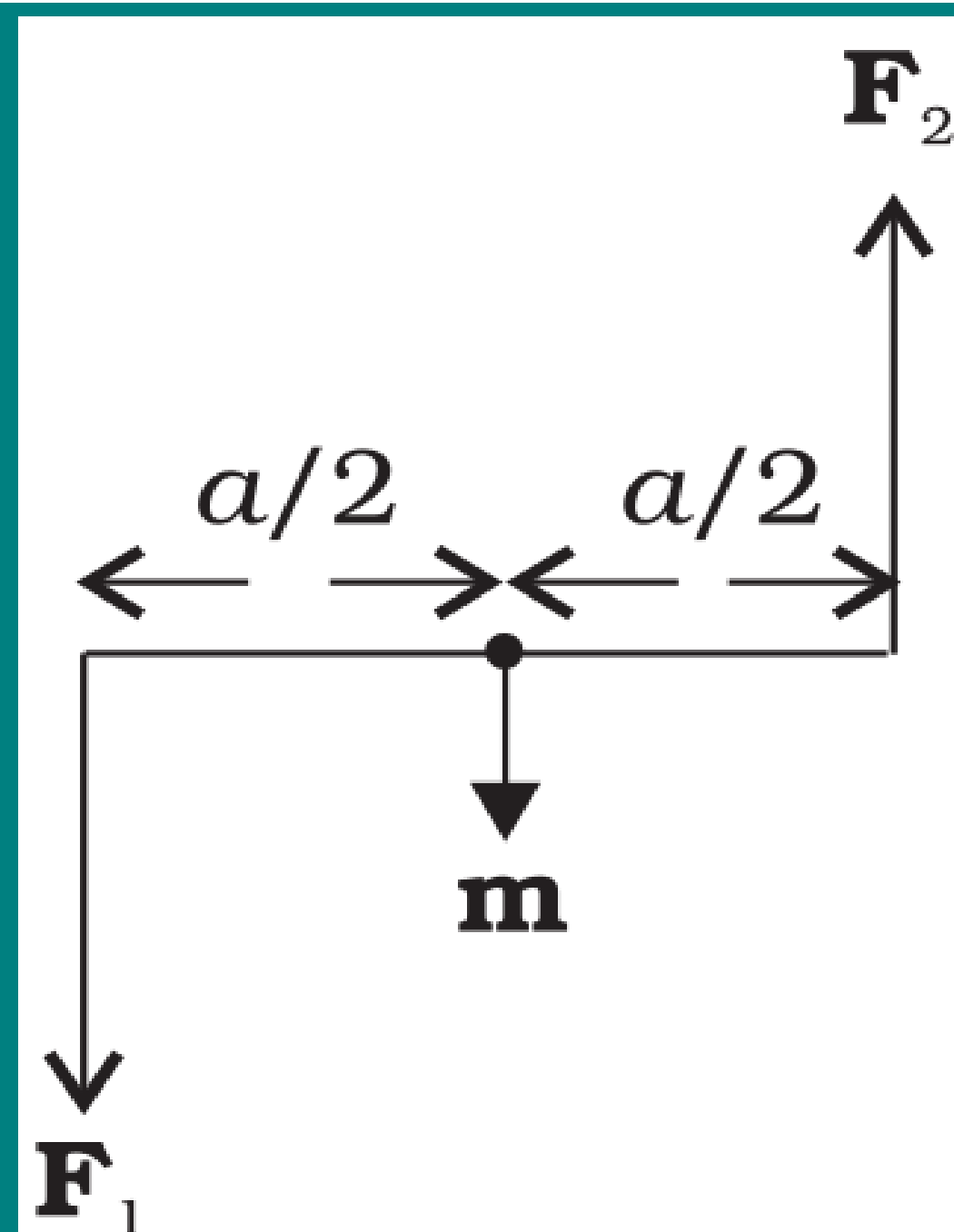
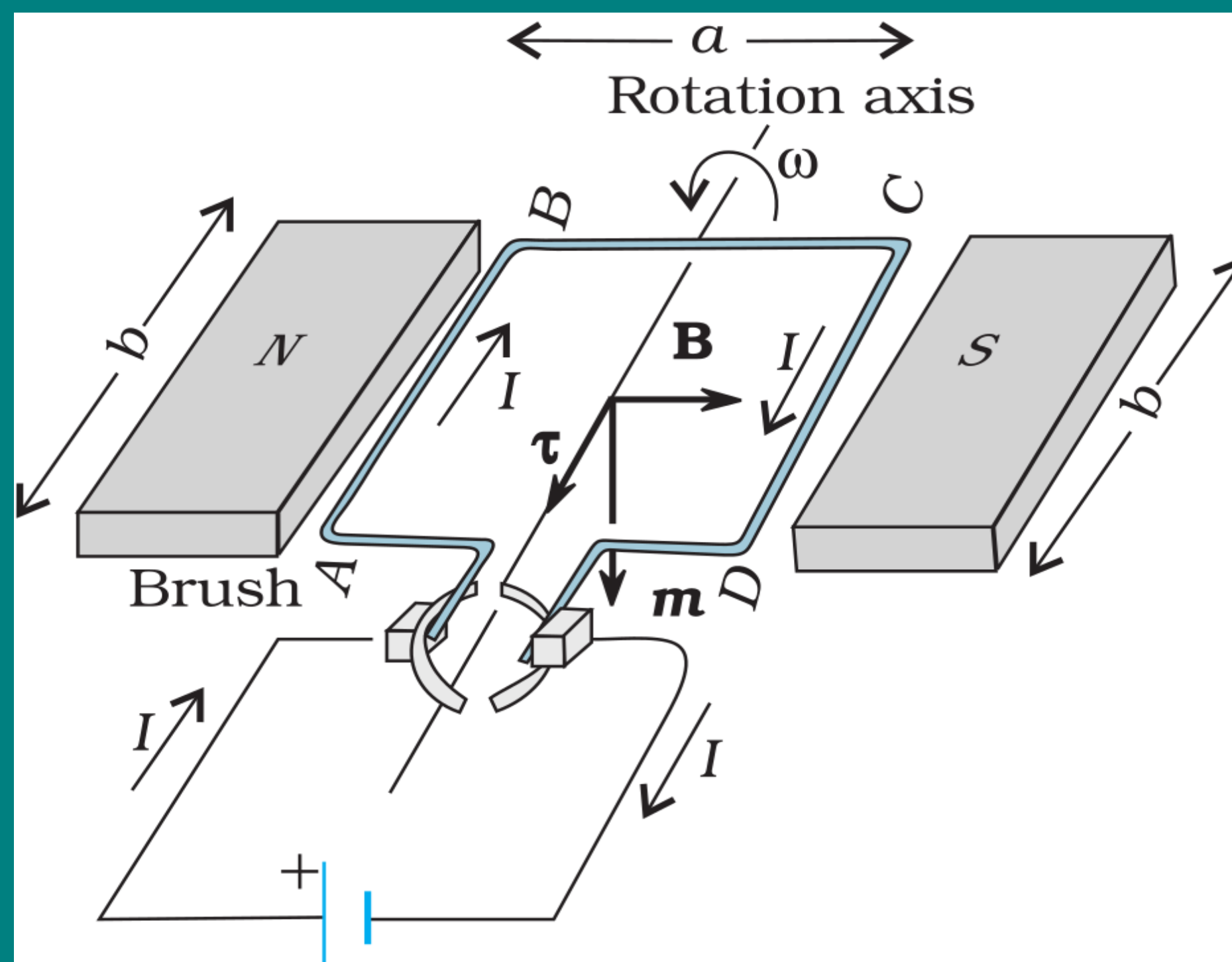


Case 1: When uniform magnetic field $\mathbf{B}$ is in the plane of the loop

- There is a **torque on the loop** due to the pair of **forces** $\mathbf{F}_1$ and $\mathbf{F}_2$.
- Magnitude of this torque is

$$\tau = F_1 \frac{a}{2} + F_2 \frac{a}{2}$$

Torque on a Current Loop In a Uniform Magnetic Field



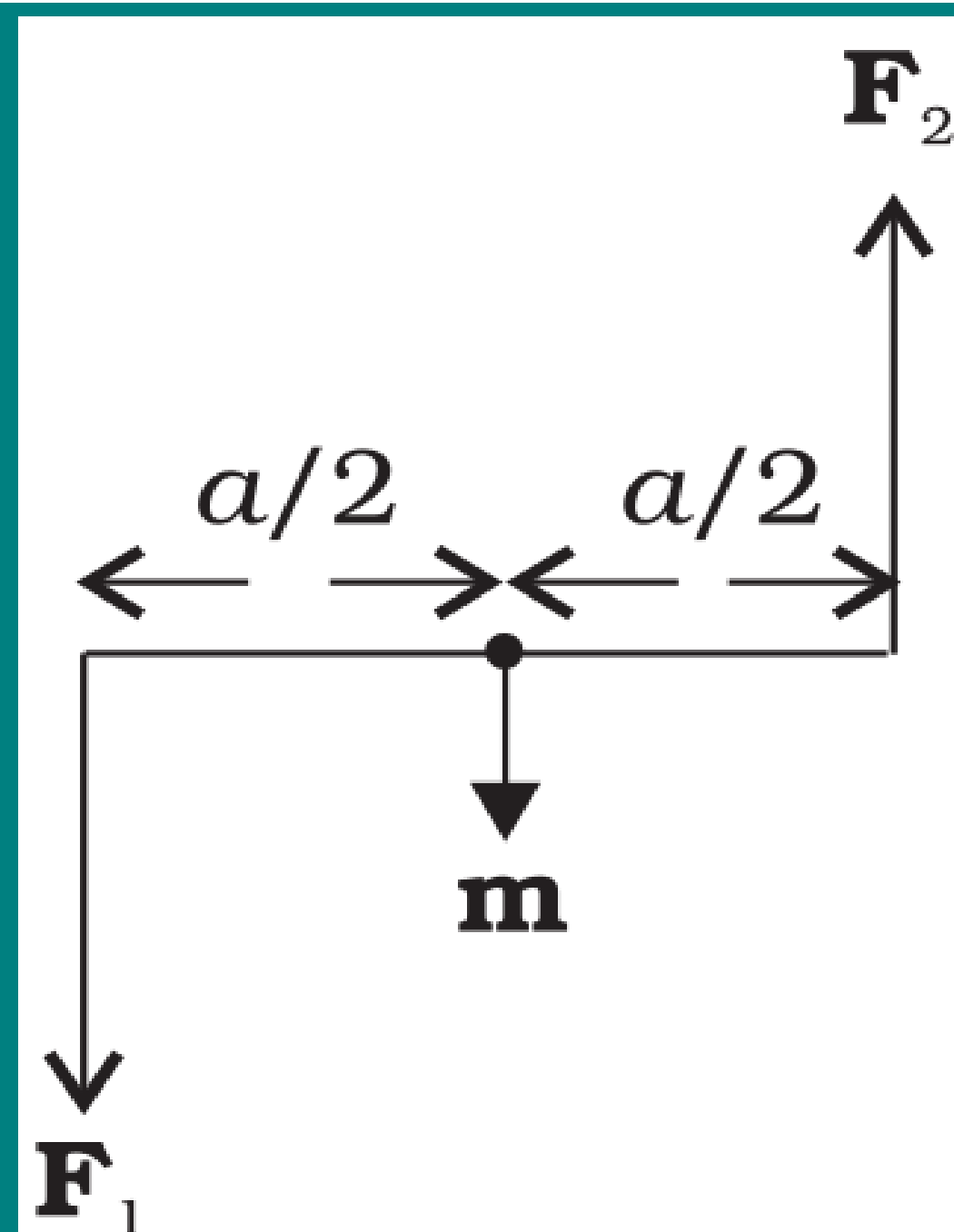
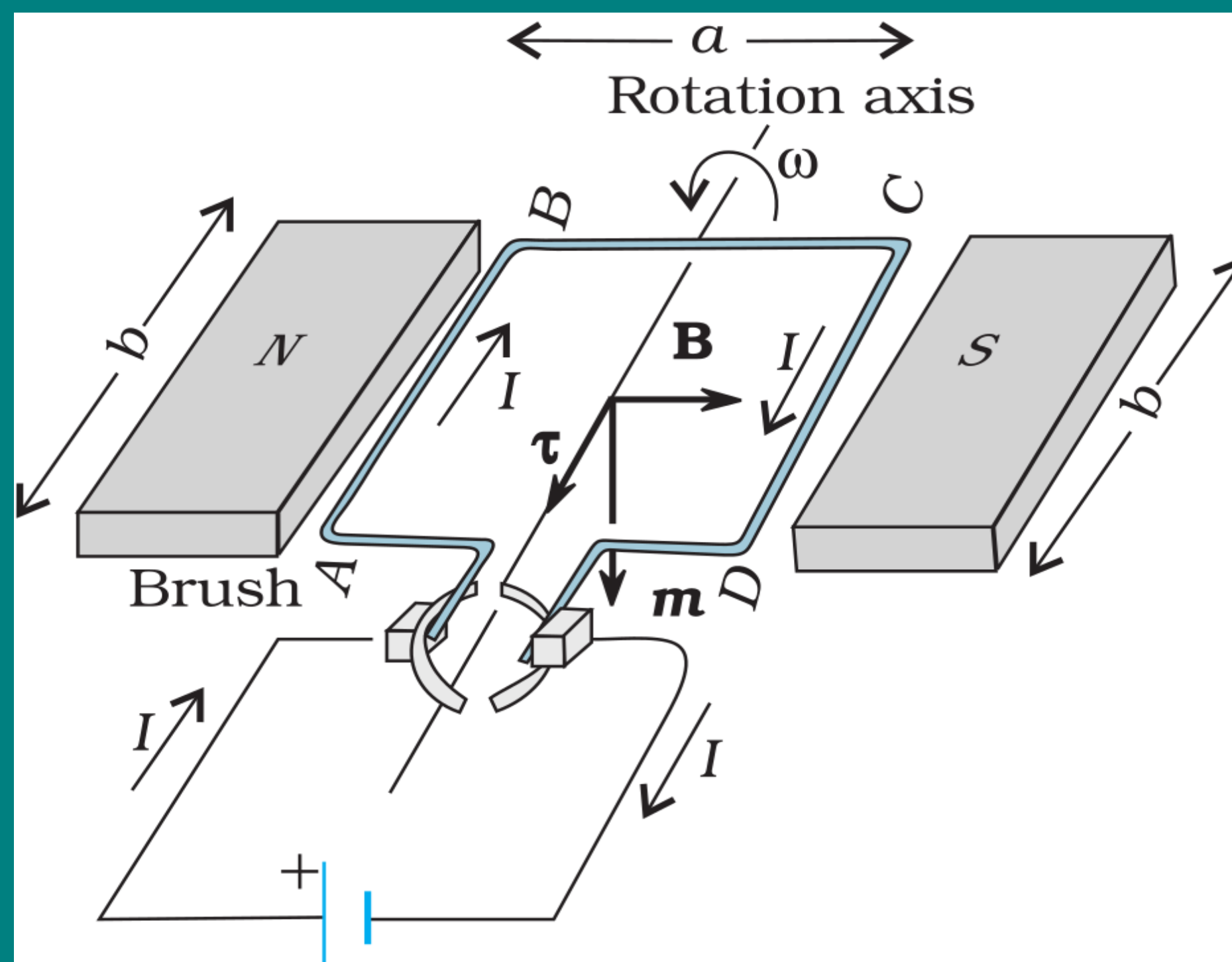
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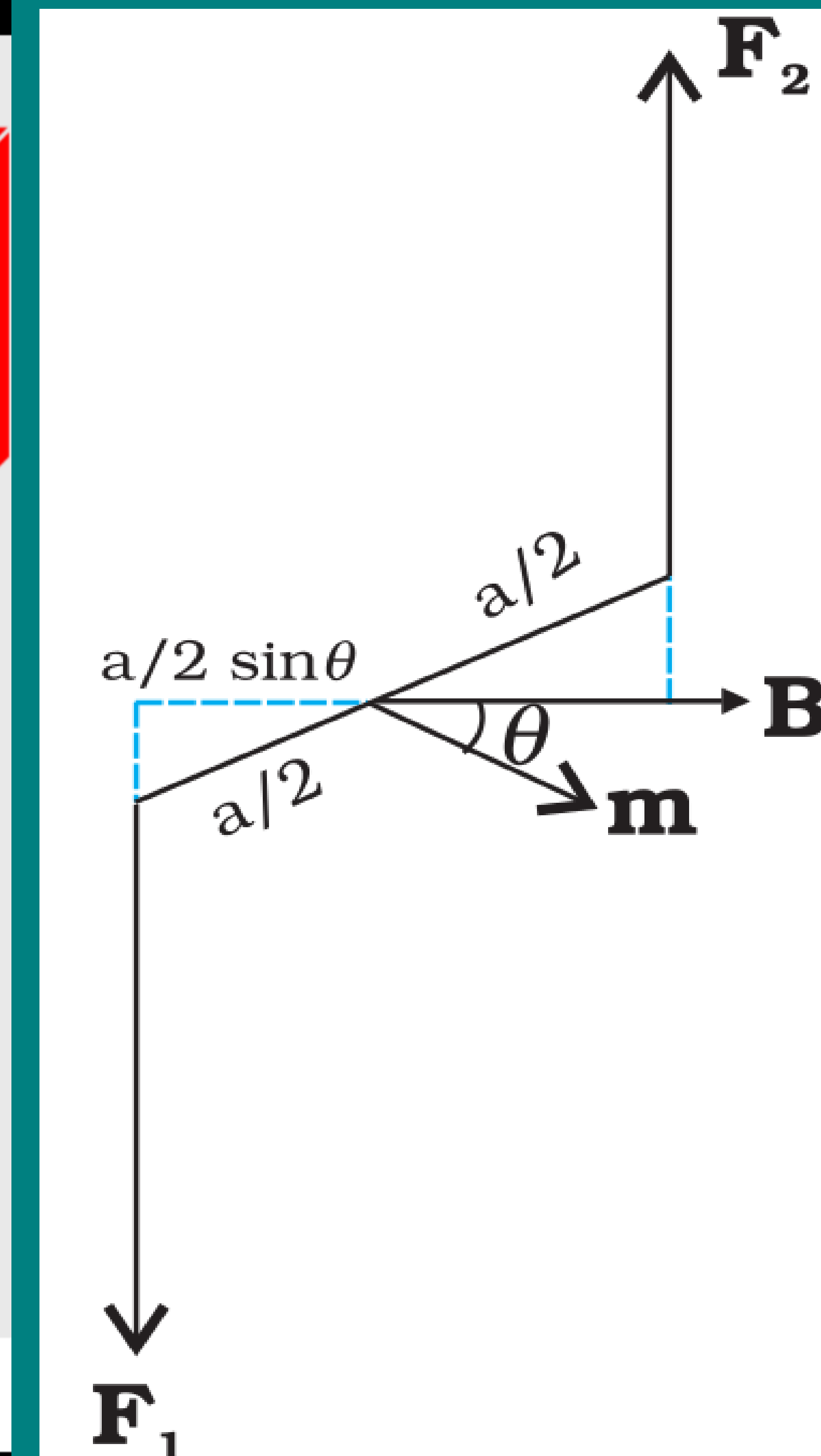
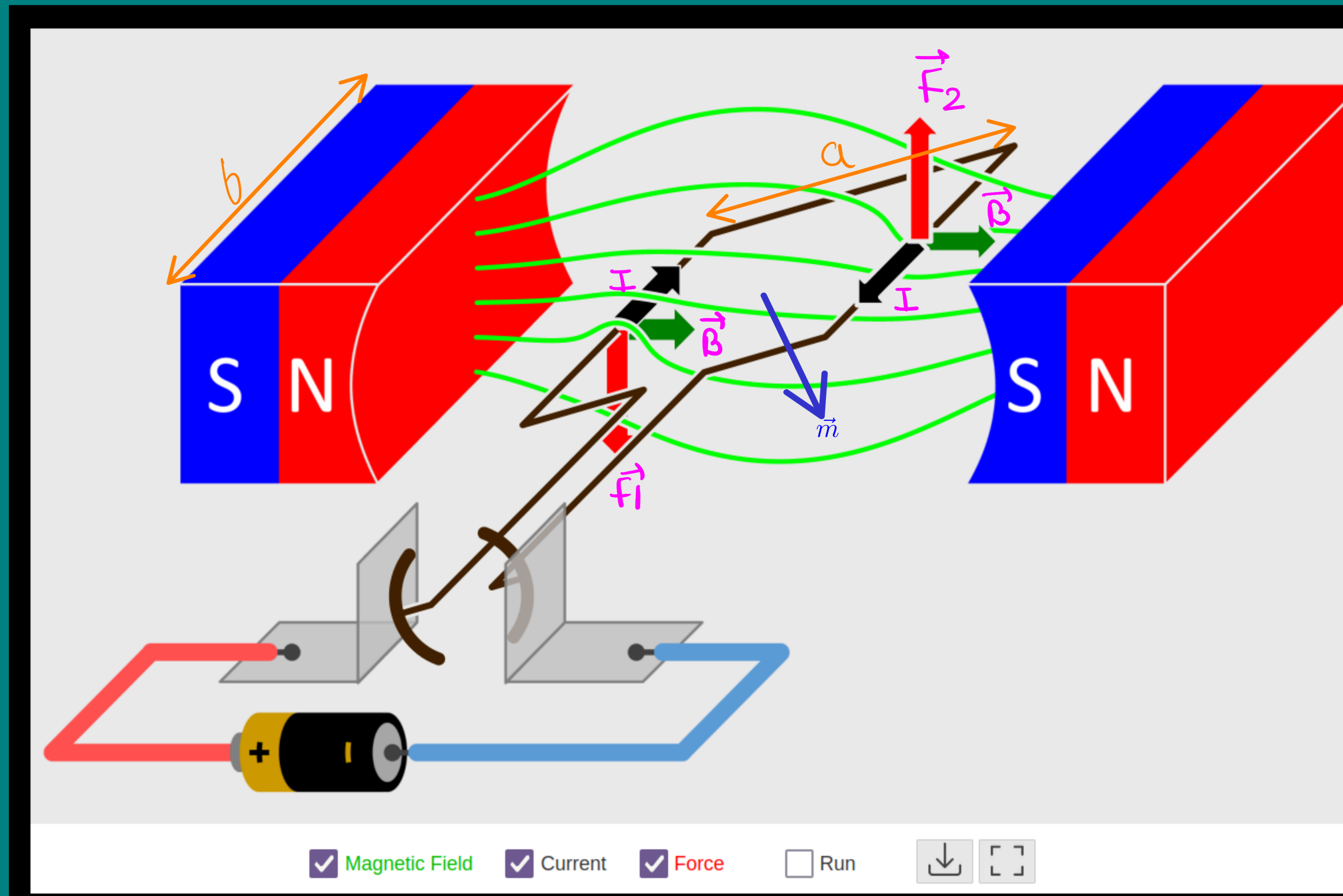
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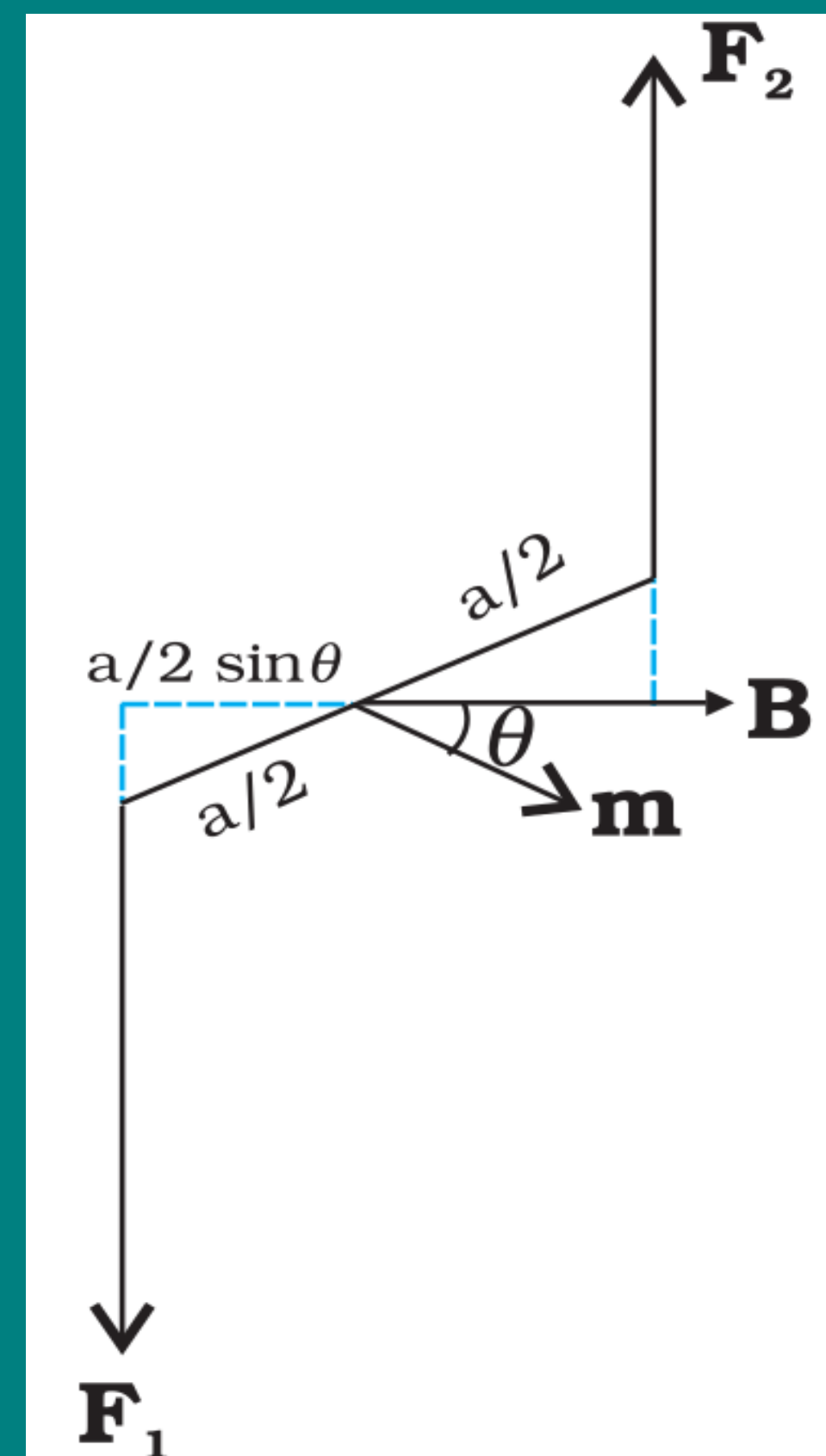
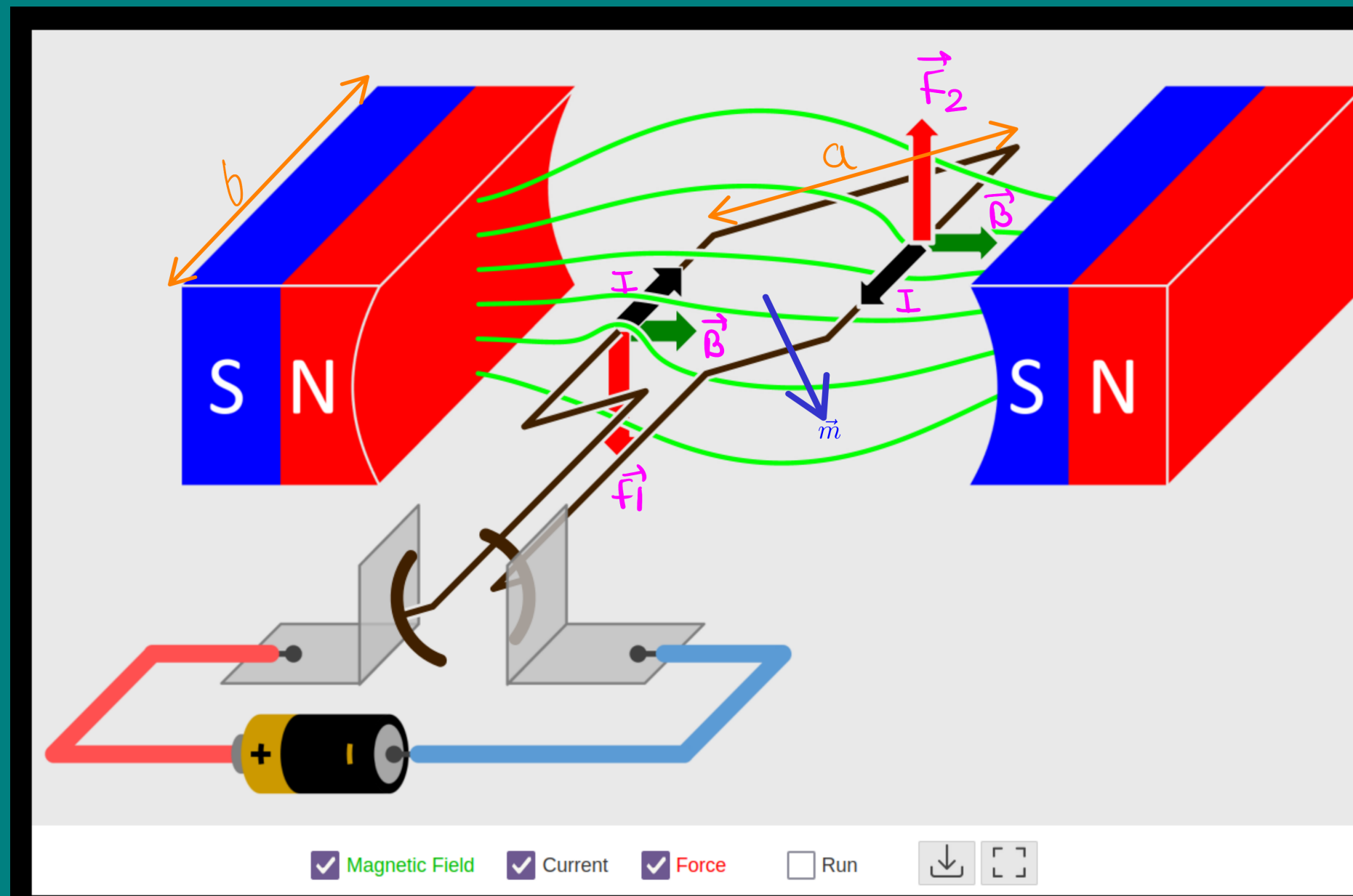
$$\tau = IbB \frac{a}{2} + IbB \frac{a}{2} = IabB$$

$$\boxed{\tau = IAB} \text{ (Here, } A = ab \text{ is the area of the loop)}$$

(2) When the plane of the loop, is not along the magnetic field

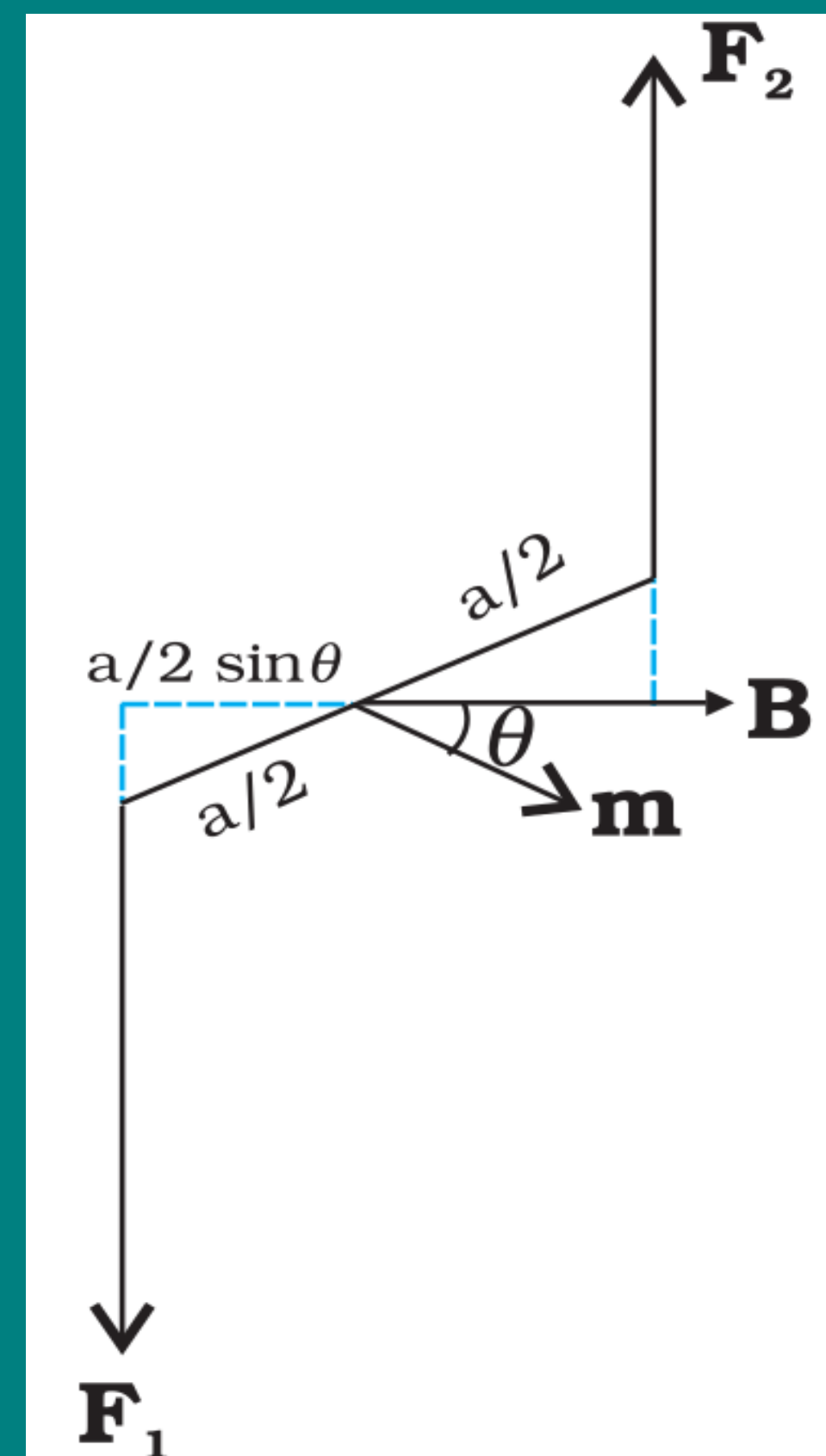
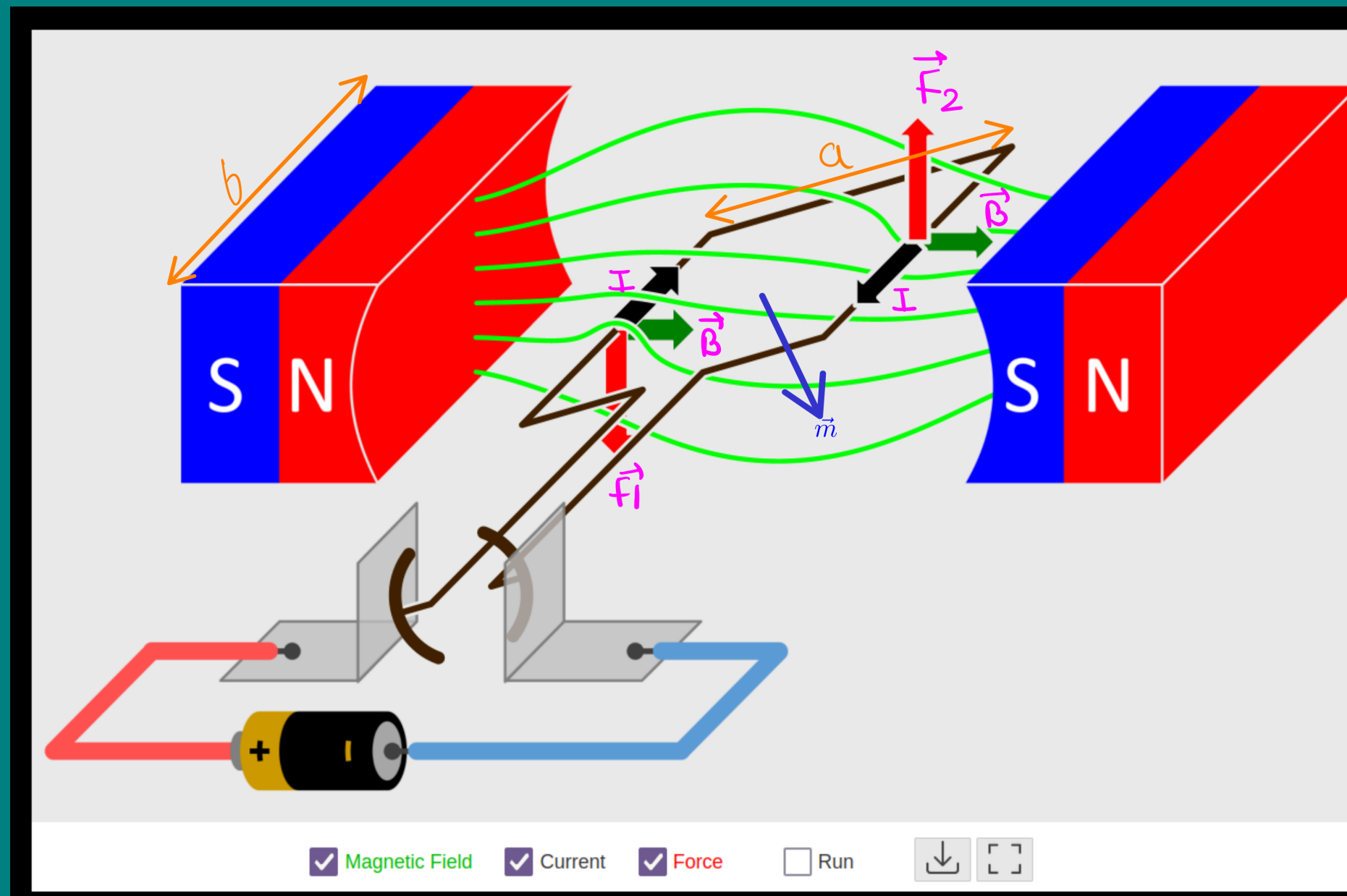


Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

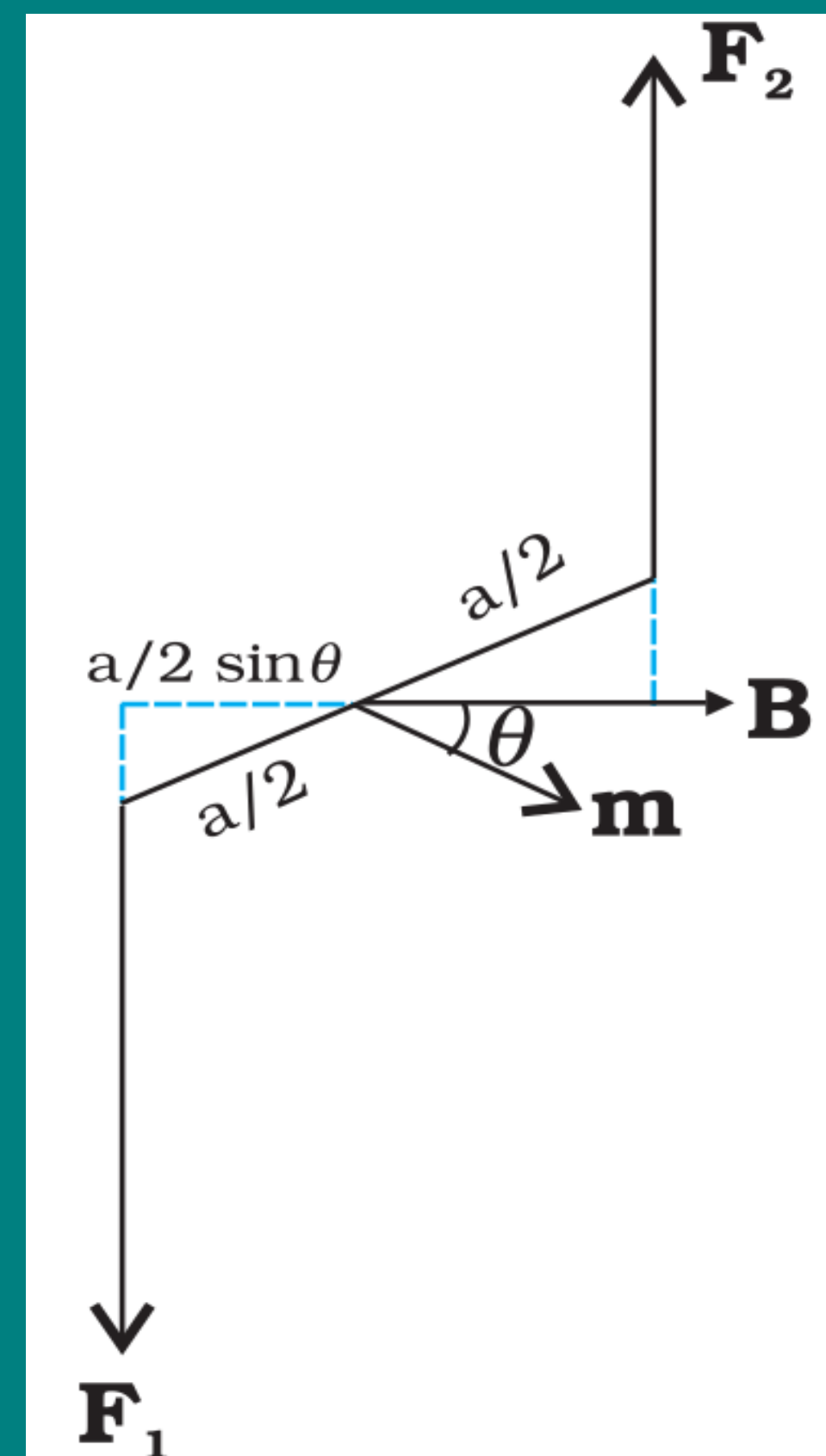
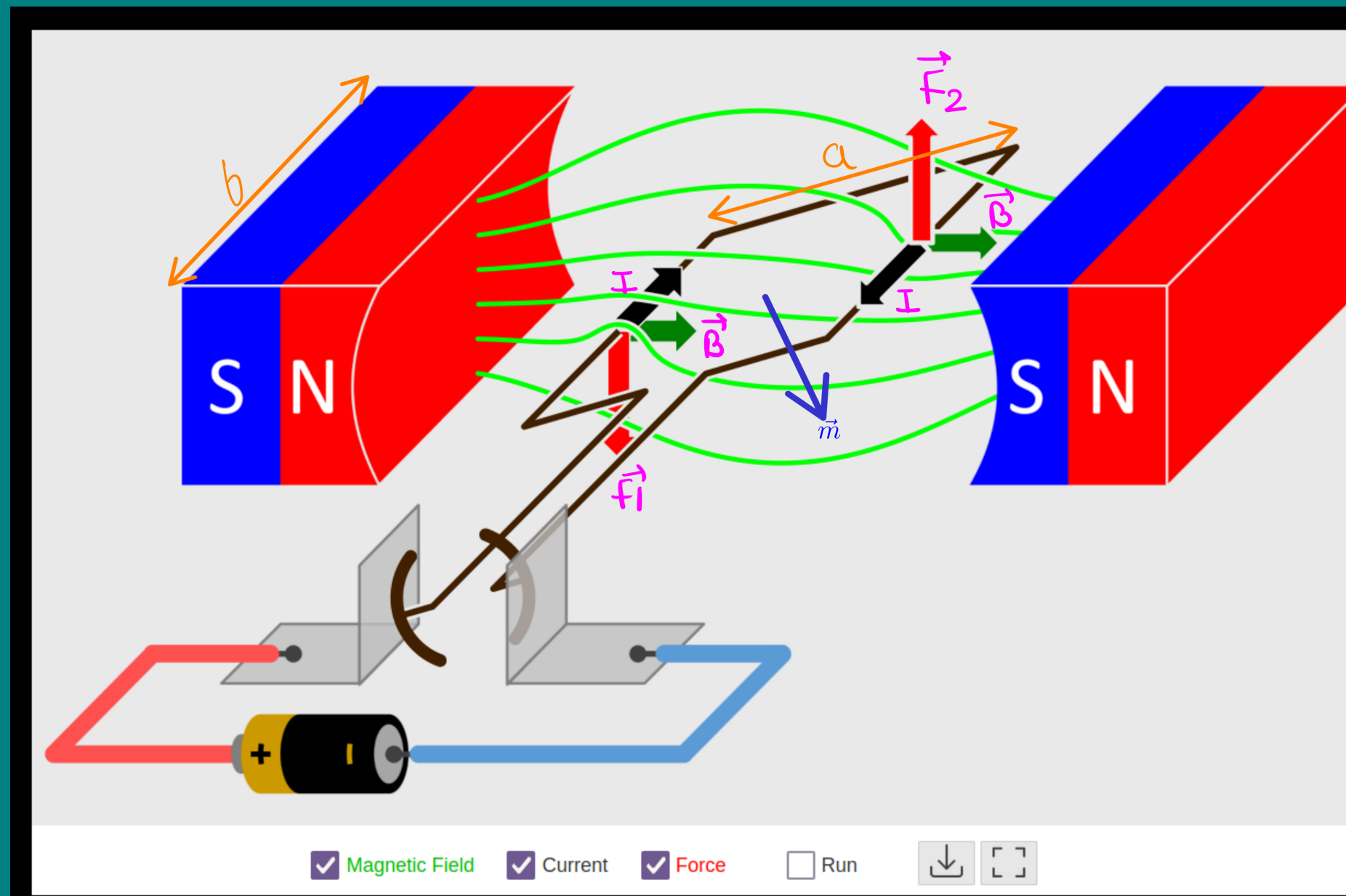
Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

- Let the **angle between the field and the normal** to the coil to be angle θ

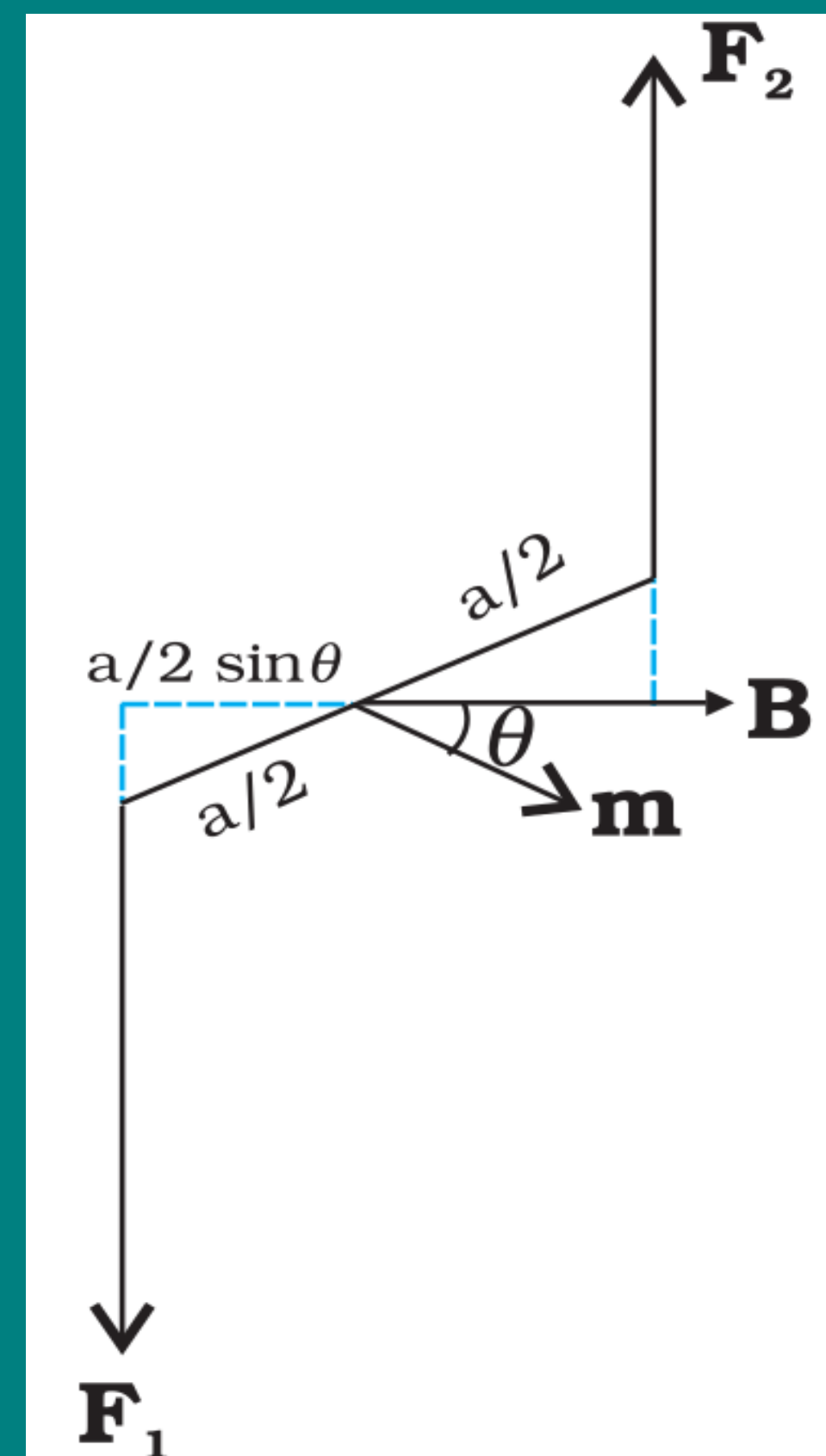
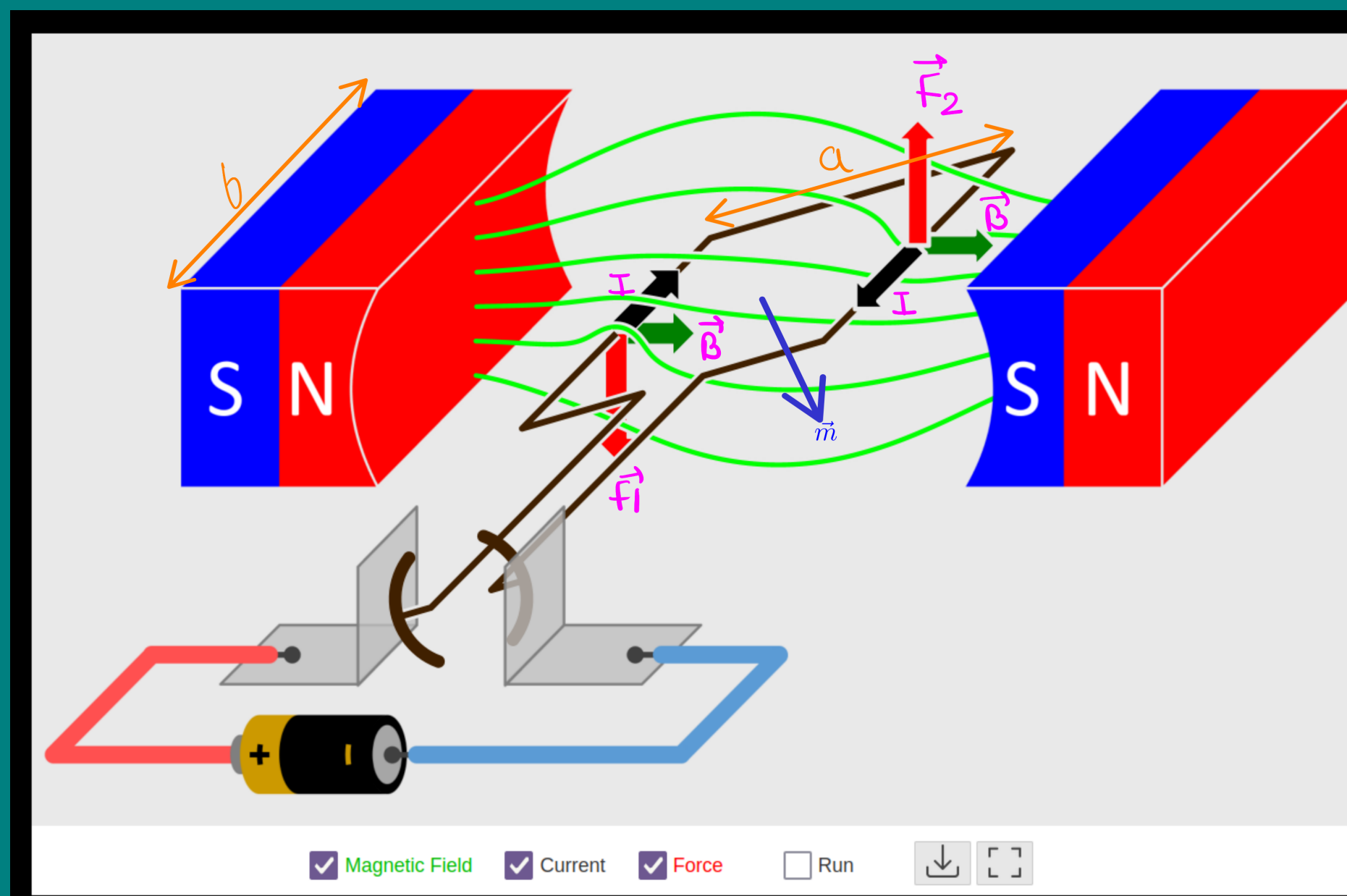
Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

- Let the **angle between the field and the normal** to the coil to be angle θ
- The **forces on the arms BC and DA are equal, opposite, and act along the axis** of the coil. Being **collinear** along the axis they cancel each other, **resulting in no net force or torque.**

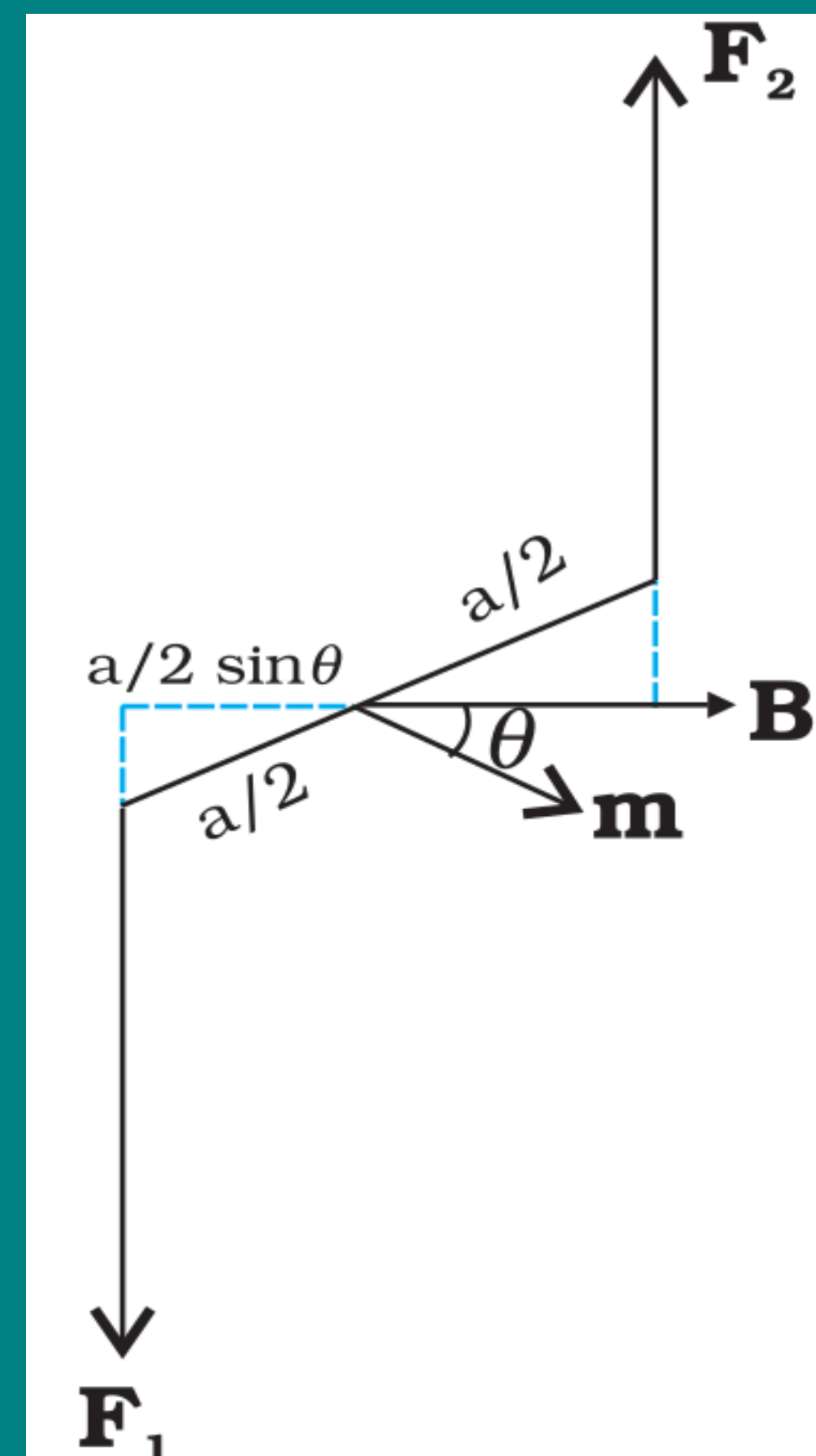
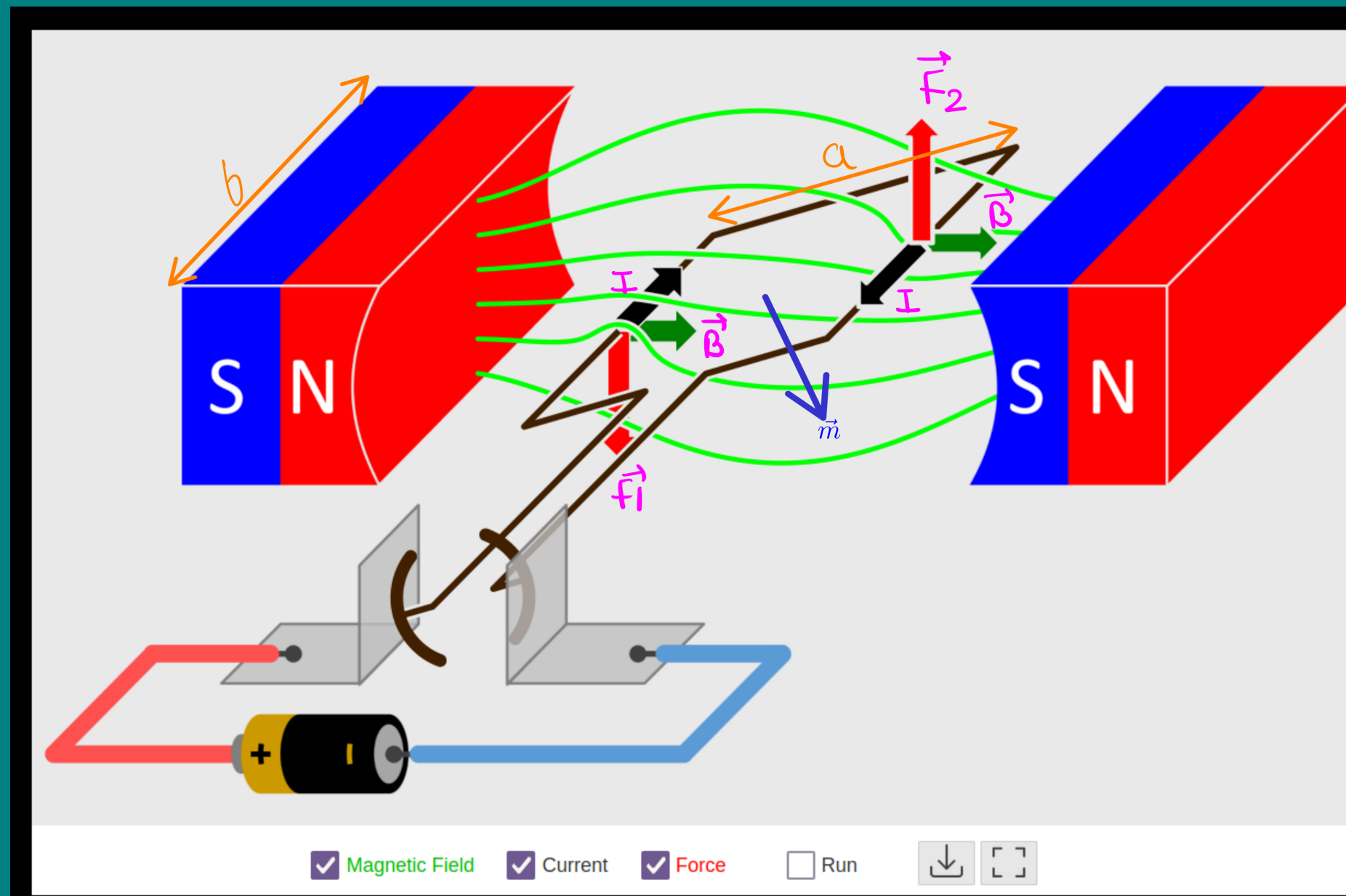
Torque on a Current Loop In a Uniform Magnetic Field



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- The forces on arms **AB** and **CD** are F_1 and F_2 . They too are **equal and opposite**, with magnitude,

Torque on a Current Loop In a Uniform Magnetic Field

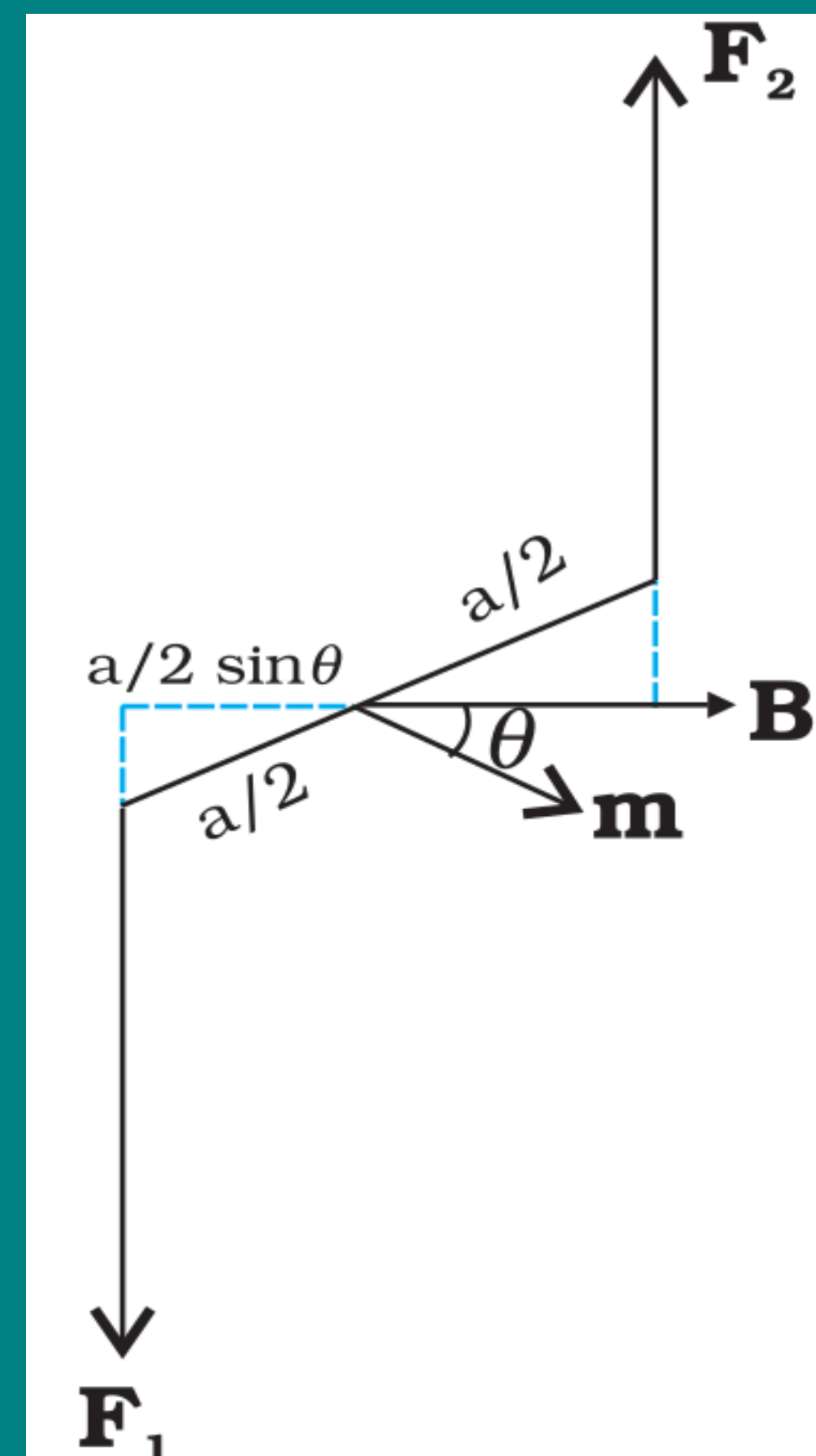
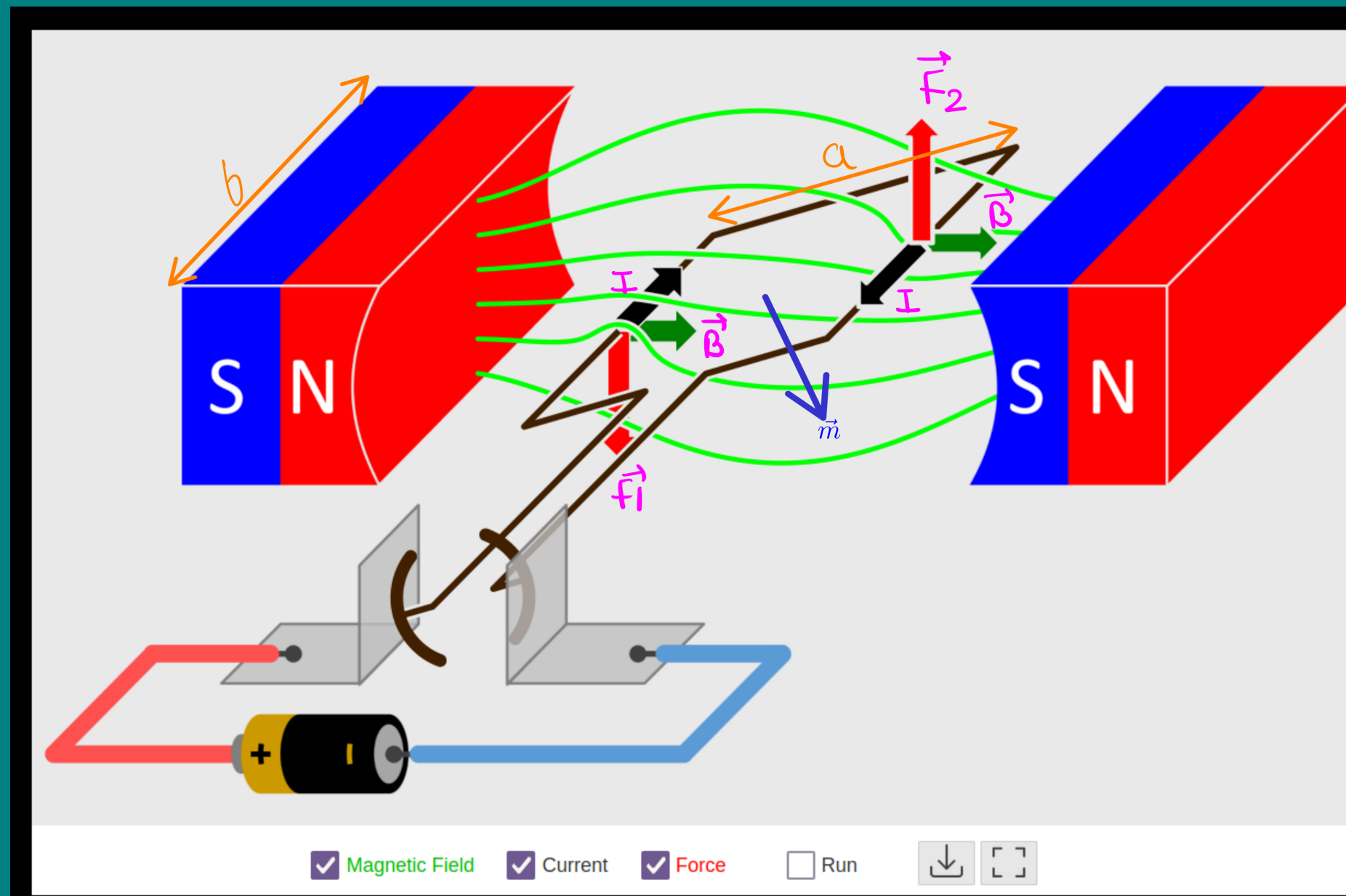


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Torque on a Current Loop In a Uniform Magnetic Field



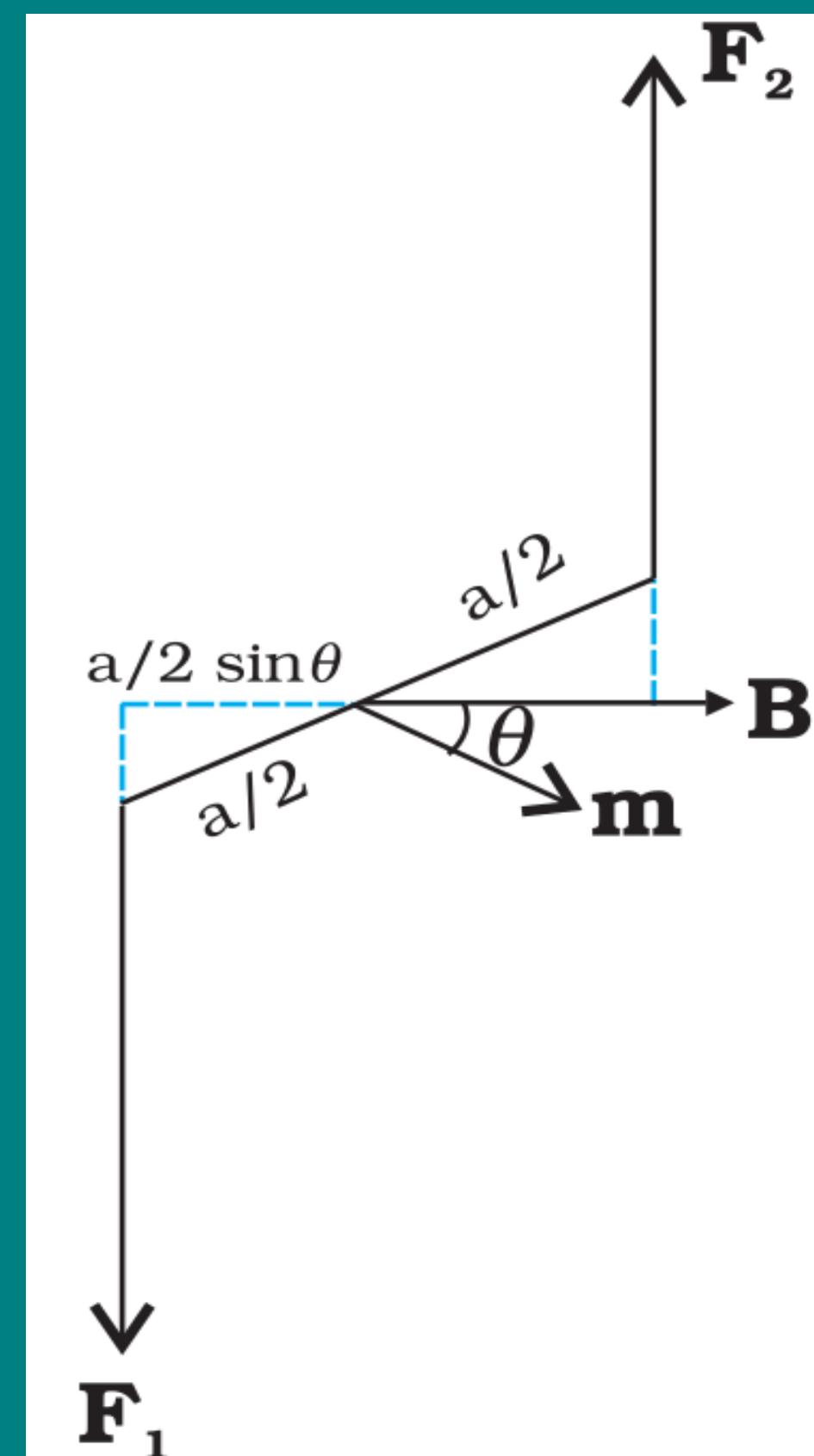
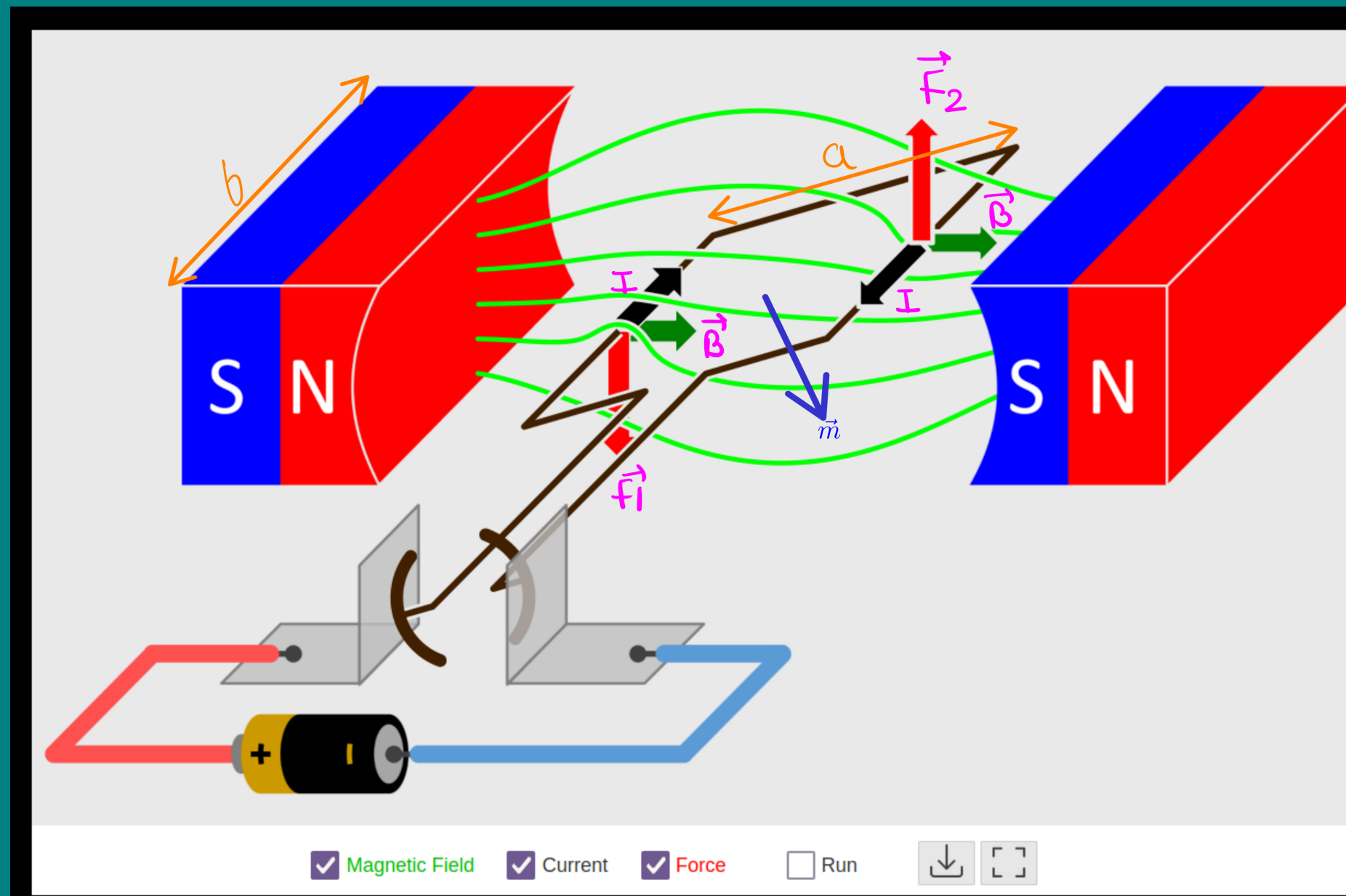
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- The forces on arms **AB** and **CD** are F_1 and F_2 . They too are **equal and opposite**, with magnitude,

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- But they are **not collinear**! This **results in a couple** as before. The torque on the loop is,

Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

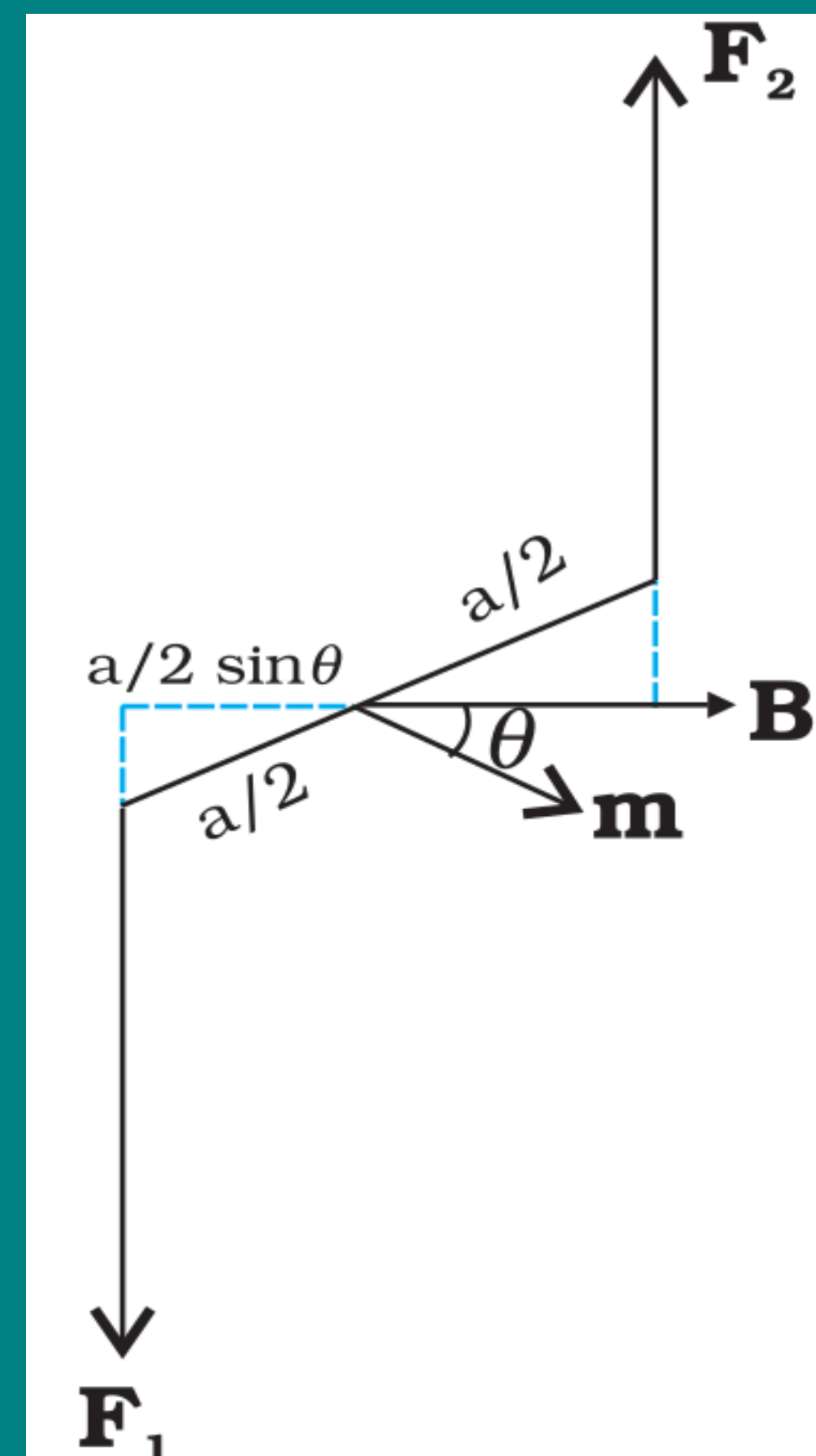
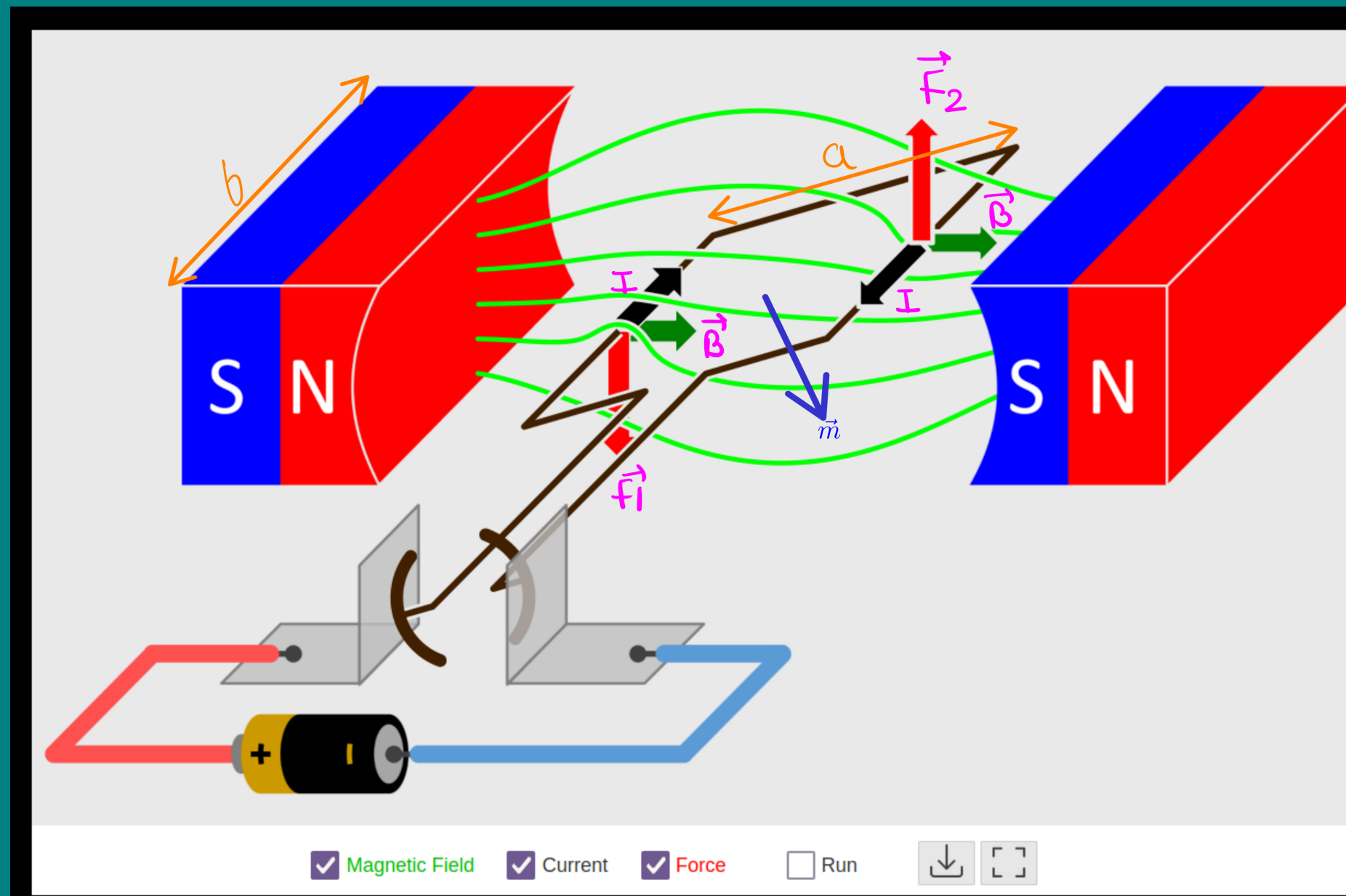
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Torque on a Current Loop In a Uniform Magnetic Field



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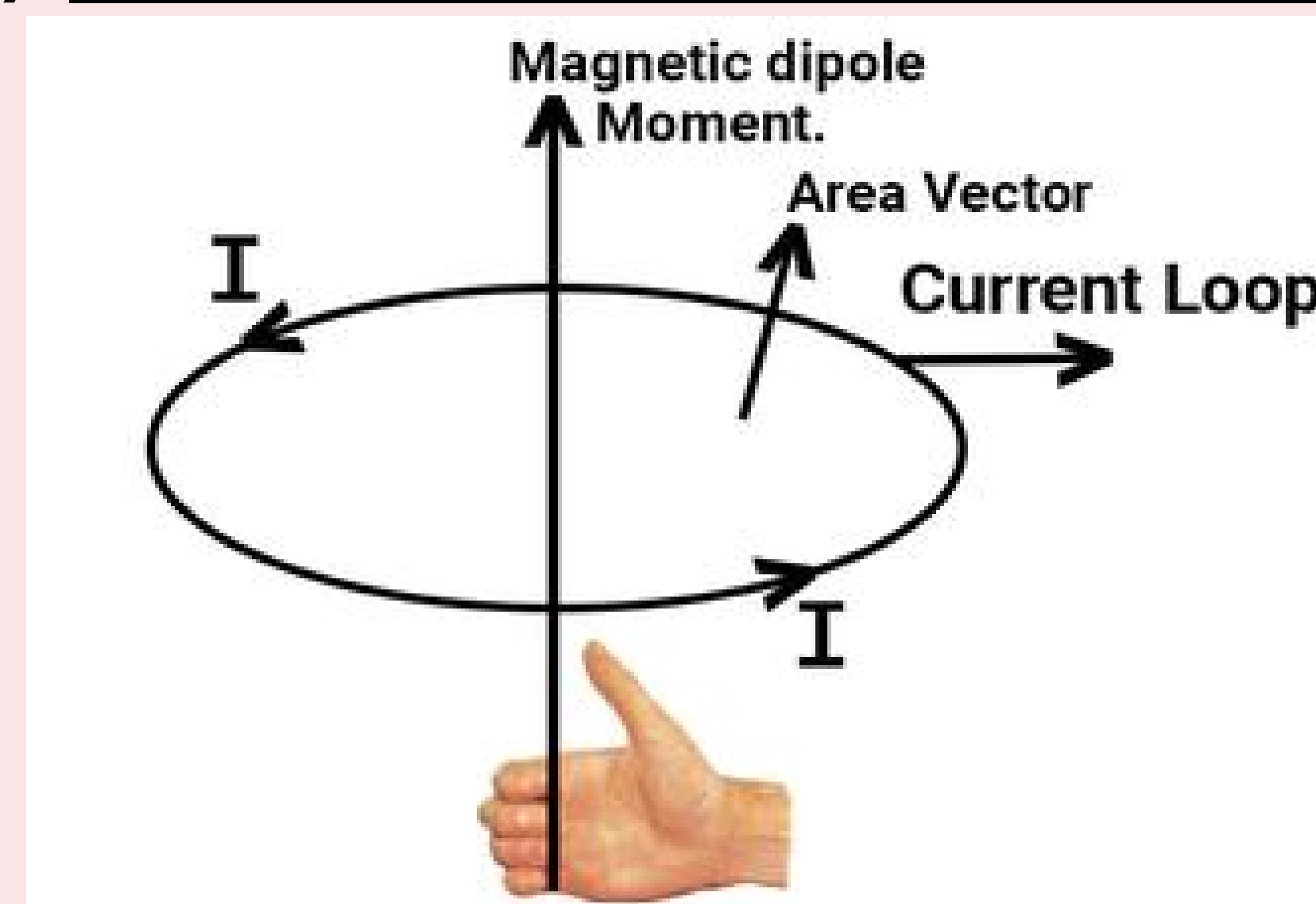
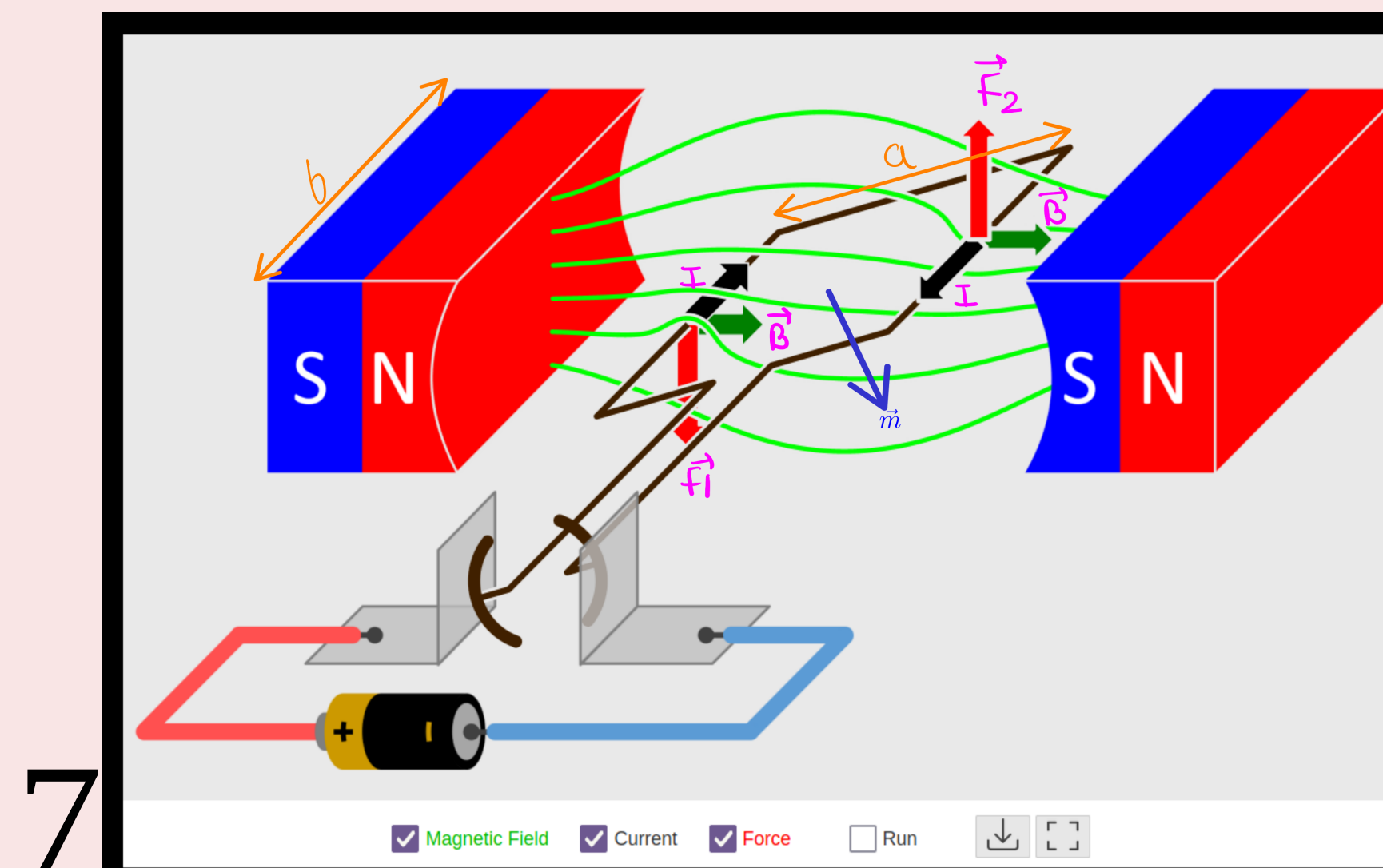
- But they are **not collinear**! This **results in a couple** as before. The torque on the loop is,

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$$\tau = \frac{a \sin \theta}{2} (F_1 + F_2)$$

Torque on a Current Loop In a Uniform Magnetic Field

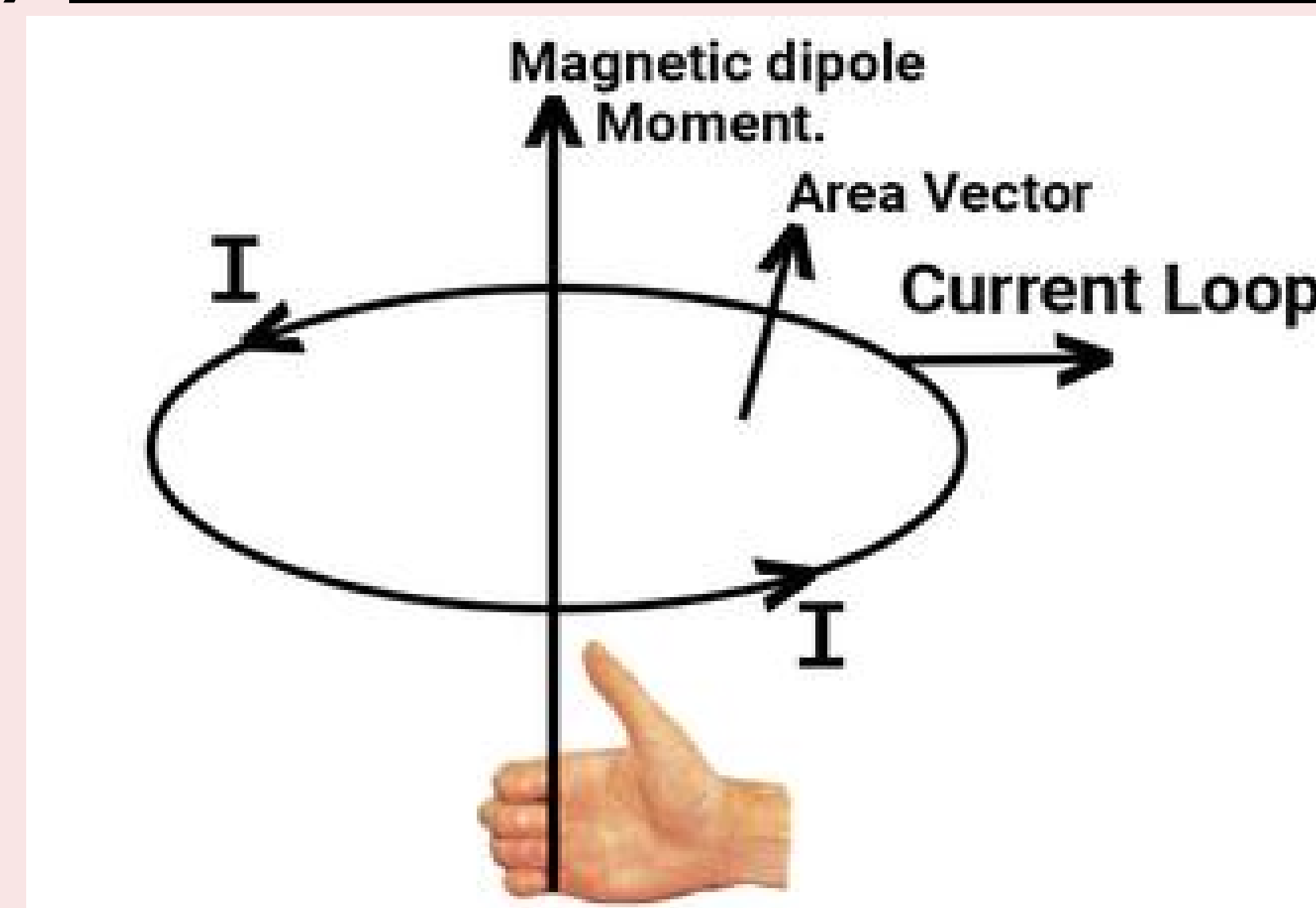
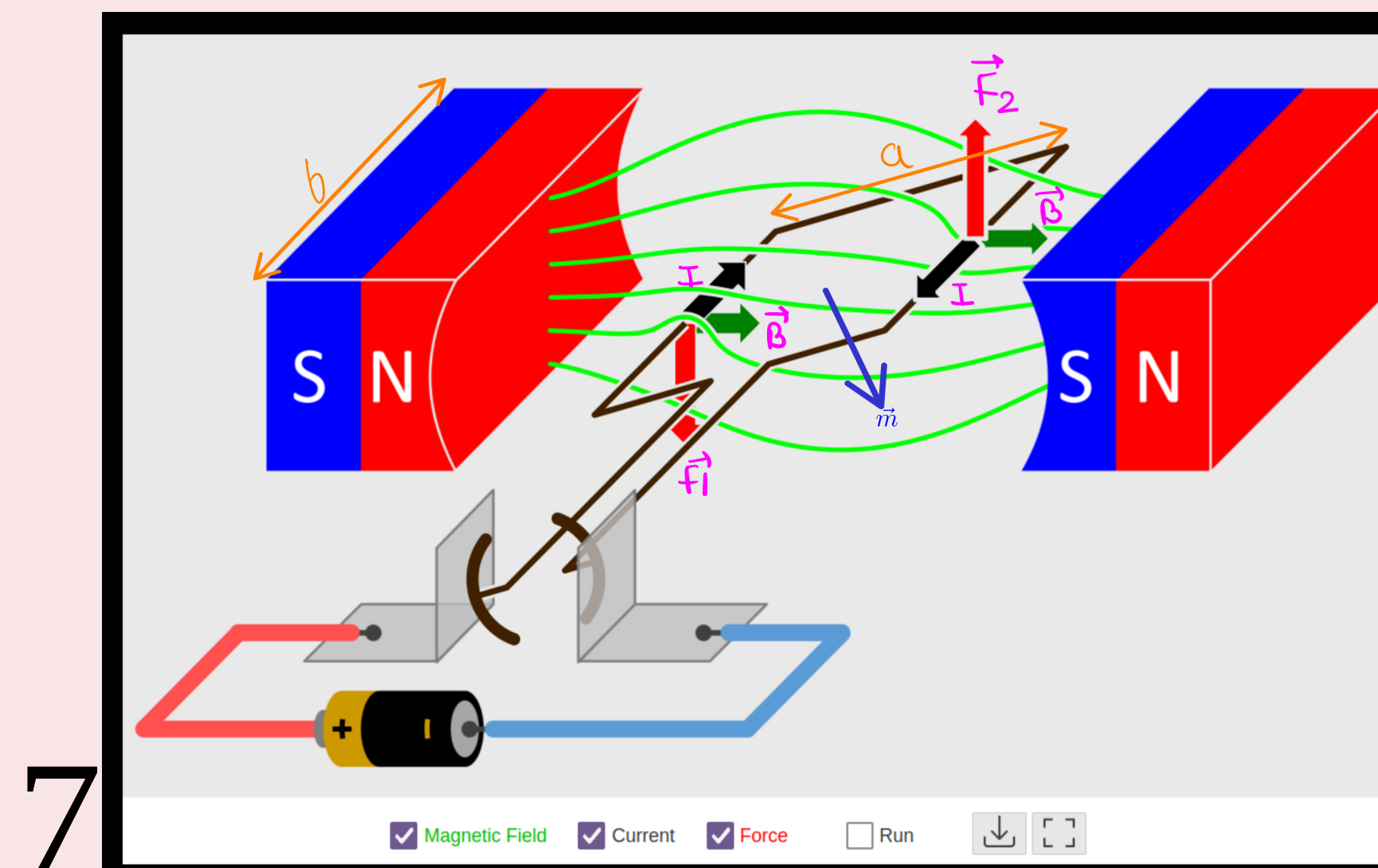
Case 2: When the plane of the loop, is not along the magnetic field



$$\tau = \frac{a \sin \theta}{2} (2IbB) = IabB \sin \theta$$

Torque on a Current Loop In a Uniform Magnetic Field

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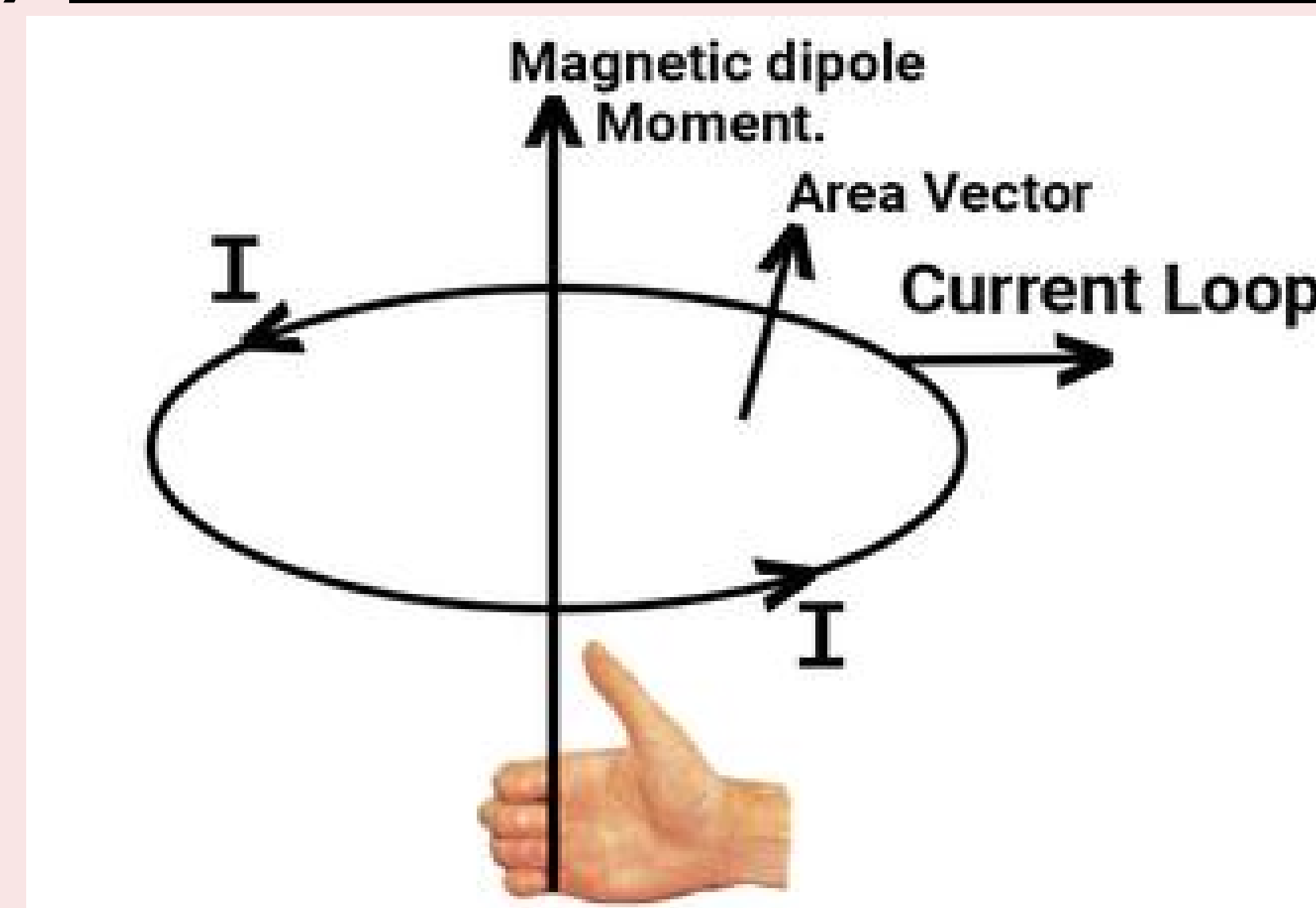
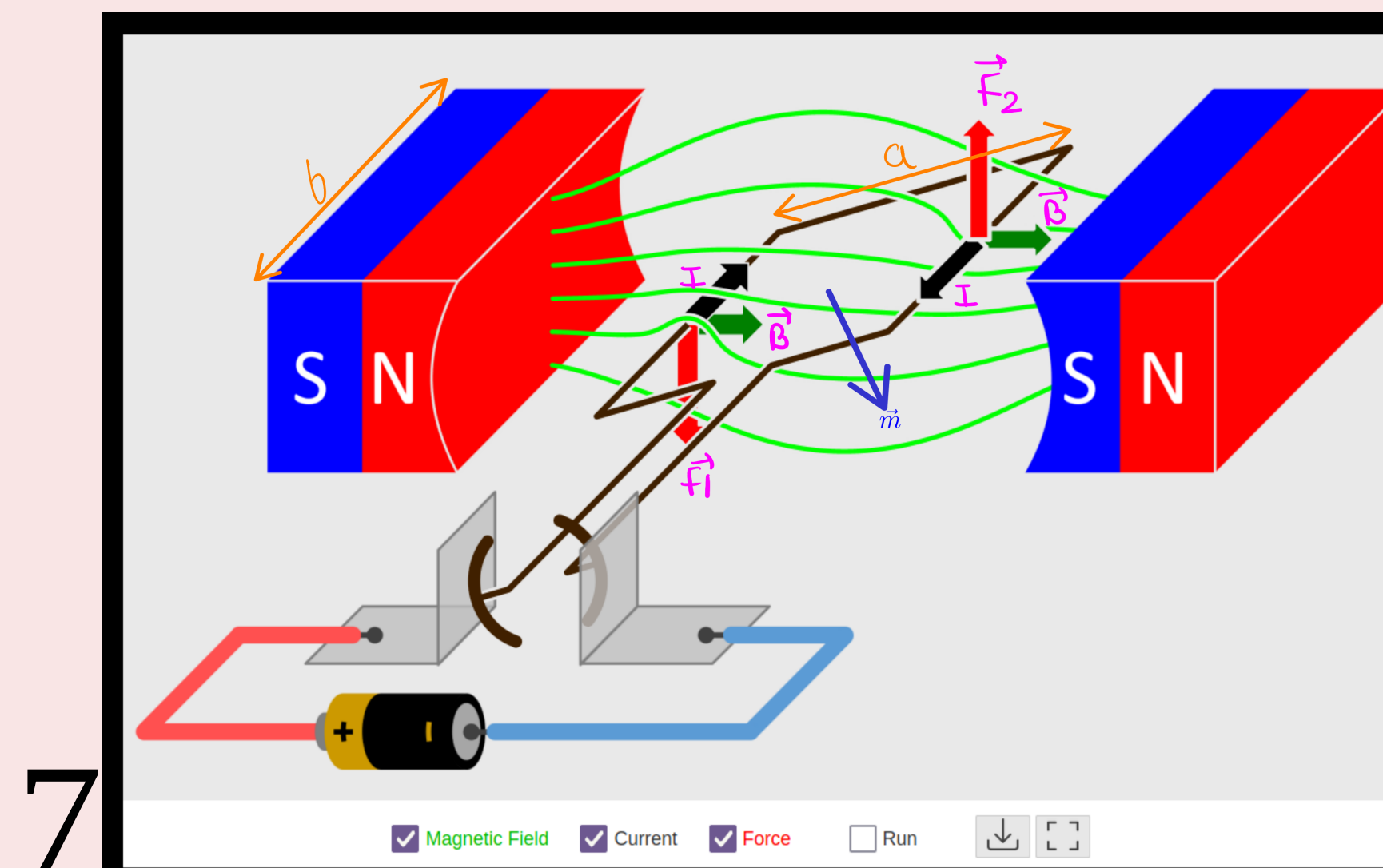


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$$\tau = IAB \sin \theta \quad (\text{Here, } A \text{ is the area of the loop})$$

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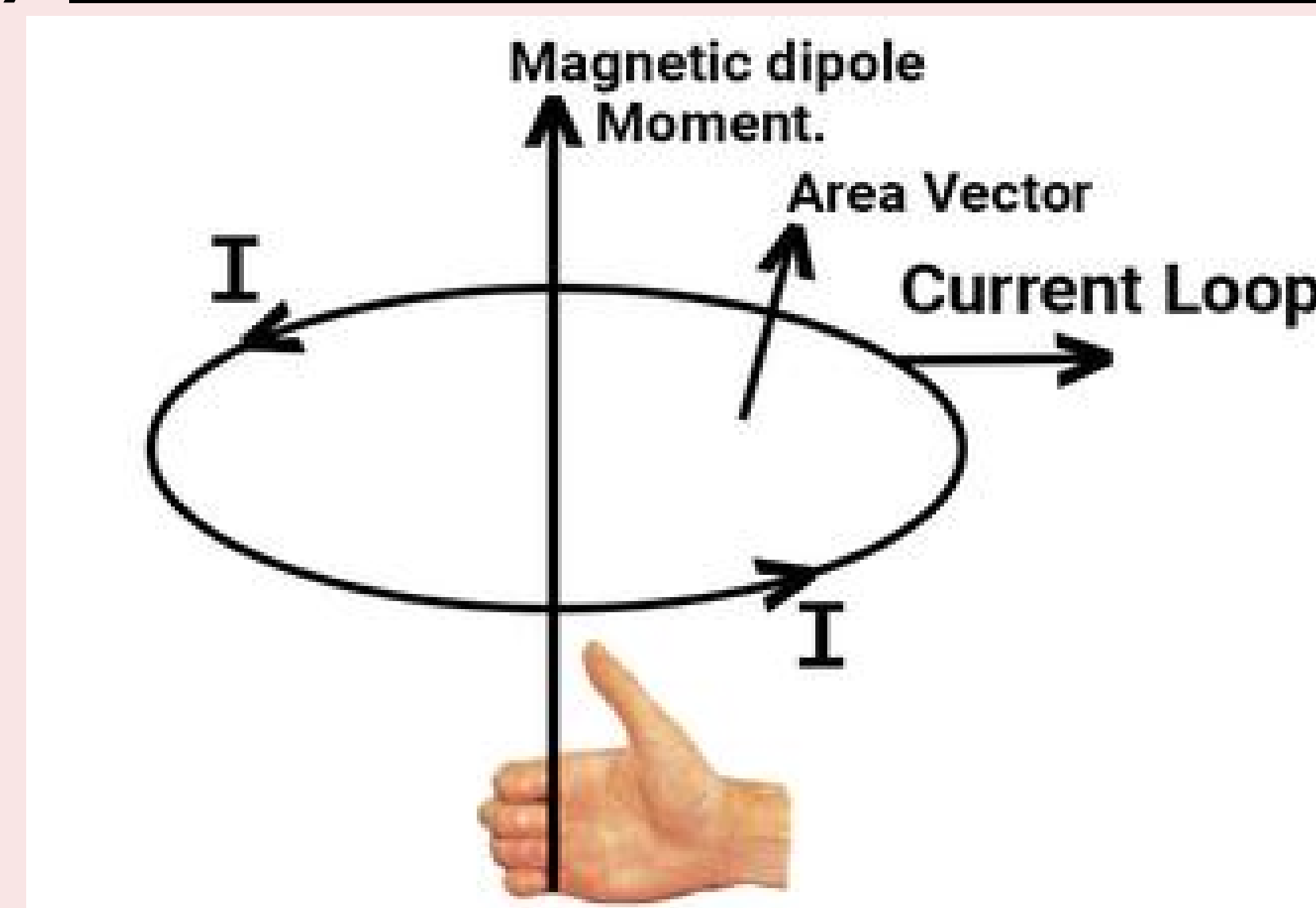
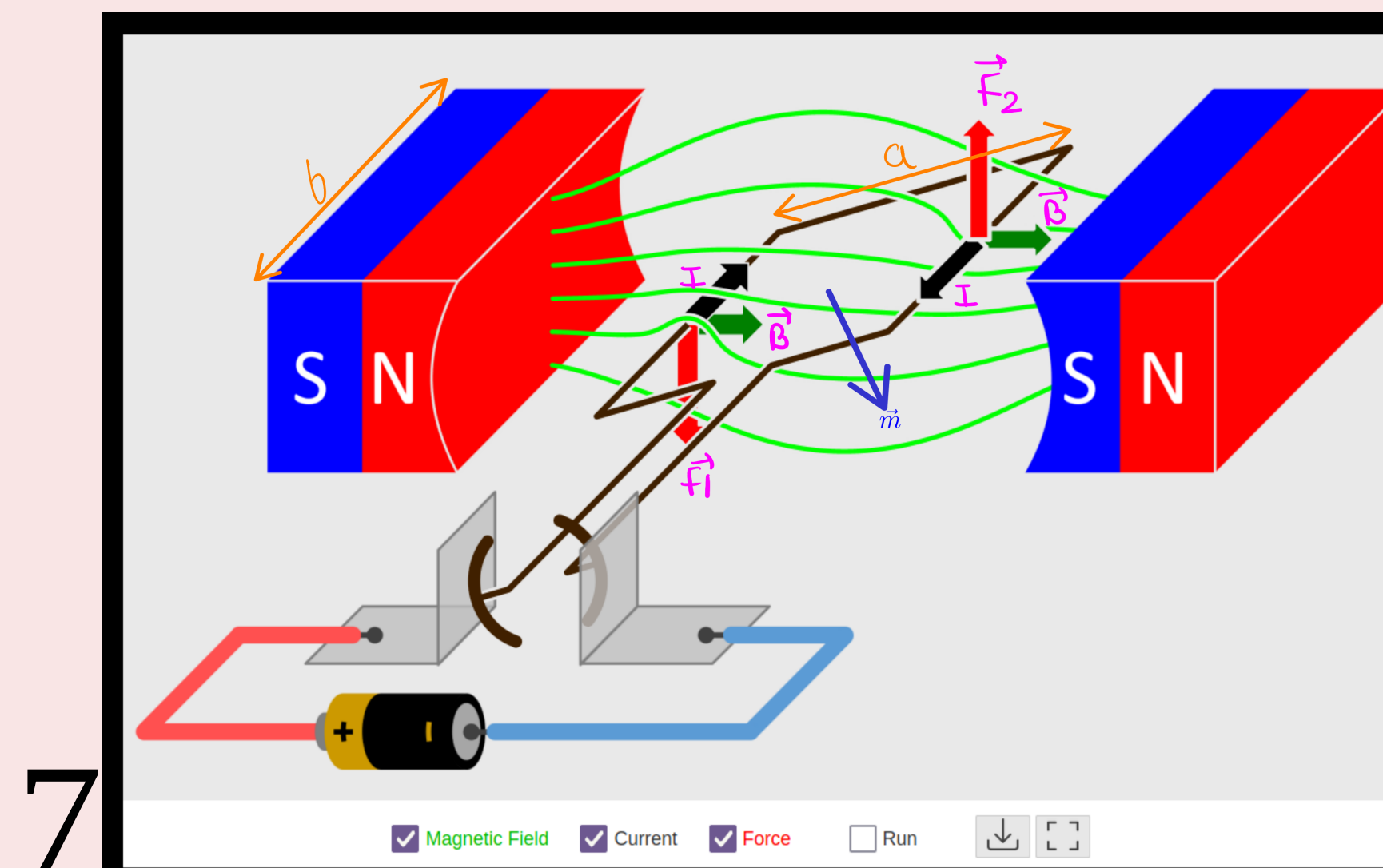
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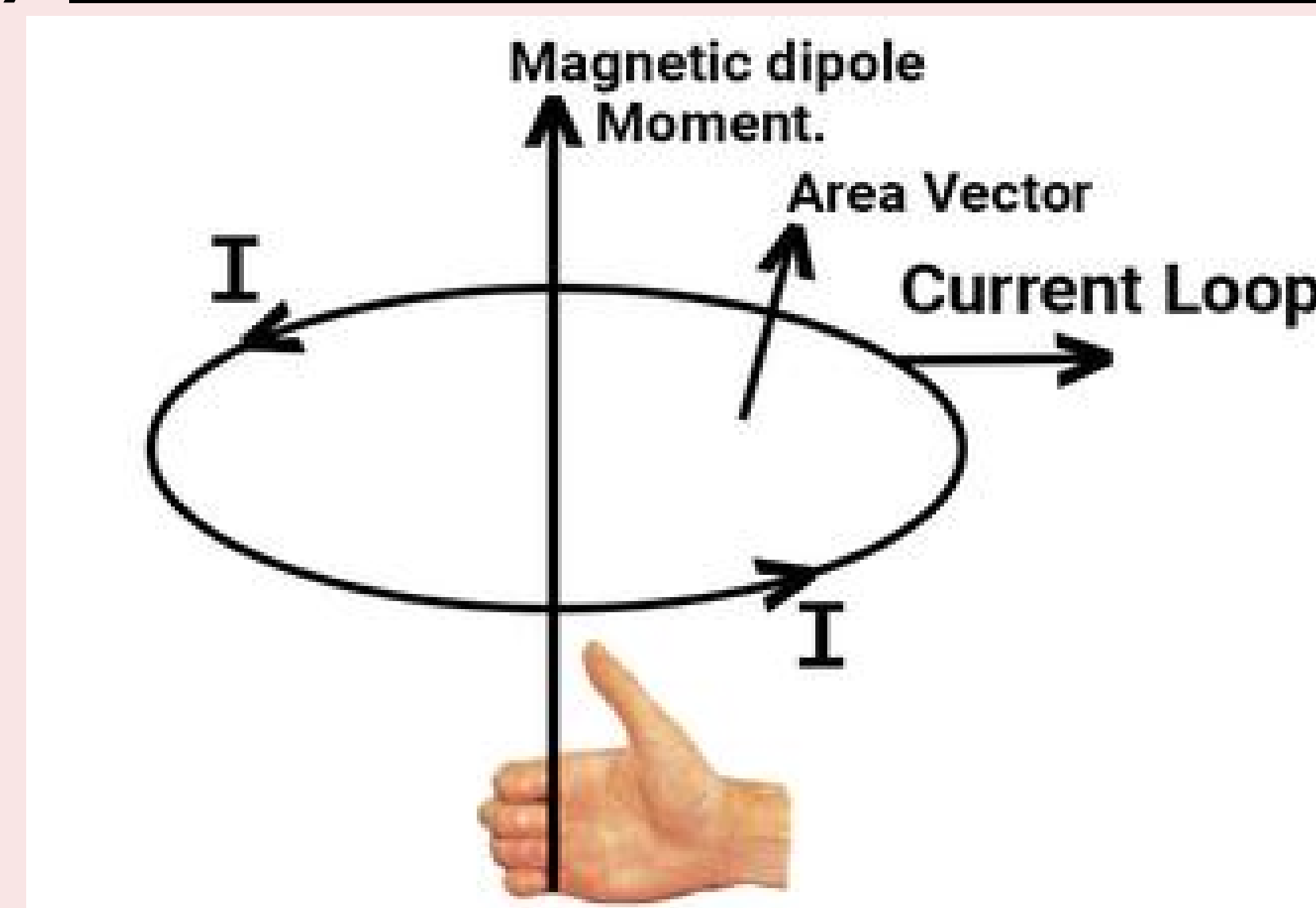
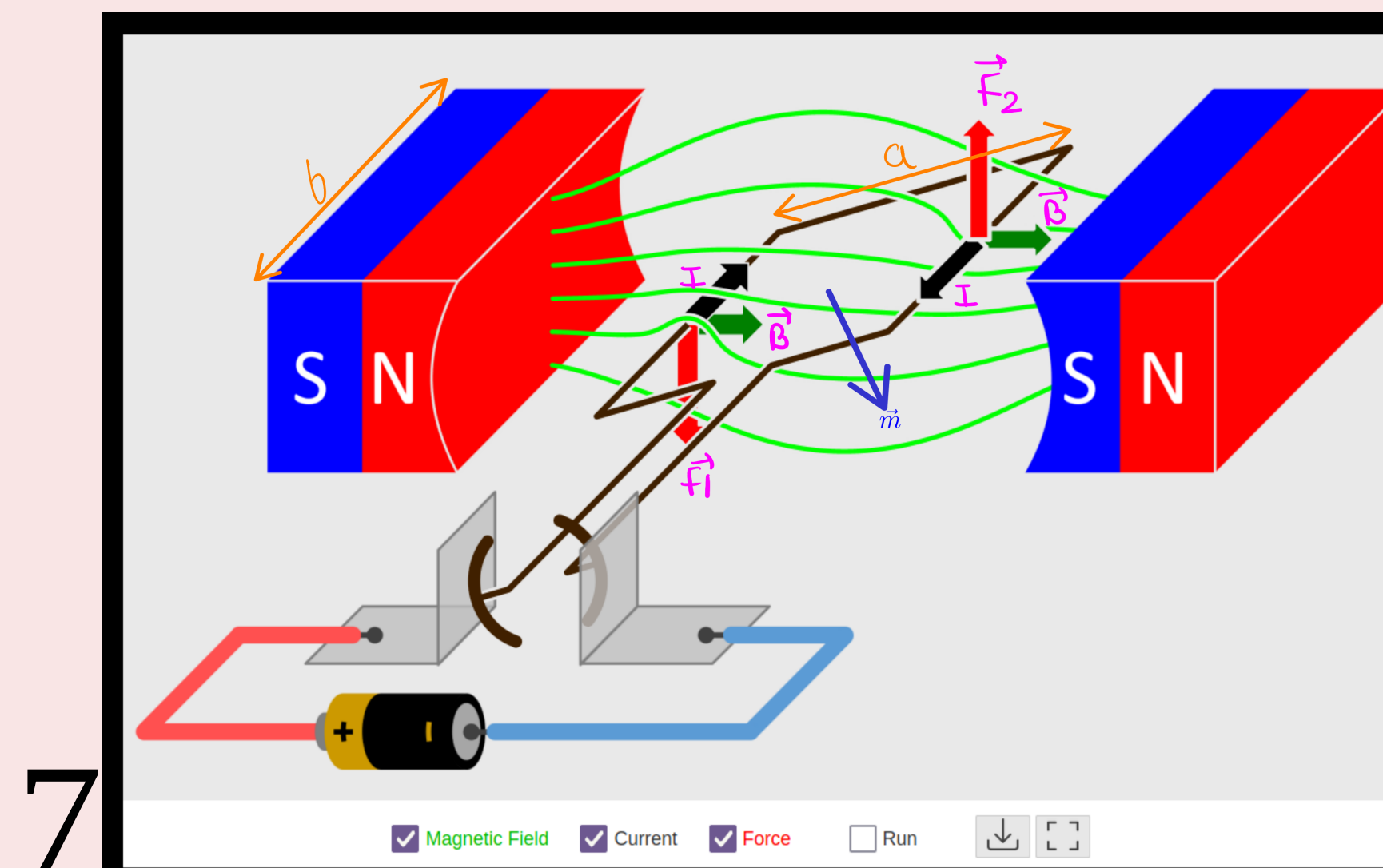
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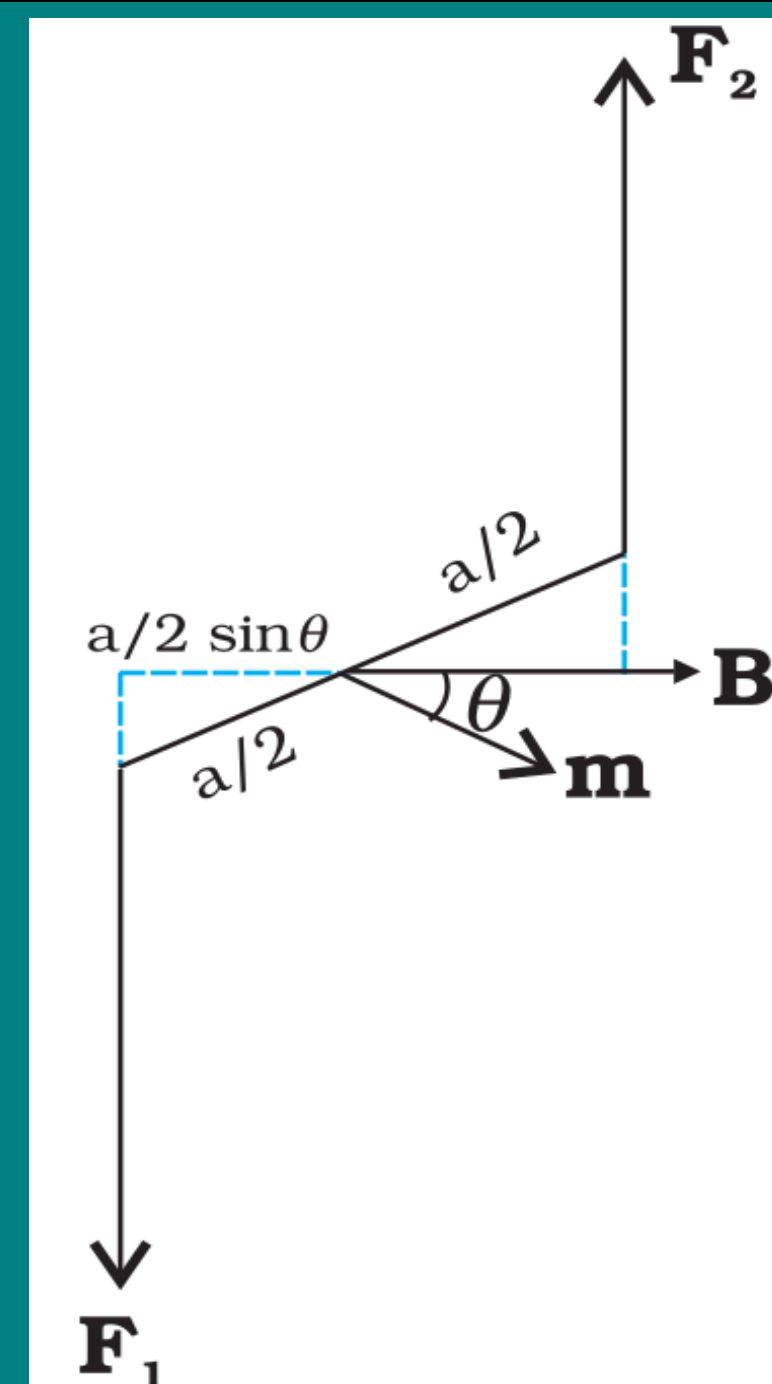
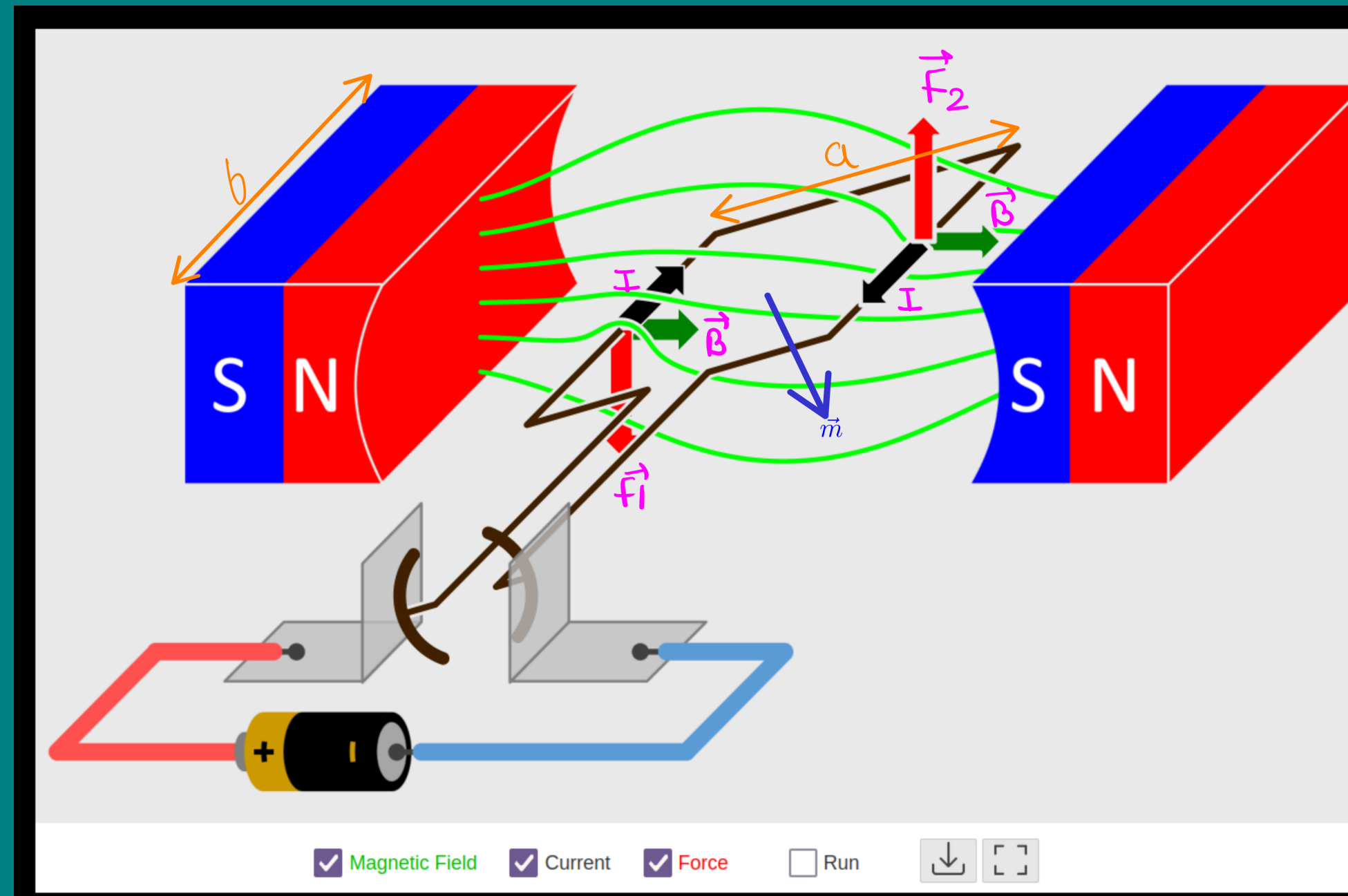
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- The **direction** of the **area vector** $\vec{A}$ is **given** by the **right-hand thumb rule**.

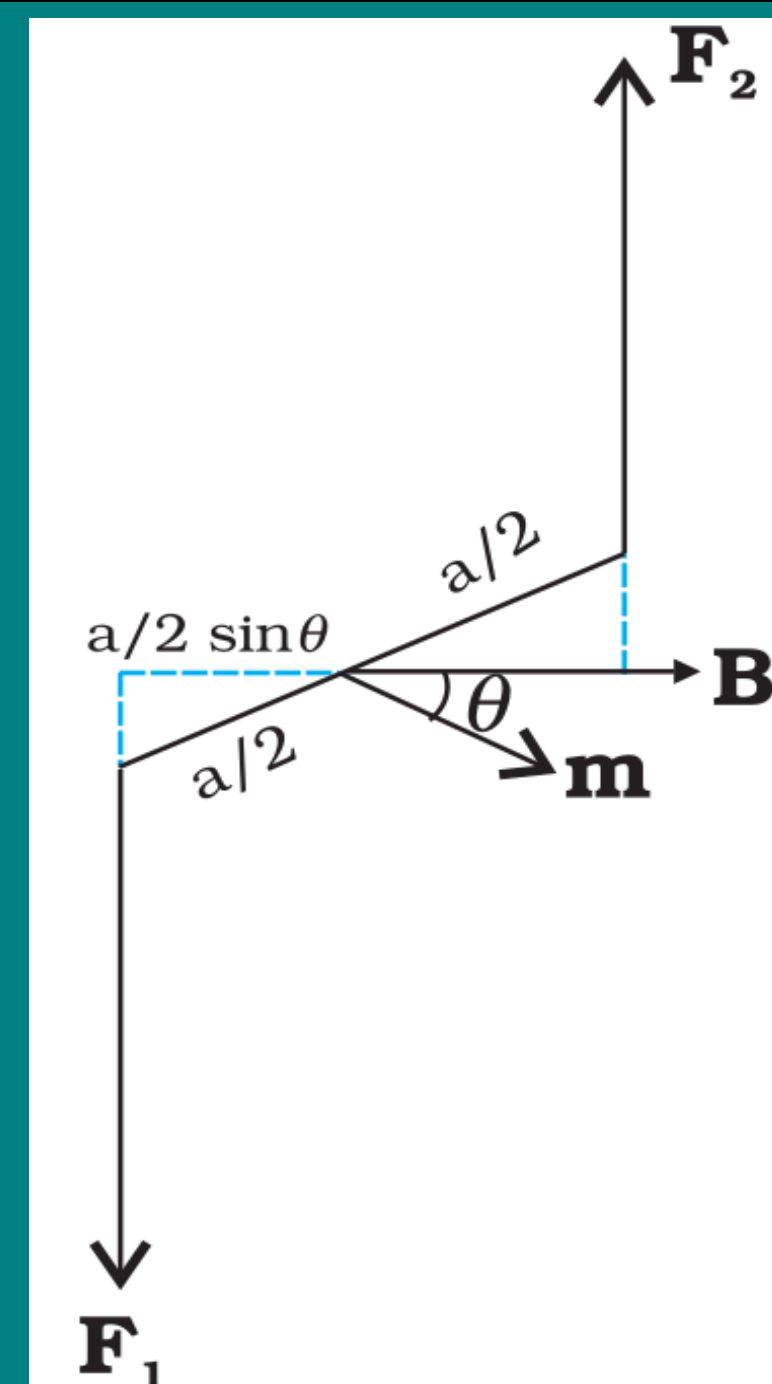
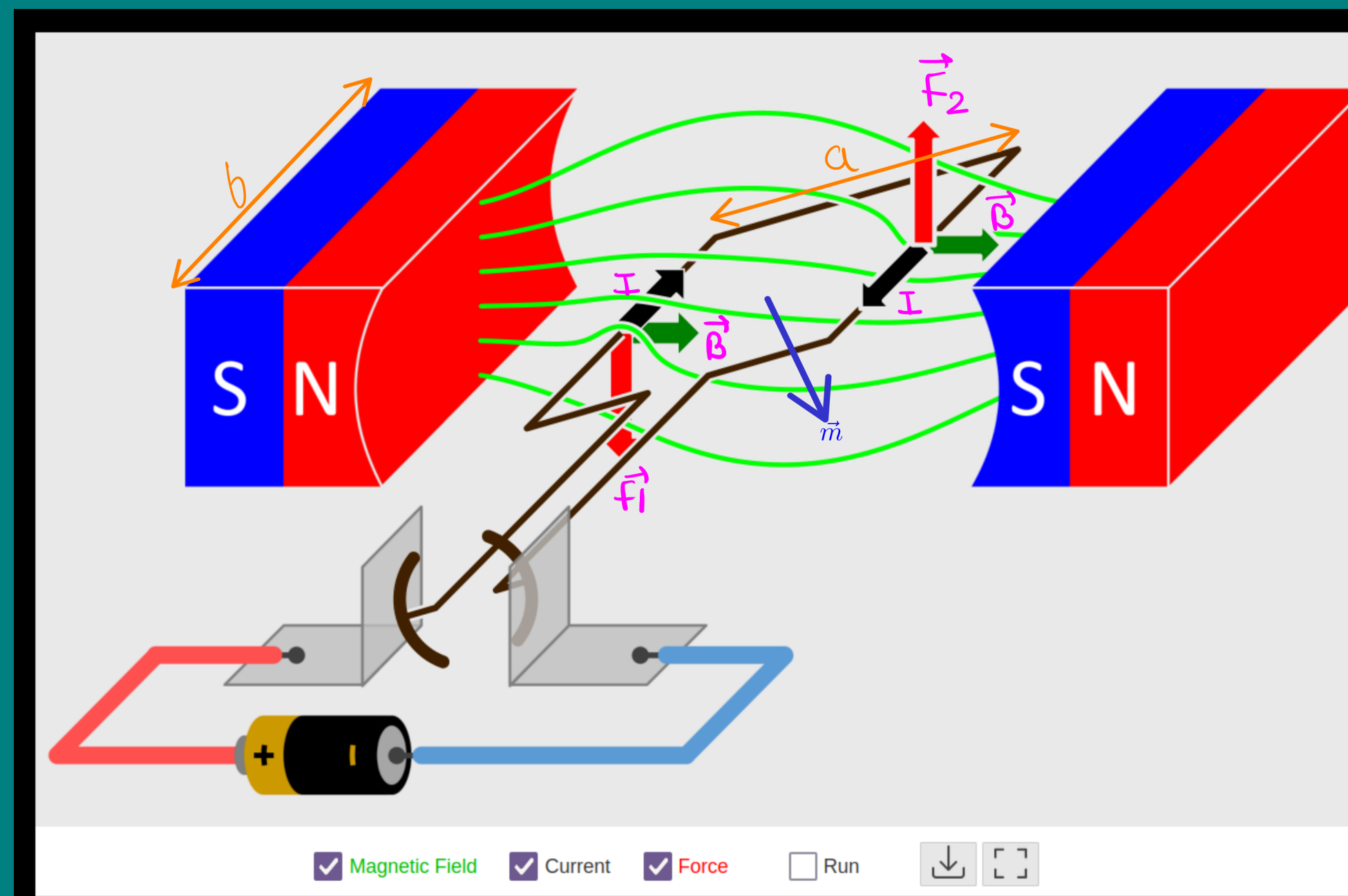
Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

- This **torque** ($\vec{\tau}$) can be **expressed** as **vector product** of the **magnetic moment** ($\vec{m}$) of the coil **and** the **magnetic field** ($\vec{B}$)

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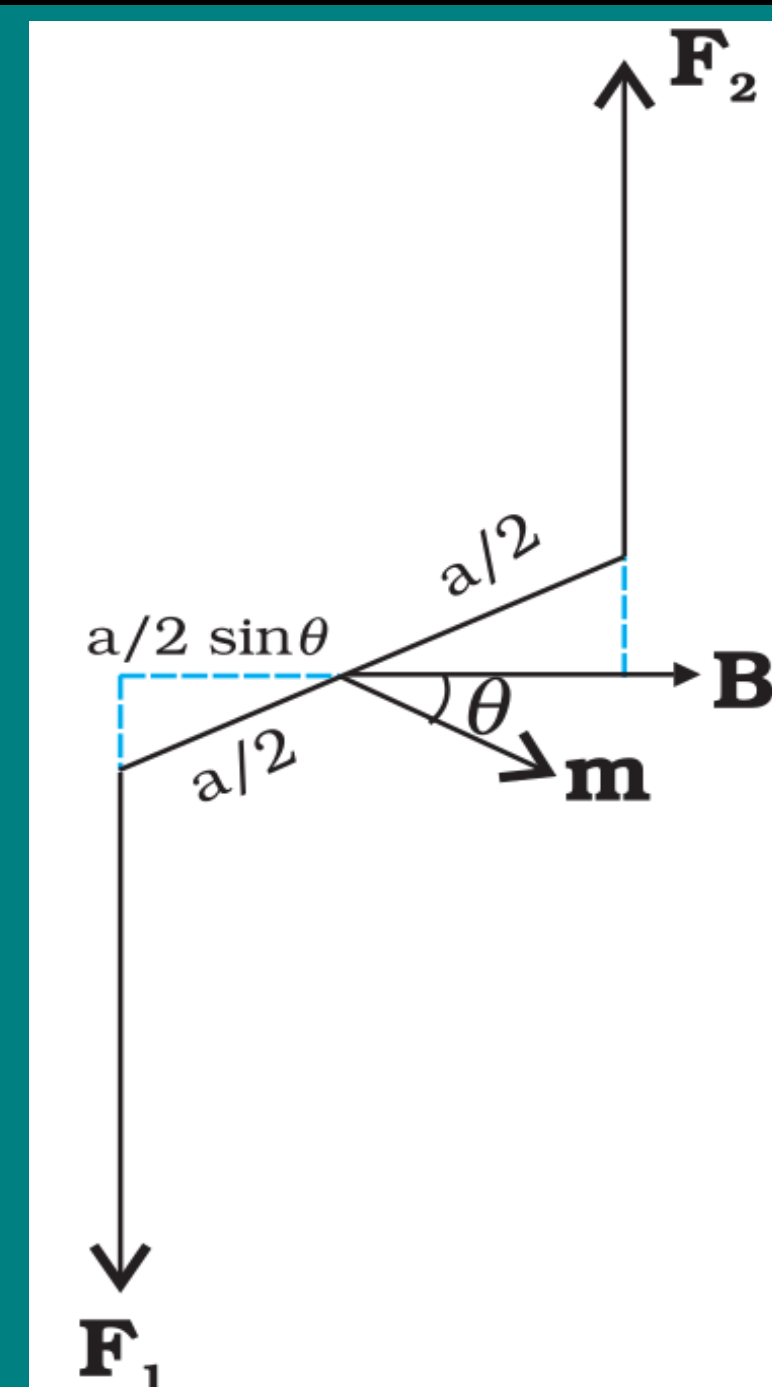
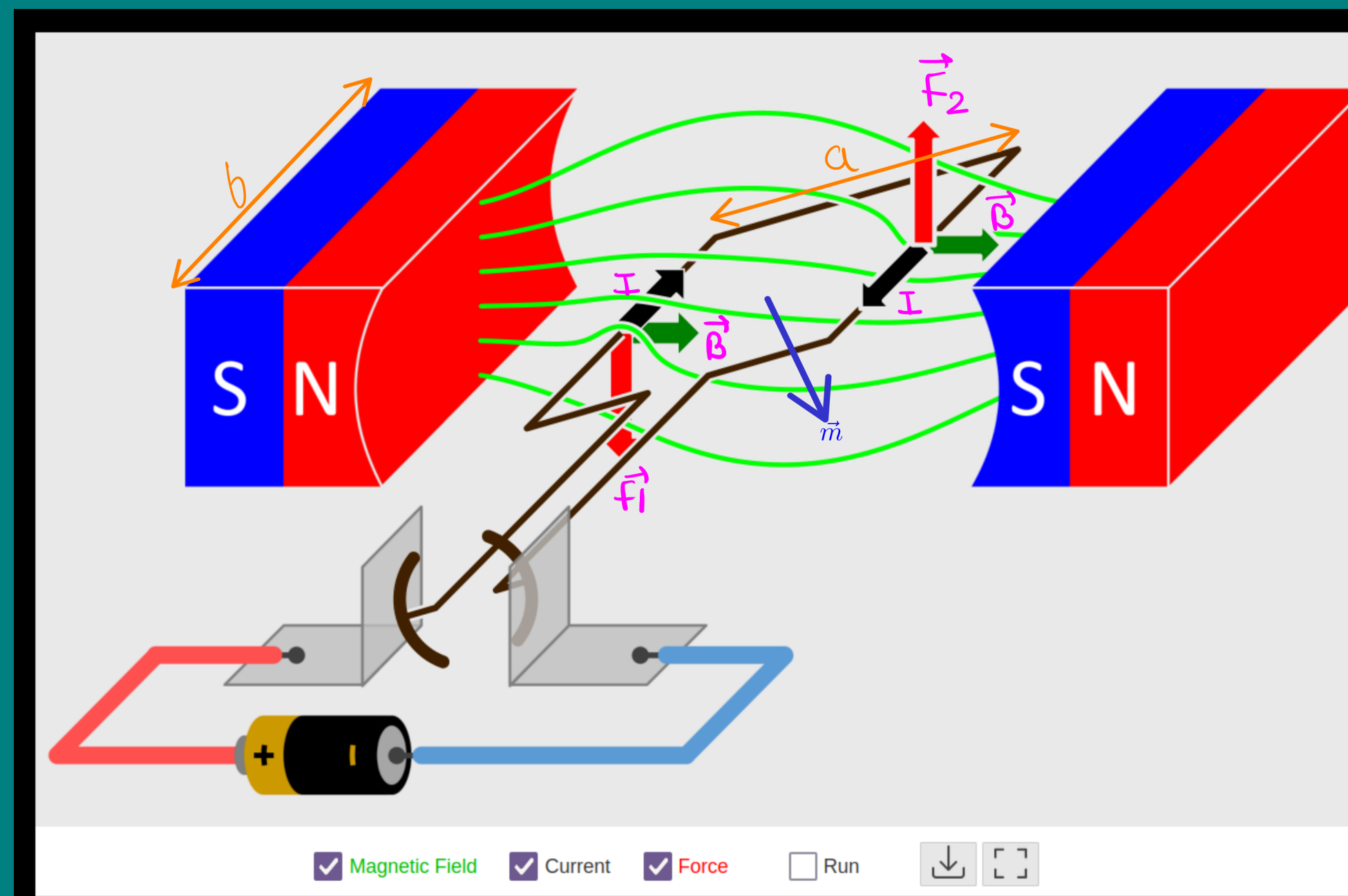


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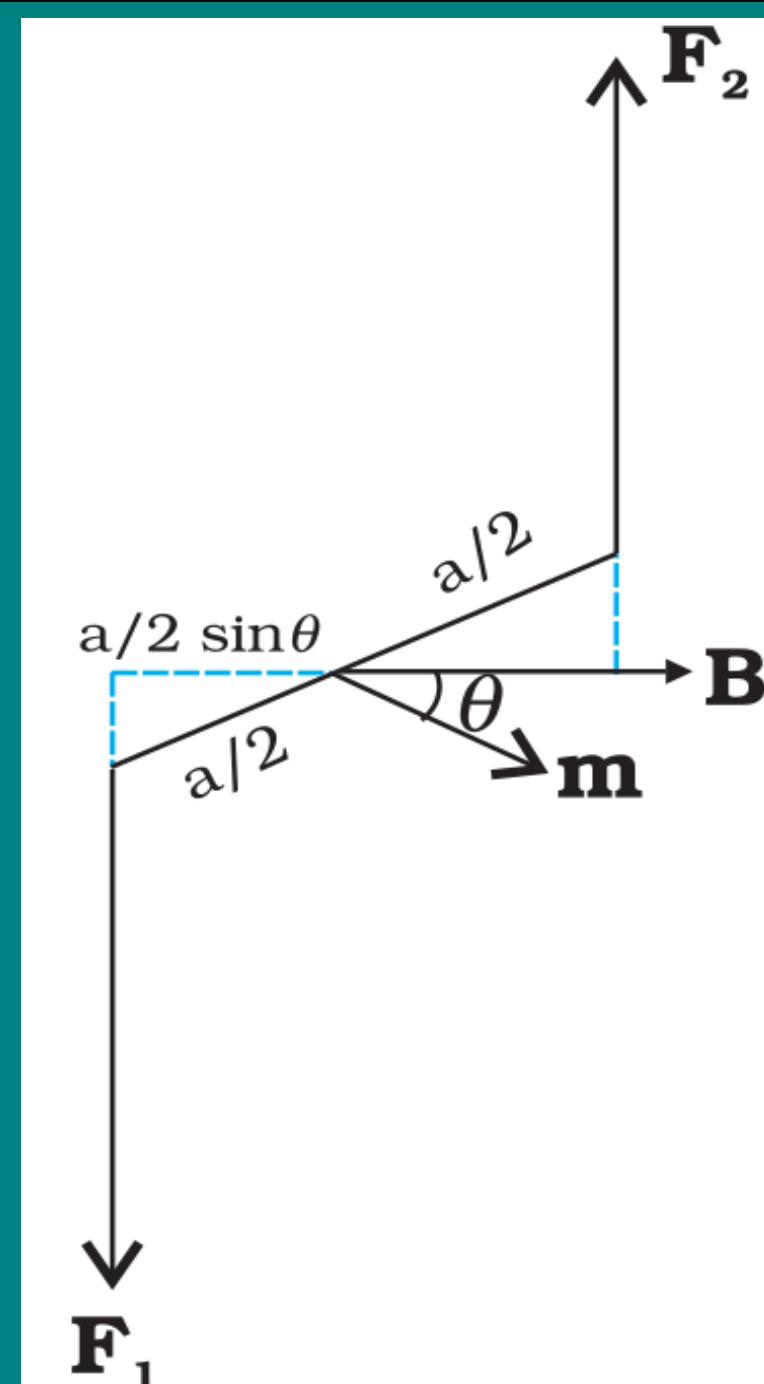
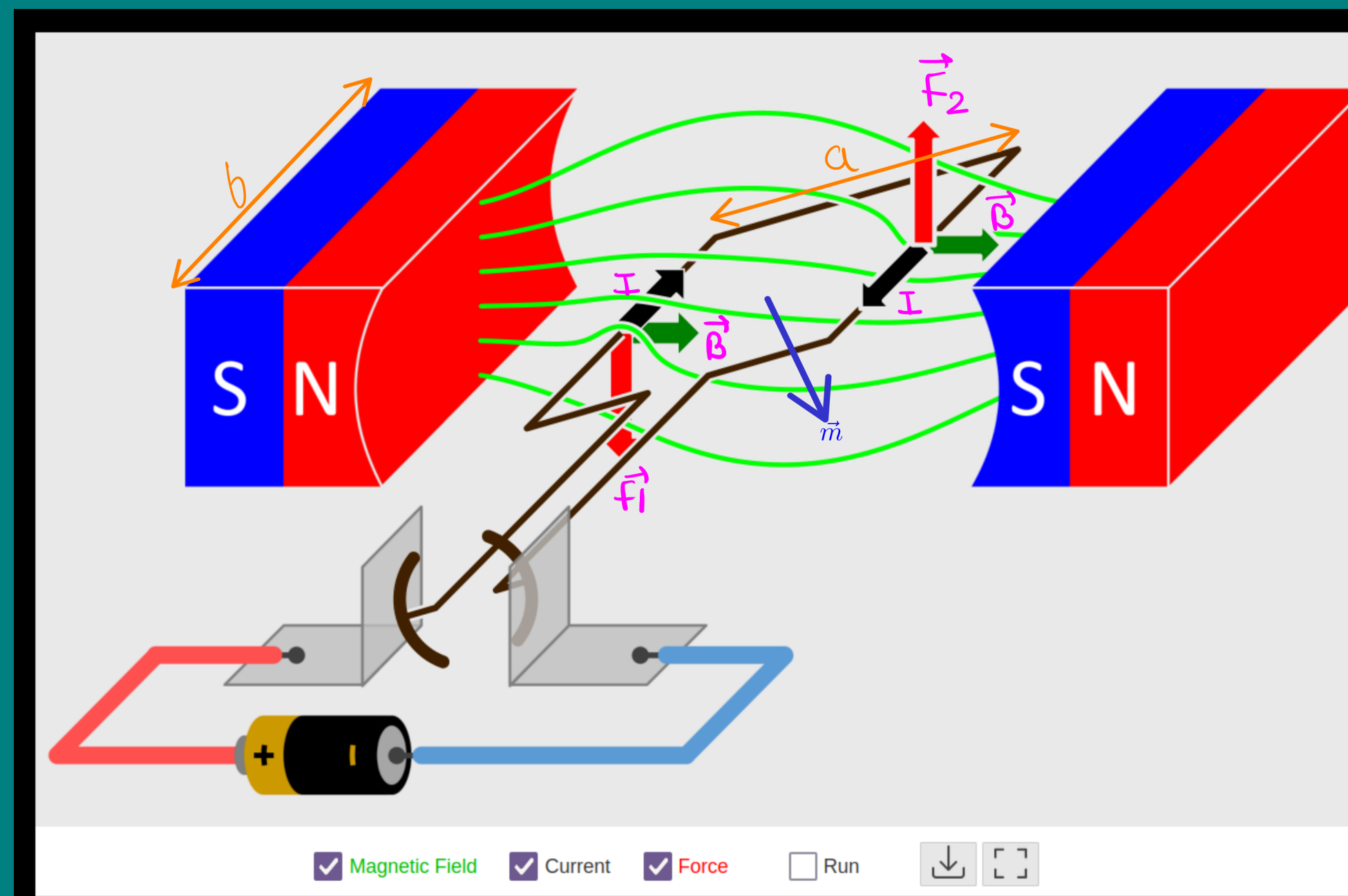
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- The **torque** ($\vec{\tau}=0$) **vanishes** when $\vec{m}$ is either **parallel or antiparallel** to the **magnetic field** $\vec{B}$

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- When $\vec{m}$ and $\vec{B}$ are **parallel** ($\theta = 0^\circ$) the **equilibrium is a stable one**. **Any small rotation** of the coil **produces a torque** which **brings it back** to its **original position**.

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- When they are **antiparallel** ($\theta = 180^\circ$), the **equilibrium is unstable**.
- If the **loop has N** closely wound **turns**, the expression for **torque** is

$$\vec{\tau} = N I \vec{A} \times \vec{B}$$

Example 4.11

A **100 turn** closely wound circular coil of **radius 10 cm** carries a **current of 3.2 A**.

(a) **What is the field at the centre of the coil?**

(b) **What is the magnetic moment of this coil?**

The coil is placed in a vertical plane and is free to rotate about a horizontal axis which coincides with its diameter. A **uniform magnetic field of 2 T** in the horizontal direction exists such that **initially the axis of the coil is in the direction of the field**. The **coil rotates** through an **angle of 90°** under the influence of the magnetic field.

(c) **What are the magnitudes of the torques on the coil in the initial and final position?**

(d) **What is the angular speed** acquired by the coil when it has rotated by 90° ? The **moment of inertia** of the coil is 0.1 kg m^2 .